CONTRACTIBILITY OF THE COMPLEX OF INCOMPRESSIBLE SEIFERT SURFACES: THE KNOT CASE OF KAKIMIZU'S PROBLEM

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Abstract. Let $K \subset S^3$ be a non-trivial knot and let $\mathrm{IS}(K)$ be the simplicial complex whose vertices are the ambient isotopy classes of incompressible Seifert surfaces in the exterior $E(K)$, a finite set of distinct vertices spanning a simplex exactly when its classes admit simultaneously pairwise disjoint representatives. Kakimizu proved that $\mathrm{IS}(K)$ is connected; whether it is contractible was asked by Przytycki and Schultens, who also identified the obstruction, namely that the projection used in the connectedness proof is not known to be well defined on isotopy classes. We prove that $\mathrm{IS}(K)$ is contractible, and likewise every truncation $\mathrm{IS}_\ell(K)$ on the vertices of genus at most ℓ. The argument factors through a purely combinatorial theorem: every non-empty connected flag *exchange complex* is contractible, where an exchange complex carries a complexity function subject to two axioms which require only that an exchanging vertex *exist*, never that one be selected. That is what circumvents the obstruction. We locate the combinatorial theorem precisely against the existing literature — it is not implied by dismantlability, and, when all descending links are finite, it follows from Zaremsky's descending-link criterion by a reindexing (§7.4) — and we record what remains genuinely open. Two extensions are proved: the truncations above, and the case of links satisfying a linking condition that forces every spanning surface to be connected.

1. Introduction

1.1. **The problem.** Let $K \subset S^3$ be a knot with exterior $E(K) = S^3 \setminus \mathrm{Int}\, N(K)$. A *Seifert surface* for K is a compact connected orientable surface properly embedded in $E(K)$ whose boundary is a longitude of K; it is *incompressible* if the inclusion induces an injection on fundamental groups. Kakimizu [12] introduced two simplicial complexes attached to K: the complex $\mathrm{IS}(K)$ whose vertices are the ambient isotopy classes of incompressible Seifert surfaces, and the full subcomplex $\mathrm{MS}(K)$ on the classes of minimal genus. In both, a finite set of distinct vertices spans a simplex exactly when its classes admit simultaneously pairwise disjoint representatives.

Kakimizu proved that these complexes are connected. We shall use his statement in the form in which he printed it [12, Theorem A, p. 226]:

Let L be a non-split oriented link. Then both $\mathrm{IS}(L)$ *and* $\mathrm{MS}(L)$ *are connected.*

The two cases are not proved by a common argument: the proof of the underlying engine [12, Theorem 2.1, p. 228] splits on p. 229 into a minimal-genus case and an incompressible case, the latter run through McMillan's simple moves. This matters here, because it is the incompressible half that we need and it is the half that is easy to lose sight of when the result is quoted for MS alone. We also use Kakimizu's sharper statement [12, Proposition 3.1(1), p. 231] that the edge-path metric of $\mathrm{IS}(L)$ coincides with the distance d defined through the infinite cyclic cover.

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Whether $\mathrm{IS}(K)$ is contractible was asked by Przytycki and Schultens [15]. We attribute the question to them rather than to [12], where it does not appear. Przytycki and Schultens proved that $\mathrm{MS}(K)$ is contractible, and in the same paper isolated the reason their method does not reach $\mathrm{IS}(K)$: their projection is defined by choosing, for a surface S and a vertex v, a distinguished representative of v disjoint from S, and for incompressible surfaces of unbounded genus that choice is not known to descend to isotopy classes.

1.2. **Results.** Our main result is the following.

Theorem A. *Let $K \subset S^3$ be a non-trivial knot. Then* $\mathrm{IS}(K)$ *is contractible.*

The hypothesis excludes only a case that is settled and trivial: IS(unknot) is a single point, by Proposition 3.2, hence contractible. We nevertheless keep the exclusion, because every statement of §5 is proved for a non-trivial knot and we prefer not to carry a hypothesis in one place and drop it in another.

The proof factors through a combinatorial statement that involves no three-dimensional topology.

Definition 1.1 (Exchange complex)**.** Let X be a connected flag simplicial complex, let W be a well-ordered set, and let $c\colon X^{(0)} \to W$. For vertices u, w at distance 2 in the 1-skeleton write $\mathcal{I}(u, w)$ for the set of common neighbours of u and w. Call $x \in \mathcal{I}(u, w)$ an *apex* if x is equal or adjacent to every member of $\mathcal{I}(u, w)$; equivalently, the full subcomplex on $\mathcal{I}(u, w)$ is a cone with apex x. The pair (X, c) is an *exchange complex* if

- (N1) every two-interval has an apex; and
- (N2) whenever $\mathrm{dist}(u, w) = 2$, *some* apex x of $\mathcal{I}(u, w)$ satisfies $c(x) < \max\{c(u), c(w)\}$.

This is Definition 6.1, repeated verbatim. Note that $\mathcal{I}(u, w) \neq \emptyset$ is automatic when $\mathrm{dist}(u, w) = 2$ and is therefore not an axiom; what (N1) asks is that the interval be a cone.

Theorem B. *Every non-empty connected flag exchange complex is contractible.*

The point of the two axioms is what they do *not* require. They ask that an exchanging vertex *exist*; they never ask that one be *selected*. This is precisely the obstruction identified in [15], and avoiding a choice is what lets the argument run on $\mathrm{IS}(K)$ rather than on $\mathrm{MS}(K)$. The price is paid in §8: the same feature that makes Theorem B applicable makes its equivariant analogue unavailable, and we explain there why that is not a defect of our proof but a theorem about the situation.

Two further results are proved. The first is free: it costs no additional hypothesis and no additional citation.

Theorem C. *For every integer $\ell \geq g(K)$ the truncation* $\mathrm{IS}_\ell(K)$*, the full subcomplex of* $\mathrm{IS}(K)$ *on the vertices of genus at most ℓ, is isometrically embedded in* $\mathrm{IS}(K)$ *and is contractible. The same holds for the full subcomplex on the vertices of genus at most ℓ and relative area at most a_0, for every $a_0 > 0$.*

Taking $\ell = g(K)$ recovers the contractibility of $\mathrm{MS}(K)$ [15, Theorem 1.1, p. 1490], a statement made there for spanning surfaces of minimal Thurston norm, which for a knot in S^3 is the minimal-genus condition. The second family in Theorem C does not appear in the literature under any name. The connectedness of $\mathrm{IS}_\ell(K)$ is part of Theorem C, obtained there from Corollary 6.7; it has an independent source in Chen and Shen [4, Proposition 5.15, p. 9 of v4], which we do not use.

The second extension concerns links, and requires a condition. Call a link $L = L_1 \cup \cdots \cup L_n$ *linking-indecomposable* if for every partition $\{1, \ldots, n\} = I \sqcup I^c$ into two non-empty parts there is an index m with $\mathrm{lk}(L_m, L_J) \neq 0$, where L_J is the sublink indexed by the part *not* containing m. The condition is vacuous for $n = 1$. Its role is to force every spanning surface to be connected, so that Kakimizu's vertices — which are allowed to be disconnected, and are required only to be incompressible componentwise — coincide with ours.

Theorem D. *Let L be a non-split, linking-indecomposable link in S^3 which is not the unknot. Then* $\mathrm{IS}(L)$ *is contractible.*

1.3. **Where the combinatorial theorem sits.** Theorem B is a statement about arbitrary flag complexes. We locate it against the three results with which it might be confused; the details are in §7.4.

Dismantlability. Deleting a dominated vertex, the mechanism behind [15, §7], requires the descending link of that vertex to be a *cone*. Axioms (N1) and (N2) give only that it is *contractible*, which is strictly weaker: §7.3 exhibits a seven-vertex exchange complex whose descending link is a hexagon with three chords, contractible but not a cone. So Theorem B is not a corollary of the dismantlability theory, and the route through [2] is closed.

The descending-link criterion. Zaremsky's form of the Bestvina–Brady criterion [20, Lemma 2.3] assumes no local finiteness, so the failure of local finiteness for $\mathrm{IS}(K)$, recorded by Kakimizu on [12, p. 231], is irrelevant to the comparison. Write (F) for the condition that every descending link $L_c(v) = \{u \in \mathrm{lk}(v) : c(u) < c(v)\}$ is finite. We prove in §7.4 that a countably infinite exchange complex satisfying (F) admits a bijection $h\colon X^{(0)} \to \mathbb{N}$ with $\mathrm{lk}^{\downarrow}_h(v) = L_c(v)$, and that Theorem B then follows from [20, Lemma 2.3] at $t = 0$ with an induction on the number of vertices; without (F) that reindexing fails, at a step identified there. The residual content of Theorem B is therefore confined to exchange complexes with an infinite descending link, and we do not know whether $\mathrm{IS}(K)$ is one.

The ordering theorem. The complexity-convex geodesics and the isometric embedding of the truncations are both special cases of [3, Lemma 9.13], whose hypotheses are strictly weaker — no apex, no flagness, no local finiteness. We nevertheless give the argument, as Corollary 6.7, so that §6 is self-contained and [3] is a comparison rather than a link in the chain leading to Theorems A–D.

1.4. **The shape of the argument.** The proof of Theorem A verifies that $\mathrm{IS}(K)$ is an exchange complex for the complexity

$$c(v) = \big(g(v),\, A(v)\big) \in \mathbb{N} \times \mathbb{R}_{>0},$$

ordered lexicographically, where $g(v)$ is the genus and $A(v)$ the infimum of area over the class. Axiom (N1) is supplied by three things together: the connectedness of $\mathrm{IS}(K)$, which is Kakimizu's Theorem A and is what makes a two-interval non-empty; flagness; and Proposition 5.11(1), which manufactures an apex of the interval out of the exchange lemma. Axiom (N2) is the substance, and is supplied by an exchange lemma for incompressible surfaces (§5) proved by minimal-surface methods: given two vertices at distance 2, one produces a common neighbour of strictly smaller complexity by cutting and recombining area minimisers in the two classes along their intersection.

That construction needs four inputs about area minimisers in $E(K)$. Three are cited (§2): the universal disjointness of minimisers [16, Theorem 4, p. 885; Theorem A.5, p. 899], and the attainment of the area infimum in each of the two competing classes [8, Theorem 6.12, pp. 112–113], [16, Theorem 2 = Corollary A.4]. The fourth is the boundary regularity of a relative area minimiser, labelled (R1): that some minimiser in the given proper isotopy class is smooth up to $\partial E(K)$ and meets $\partial E(K)$ exactly in its own boundary. It too is cited: read with the definition on [16, p. 897], the corollary just quoted for the sliding class delivers the minimiser as the image of a *smooth proper embedding*, which is (R1) verbatim. No step of this paper therefore stands on anything unproved and uncited. We nevertheless prove it again in §2.2 as Theorem 2.10, and the second proof is self-contained in a way the citation is not: it takes from [8] only the existence statement, that the area infimum over the piecewise smooth surfaces isotopic to F rel ∂F is attained, and derives everything else from the first variation of area and Hopf's boundary point lemma. The reason for doing so is that the citation's own proof leaves a step out, and we say which (Remark 2.5).

The point deserves a sentence, because it is easy to mistake (R1) for something already in print. Corollary A.4 of [16, p. 898] passes from a minimiser among stable minimal surfaces to a minimiser in the whole sliding class by treating the fixed-boundary minimiser of [8, Theorem 6.12] as a member of $\mathcal{M}(S)$, that is, as a smooth proper embedding; but Theorem 6.12 as stated produces a minimiser only in the piecewise smooth category. Theorem 2.10 supplies the missing regularity, so that Corollary A.4 may be used as stated.

Two analytic contributions are genuinely ours and are not repetitions of textbook facts. The first is the self-contained regularity argument just described. Its engine is one lemma, that a minimal graph over a plane sector of opening angle at least π, lying in M and touching ∂M along one edge, cannot be tangent to ∂M at the vertex; that lemma also yields the transversality statement (R2) in three lines, in place of a divergence-form comparison and an explicit barrier, and it leaves §2 using nothing from [7, Ch. 8]. The second is the collar conditions (34) and (33) of §5, which control the behaviour of the intersection pattern at a boundary tangency. Freedman, Hass and Scott note on [6, p. 636] that non-transverse boundary behaviour occurs; they do not supply a structure theorem for it, and §5.3 does.

Organisation. The paper falls into two halves that meet only at Definition 6.1.

The geometric half is §§2–5. §2 fixes the metric, states the two citations (U) and (E), and proves the regularity statements (R1) and (R2). §3 pins down the definitions of vertex, edge and simplex against [12], and proves flagness and the two compression lemmas. §4 introduces the complexity $c = (g, A)$, proves that the values it attains are well-ordered, and gives the cyclic-cover construction that manufactures the candidate apices. §5 is the exchange lemma itself, and is the longest section: rounding and pushing off in §5.1, the transverse case in §5.2, and the two non-transverse configurations in §§5.3–5.5.

The combinatorial half is §§6–7, and uses no three-dimensional topology at all: §6 sets out the axioms and their heredity, §7 proves Theorem B and locates it against dismantlability and the descending-link criterion. §8 puts the halves together and proves Theorems A, C and D, and then says where the method stops.

A reader who wants only the combinatorics may begin at §6; a reader who wants only the geometry may stop after §5, whose conclusion, axiom (N2) for IS(K), is Proposition 5.11.

1.5. **Open questions.** The first is not on the same footing as the others, and we put it first for that reason: it is the one input of this paper that is asserted in the literature rather than demonstrated there or here.

> **Problem.** *Let F be an incompressible surface properly embedded in a compact P^2-irreducible Riemannian 3-manifold M with strictly convex boundary, with ∂F a smooth closed 1-manifold in ∂M. Is the infimum of area over the piecewise smooth surfaces properly isotopic to F rel ∂F attained within that class?*

By Proposition 2.9 this is equivalent to asking whether the infimum is attained by a *smooth embedded minimal surface*, so it is a question of a familiar kind and not one about corners. It is asserted by [8, Theorem 6.12]; the printed proof of that theorem, which is the sentence on [8, p. 112] extending the closed case, establishes the interior regularity of the limit and does not reach its behaviour at ∂M. Everything geometric in this paper, and everything resting on the appendix to [16], rests on the answer being yes. Remark 2.2 gives the evidence that it is: the degeneration that makes the closed statement fail is excluded once $\partial F \neq \emptyset$, and we check that directly for a knot or link exterior.

Two things narrow the problem, and we record them because they say where to push. The first is that the boundary *convergence* is not what is missing: it is already available from [8, Lemma 6.9] read with the two faces of a half-ball exchanged from the way that paper reads them, since it is the intersection with ∂M, and not the intersection with the spherical face, that a minimising sequence holds fixed. Remark 2.2 writes out the assignment. What is missing is that the replacement performed inside such a half-ball keep the surface in the class; and that comes down to a local question about the Plateau problem, namely whether the least area disc spanning a piecewise smooth Jordan curve with corners — one of whose arcs lies on the mean-convex ∂M — is itself piecewise smooth up to that curve.

The second is a list of routes around that local question which do *not* work, recorded so that the next person need not walk them again.

(a) Replacing by a piecewise smooth *near*-minimiser instead of the exact least area disc. This needs the infimum of area over piecewise smooth embedded discs to agree with the infimum over all finite-energy maps, which is the local question restated.
(b) Localising in a region containing a whole component of ∂F, so that the frontier misses ∂F and no corner arises. The corner does disappear, but the piece cut out is then an annulus and not a disc, and the replacement lemmas of [8, §6] are lemmas about discs.
(c) Localising in a region whose frontier meets ∂M tangentially, so that $\partial\Omega$ and the cut curve are C^1. The tangency can be arranged, but a frontier tangent to ∂M is not known to be sufficiently convex in the sense of [8, p. 110], and the cut curve is then only C^1 where the corner theory wants $C^{1,\alpha}$.
(d) Replacing only the disc components that miss ∂M, whose boundaries are circles on the spherical face and carry no corner. Then nothing is replaced in the collar of ∂F, and there is no convergence there.
(e) Weakening Definition 2.1 to allow finitely many exceptional boundary points. The first variation and the gluing survive, a finite set being null; but an exceptional point in the interior then needs a removable singularity theorem, and an exceptional point on ∂F returns to the local question.

- We do not know whether $\mathrm{IS}(K)$ satisfies (F). Equivalently, we do not know whether a vertex of $\mathrm{IS}(K)$ can have infinitely many neighbours of smaller genus. If it cannot, the combinatorial half of this paper reduces to [20, Lemma 2.3].
- Theorem D does not extend to general links: the step that fails is identified in §8.4, and we give no counterexample, only an obstruction.
- The equivariant version, for a knot with symmetry group G acting on $\mathrm{IS}(K)$, is *blocked* rather than merely unproved; the obstruction is cited in §8.4.
- There is no analogue for a general Haken manifold, because the required connectedness of the corresponding complex has no source in the literature; see §8.4.

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2. Geometric inputs

This section fixes the Riemannian data on $E(K)$ and assembles the facts about area minimisers that the exchange lemma of §5 consumes. §2.1 fixes the metric and states the two citations, the universal disjointness (U) and the attainment of the area infimum (E). §2.2 proves the regularity statements (R1) and (R2): that a minimiser is smooth up to $\partial E(K)$, meets it exactly in its own boundary, and meets it transversally. (R1) is also contained in the statement of the second citation, as Remark 2.5 explains; the proof given here takes less from the literature than that reading does, and that is its purpose.

2.1. **The metric, and the two citations.** Fix on $E(K)$ a *relative Riemannian metric*, that is, one making $\partial E(K)$ strictly convex. Such a metric exists: on a collar $[0, \varepsilon)_r \times \partial E(K)$, with r the inward coordinate and $\partial E(K) = \{r = 0\}$, take

$$dr^2 + \varphi(r)^2\, g_{\partial E(K)}, \qquad \varphi(0) = 1, \quad \varphi'(0) < 0, \tag{1}$$

say $\varphi(r) = 1 - r$, and extend smoothly; strict convexity is the only source of a barrier anywhere in this paper. Two consequences of (1) are used repeatedly and are recorded now. Along the collar $\nabla_X \partial_r = (\varphi'/\varphi)X$ for X tangent to the level sets, so

$$\operatorname{Hess} r = \varphi\varphi'\, g_{\partial E(K)} \;<\; 0 \qquad \text{on the level sets,} \tag{2}$$

and in particular r is strictly concave along any geodesic with initial velocity tangent to a level set: a geodesic joining two points of $\partial E(K)$ therefore has $r > 0$ at its interior points. Dually, the second fundamental form of $\{r = 0\}$ with respect to the *inward* normal ∂_r is $-\varphi\varphi' g_{\partial E(K)} = g_{\partial E(K)} > 0$, so $\partial E(K)$ has mean curvature $2 > 0$ with respect to the inward normal, which is the sign convention of [8, p. 110]. We also write $\widetilde{E}$ for the manifold obtained by extending the collar to $r \in (-\varepsilon, \varepsilon)$ with the same formula (1); thus $E(K) \subset \widetilde{E}$ isometrically and $\partial E(K)$ is a two-sided smooth surface in $\operatorname{Int} \widetilde{E}$.

Fix also a *real-analytic* foliation J of the boundary torus by preferred longitudes; smoothness of J is all the citations below require [16, Definition 3, pp. 884–885], and real-analyticity, free here, is used in §5. A properly embedded $F \subset E(K)$ with $\partial F \subset J$ is *relatively least-area* if $\operatorname{Area}(F)$ is least in its proper isotopy class subject to the boundary lying in J: the boundary slides from leaf to leaf and is not pinned pointwise.

Definition 2.1 (Piecewise smooth surfaces)**.** A compact surface F properly embedded in a Riemannian 3-manifold X is *piecewise smooth* if there is a finite graph $\Gamma_F \subset F$, embedded as a finite union of smooth arcs and circles, *properly* — so that $\Gamma_F \cap \partial F$ is the finite set of endpoints of its arcs, and no arc runs along ∂F — such that $F \setminus \Gamma_F$ is a smooth submanifold of X and the closure of each of its components — its *pieces* — is a compact surface with corners, smooth in its interior and embedded of class C^1 up to its own boundary, the two edges at each corner having distinct tangent rays. A smooth surface is piecewise smooth, with $\Gamma_F = \emptyset$. Such an F admits a smooth triangulation compatible with Γ_F, and *isotopy* of piecewise smooth surfaces is understood throughout in the piecewise smooth category, in which the classical theorems of Alexander — that a locally flat 2-sphere in a 3-ball bounds a ball on each side, and that a homeomorphism of a ball fixing the boundary pointwise is isotopic to the identity rel the boundary — are available.

The clause about the pieces is stated with C^1 and not with C^∞ deliberately. The reading of "piecewise smooth" on which §2.2 depends should be as weak as the argument allows, and C^1 is what the argument allows: the elliptic boundary regularity used in Proposition 2.9 upgrades C^1 to C^∞ by itself, so demanding C^∞ of the pieces would be demanding something the proof does not need. The two normalisations — that Γ_F is properly embedded and that a corner has two distinct tangent rays — pull the other way, and are stated because §2.2 does consume them:

without the first, an arc of Γ_F could run inside ∂F, and without the second a piece could end in a cusp. Both are recorded in Remark 2.2, with the rest of what the class is being read as carrying.

Remark 2.2 (Whose definition this is). The term is Hass and Scott's: the class $\mathcal{F}$ of (E)(i) below is, in their words, "the set of piecewise smooth surfaces in M which are isotopic to F rel ∂F" [8, Theorem 6.12, p. 112]. *They do not print a definition of the term*, and since §2.2 feeds the minimiser F' they produce straight into an argument that consumes the stratified structure of Definition 2.1 — and consumes, in particular, the clause that $F \setminus \Gamma_F$ is a *smooth submanifold* of X, which is what carries that argument across the points of λ that $\Gamma_{F'}$ misses — the reading matters and we state it rather than leave it implicit.

Definition 2.1 is the reading their own construction supports. In the proof of [8, Theorem 5.1, pp. 106–107], from which Theorem 6.12 is obtained, the members of a minimising sequence are made transverse to the frontiers of finitely many balls and the outermost discs so cut off are replaced by least area discs with the same boundary; what is manipulated there is a surface smooth off a finite union of circles, which is Definition 2.1. That is evidence for the reading, not a proof of it: the infimum I is taken over $\mathcal{F}$ before any modification, so a strictly larger $\mathcal{F}$ would lower I.

We therefore use (E)(i) with $\mathcal{F}$ read as Definition 2.1, and we can say exactly how much of the reading is consumed. *Not* the minimality: the proof of Lemma 2.6 uses only that $\mathcal{F}$ is closed under diffeomorphisms of M fixing ∂M pointwise and that F' minimises over it, and both hold whatever the term means. What is consumed is the single clause

$$\text{the surface } F' \text{ of (E)(i) satisfies Definition 2.1,} \tag{3}$$

and that clause is the membership assertion *of the cited theorem*: Hass and Scott conclude "there is a surface F' *in* $\mathcal{F}$ with area I", and $\mathcal{F}$ is by their own description a class of piecewise smooth surfaces. We are taking a published theorem at its word about the class its output lies in, having said what we take the word to mean. If a strictly larger class were intended — one whose members are not required to be smooth up to their own boundary — then (3) would not follow from their statement and Proposition 2.9 would have nothing to act on. That is the precise exposure, it is confined to (3), and everything else in §2.2 is proved from the first variation of area and Hopf's boundary point lemma.

Two things put (3) in proportion. It asks as little as the method permits: what Proposition 2.9 needs of the pieces is C^1 up to their own boundary, and Definition 2.1 asks exactly that. Beyond regularity it asks two normalisations of the stratification — that $\Gamma_{F'}$ meet λ in finitely many points rather than running along it, and that no piece end in a cusp. Neither is free, and we do not know how to delete either without the existence theorem discussed below: an arc of $\Gamma_{F'}$ lying inside λ cannot simply be erased, since the definition would then demand smoothness along it. Both are, however, absent from every surface the constructions of [8] manipulate, those being smooth off a finite union of circles. And it is strictly weaker than what [16, Corollary A.4] needs, whose non-emptiness step, recalled in Remark 2.5, requires the same minimiser to be *smooth* up to ∂M; §2.2 is the passage from the one to the other, so this paper's exposure is strictly smaller than that of the literature it stands on.

What would remove (3) is not a regularity theorem — that is what §2.2 supplies — but an *existence* theorem, the one input of this paper asserted in the literature rather than demonstrated there or here:

$$\textit{the infimum of area over the surfaces of Definition 2.1 properly isotopic to } F \textit{ rel } \partial F \textit{ is attained within that class.} \tag{4}$$

Theorem 6.12 of [8] asserts (4), with $\mathcal{F}$ so read. Its proof is the sentence on [8, p. 112] that the existence results of §§4–5 extend to a manifold with sufficiently convex boundary once Lemmas 6.9

and 6.11 replace Lemmas 3.3 and 3.6, together with the record, on the same page, that in those applications the 1-manifold Γ is empty. With Γ empty the convergence of [8, Lemma 6.9] is smooth in $X \setminus Y$ with Y all of ∂X, so it says nothing at ∂M; and the surfaces of §§4–5 are closed, so nothing there meets ∂M at all. The membership clause of Theorem 6.12 is therefore, as far as we can tell, a statement whose printed proof does not reach it.

Where a proof would have to push is not, however, the convergence. *That* is already in [8], in Lemma 6.9 read with the two faces of a half-ball exchanged from the way they read them. Take $X = \overline{B}(p, \rho) \cap M$ for $p \in \lambda$ and put $Y := \partial B(p, \rho) \cap M$, strictly convex for small ρ, and $Y' := \overline{B}(p, \rho) \cap \partial M$, so that $\Gamma := \lambda \cap X$ is a compact 1-manifold properly embedded in Y'. A minimising sequence for (4) has a *fixed* intersection with ∂M, namely Γ, and a varying one with the spherical face; so it is Y', not Y, that carries the fixed data, which is the assignment Lemma 6.9 asks for. Its conclusion is then smooth convergence in $X \setminus Y$ — that is, up to the interior of $\Gamma \subset \partial M$ — to an embedded minimal surface with boundary $\operatorname{Int} \Gamma$, the alternative that a piece of the limit lies in Y' being excluded because a strictly convex ∂M contains no minimal subsurface. Finitely many such half-balls put every point of λ in the interior of some Y'.

What is not available is the other half: that the replacement performed inside such a half-ball keeps the surface in $\mathcal{F}$. The replacement substitutes a least area disc for a disc of the surface, and the boundary of that disc is a piecewise smooth Jordan curve with two corners, where the arc on the spherical face meets the arc on ∂M. Whether the least area disc spanning such a curve is itself piecewise smooth up to it — a local question about the Plateau problem at a corner — we do not know. That, and not the boundary convergence, is what (4) appears to come down to. Five ways of getting round the local question, none of which works, are listed in §1.5.

Two things can be said in its favour. By Proposition 2.9, (4) is *equivalent* to the assertion that the infimum is attained by a smooth embedded minimal surface, so it is a statement of a familiar kind and not a technicality about corners. And the degeneration that makes the corresponding closed statement fail is excluded here: [8, Theorem 5.1] carries the alternative that a one-sided surface F'' of area $I/2$ appears with $\partial N(F'') \in \mathcal{F}$, and for a knot or link exterior that cannot happen, since ∂F would be the ∂I-bundle over $\partial F''$ inside ∂M, forcing on each boundary torus T_i either λ_i disconnected, which it is not, or $[\lambda_i] = 2[c]$ for an embedded $c \subset T_i$, which is impossible because an embedded essential curve on a torus is primitive.

Finally, the exposure is confined to §2.2: nothing else in this paper uses (E)(i), and by Remark 2.5 what §2.2 proves is in any case contained in the statement of [16, Corollary A.4].

Theorem 2.3 ((U): universal disjointness of relative area minimisers)**.** *Let E be compact and irreducible with the data above.*

(i) *Relative least-area incompressible surfaces $F_1, F_2 \subset E$ whose proper isotopy classes contain disjoint representatives are themselves disjoint, or coincide.*
(ii) *If $f_i \colon S_i \to E$, $i = 1, \ldots, n$, are incompressible, pairwise non-isotopic and pairwise disjoint, then relative area minimisers g_i in the classes of the f_i have pairwise disjoint images.*

Clause (i) is [16, Theorem 4, p. 885] and clause (ii) is [16, Theorem A.5, p. 899]; the latter carries the number 12 in the author's preprint, and we cite the journal version throughout. The force of (U) is the universal quantifier, which lets minimisers be chosen one class at a time, before any pair is examined.

Theorem 2.4 ((E): the area infimum is attained)**.** *Let $F \subset E(K)$ be properly embedded, incompressible, $\partial F \subset J$.*

(i) (Fixed boundary.) *Let* $\mathcal{F}$ *be the set of piecewise smooth surfaces isotopic to* F *rel* ∂F *and* $I = \inf\{\mathrm{Area}(G) : G \in \mathcal{F}\}$. *Then there is* $F' \in \mathcal{F}$ *with* $\mathrm{Area}(F') = I$.

(ii) (Boundary sliding in J.) *The infimum of area over the surfaces properly isotopic to* F *with boundary in* J *is attained, and the minimiser is a member of* $\mathcal{M}([f])$ *in the sense of* [16, p. 897]*: a stable minimal surface lying in the proper isotopy class of a smooth proper embedding* $f\colon (S, \partial S) \to (E(K), \partial E(K))$ *with* $f^{-1}(\partial E(K)) = \partial S$*, whose boundary is a parametrised curve of* J.

Clause (i) is [8, Theorem 6.12, pp. 112–113], whose hypotheses hold for $M = E(K)$: compact, hence homogeneously regular; orientable and irreducible, hence P^2-irreducible; and $\partial E(K)$ sufficiently convex in the sense of [8, p. 110], the three conditions there being met by (1), by the mean curvature computation above, and by the extension $\widetilde{E} \supset E(K)$. No passage from a closed manifold is involved: Theorem 6.12 is already the theorem for a manifold with boundary and a surface with $\partial F \neq \emptyset$, and it is the closed statement that is separate, namely [8, Theorem 5.1, p. 106]. Clause (ii) is [16, Theorem 2, p. 885 = Corollary A.4, p. 898].

Here (E) without qualification means clause (ii), because a vertex of $\mathrm{IS}(K)$ is a class in which the boundary longitude may move (§3), so the competing class is the sliding one and the complexity's area coordinate $A(v)$ is its infimum.

Remark 2.5 ((E)(ii) already contains (R1)). Read with the definition on [16, p. 897], clause (ii) says that the minimiser is the image of a *smooth proper embedding* $(S, \partial S) \to (E(K), \partial E(K))$ whose preimage of $\partial E(K)$ is exactly ∂S. That is smoothness up to $\partial E(K)$ together with $\Sigma \cap \partial E(K) = \partial\Sigma$, which is what §2.2 calls (R1). So (R1) is not an extra hypothesis and not a gap: it is part of the statement of a published corollary, and every step of this paper is on that footing.

We nevertheless prove (R1) again, independently, in §2.2, and it is worth saying why rather than leaving a reader to wonder. Corollary A.4 is deduced from the compactness of $\mathcal{M}_a([f])$, so it presupposes $\mathcal{M}([f]) \neq \emptyset$, and the non-emptiness is supplied on [16, p. 897] by the sentence that there exists a least area surface in $\{g \in [f] : g|_{\partial S} = f|_{\partial S}\}$, "see [8, Theorem 6.12]", which "is necessarily a stable minimal surface". Theorem 6.12 produces a minimiser in the piecewise smooth category and asserts nothing at ∂M; treating it as a member of $\mathcal{M}(S)$ is therefore a step that the printed proof does not carry out. It is a gap in the literature, not in the corollary's statement, and §2.2 closes it, proving what is missing from (E)(i) alone. A reader who takes Corollary A.4 as printed may read §2.2 as a redundancy check; a reader who wants the non-emptiness step supplied will find it there.

2.2. **Boundary regularity of relative area minimisers.** This subsection proves (R1) from (E)(i) alone. By Remark 2.5 the statement is already contained in (E)(ii); what is done here is to supply the one step behind (E)(ii) that the literature leaves out, so that nothing in this paper rests on it. Clause (i) of (E) produces a minimiser only in the piecewise smooth category, and what the exchange lemma needs is regularity *up to* $\partial E(K)$, together with the statement that the minimiser meets the boundary exactly in its own boundary. That is the content of Theorem 2.10, which is what we call (R1). Throughout this subsection M denotes a compact orientable P^2-irreducible Riemannian 3-manifold with the collar data (1), and $\lambda \subset \partial M$ a smooth closed 1-manifold.

Throughout, $F, \mathcal{F}, I$ are as in (E)(i) and $F' \in \mathcal{F}$ has $\mathrm{Area}(F') = I$. Write $\Gamma := \Gamma_{F'}$ for a graph as in Definition 2.1 and call the closures of the components of $F' \setminus \Gamma$ the *pieces*; each is a compact smooth surface with corners, smoothly embedded up to its own boundary, and part of that boundary may lie in λ. The argument uses nothing about F' beyond this and its minimality.

Lemma 2.6 (Stationarity). *Let X be a smooth vector field on M with compact support in* $\operatorname{Int} M$*. Then*

$$\sum_P \int_P \operatorname{div}_P X \, dA \;=\; 0, \tag{5}$$

the sum being over the pieces P of F'. Consequently

(i) *every piece is a minimal surface; and*
(ii) *along the interior of every arc of Γ that lies in* $\operatorname{Int} F'$*, the two sheets of F' meeting there have opposite outward conormals, $\eta_1 + \eta_2 = 0$.*

Proof. Let $\{\varphi_t\}$ be the flow of X. Each φ_t is a diffeomorphism of M equal to the identity near ∂M, so $\varphi_t(F')$ is properly embedded, is piecewise smooth in the sense of Definition 2.1 — that definition is invariant under diffeomorphisms of M, with $\Gamma_{\varphi_t(F')} = \varphi_t(\Gamma)$ — and is isotopic to F' rel $\partial F'$. Hence $\varphi_t(F') \in \mathcal{F}$ and $\operatorname{Area}(\varphi_t(F')) \geq I = \operatorname{Area}(F')$, with equality at $t = 0$; so the derivative at $t = 0$ vanishes, and that derivative is $\sum_P \int_P \operatorname{div}_P X$, the integrand being continuous on each piece because the piece is C^1 up to its boundary. This is (5). Note which properties of $\mathcal{F}$ have been used: only that it is closed under diffeomorphisms of M fixing ∂M pointwise and that F' minimises area over it. Both hold under *any* reading of "piecewise smooth", so the clause discussed in Remark 2.2 is not consumed here.

For (i), let $q \in \operatorname{Int} P$. Since F' is compact and embedded there is a ball $B \subset \operatorname{Int} M$ about q with $F' \cap B \subset \operatorname{Int} P$. For X supported in B only one term survives in (5), and the divergence theorem on the smooth surface $\operatorname{Int} P$ turns it into $-\int_P \langle H_P, X\rangle = 0$; as X is arbitrary, $H_P = 0$ near q.

For (ii), let q be an interior point of an arc $e \subset \Gamma \cap \operatorname{Int} F'$. The surface F' is a disc near q and e is an embedded arc in it, so e separates that disc into exactly two sheets, lying in pieces P_1, P_2; and there is a ball $B \subset \operatorname{Int} M$ about q with $F' \cap B$ contained in their union. Fix X supported in B and exhaust each P_i by compact subsurfaces $P_i^\delta \subset \operatorname{Int} P_i$ with $\partial P_i^\delta \to \partial P_i$ in C^1. Since $H_{P_i} = 0$ by (i), the divergence theorem gives $\int_{P_i^\delta} \operatorname{div}_{P_i} X = \int_{\partial P_i^\delta} \langle \eta_i, X\rangle$; both sides converge as $\delta \to 0$, the left because the integrand is continuous on P_i and the right because η_i extends continuously to ∂P_i, the piece being C^1 up to its boundary. So (5) becomes $\int_{e\cap B} \langle \eta_1 + \eta_2, X\rangle = 0$ for every such X, whence $\eta_1 + \eta_2 = 0$ on $e \cap B$. □

Lemma 2.7 (Sheets with opposite conormals glue smoothly). *Let $P_1, P_2 \subset M$ be smooth minimal surfaces with boundary, meeting along a common boundary arc e and otherwise disjoint near an interior point q of e, and suppose their outward conormals satisfy $\eta_1 + \eta_2 = 0$ along e. Then $P_1 \cup P_2$ is a smooth minimal surface near q.*

Proof. At each point of e the vectors η_1, η_2 are unit, orthogonal to the tangent of e, and tangent to P_1, P_2 respectively; $\eta_2 = -\eta_1$ therefore gives $TP_1 = \operatorname{span}(\dot e, \eta_1) = \operatorname{span}(\dot e, \eta_2) = TP_2$ along e. So P_1 and P_2 have a common tangent plane T at each point of e near q.

Choose coordinates near q with $T = \{y = 0\}$. For a small enough neighbourhood, P_1 and P_2 are graphs $y = u_1$, $y = u_2$ over the two closed sides of the projection of e in $\{y = 0\}$, and $u_1 = u_2$ and $Du_1 = Du_2$ on that projection; so the function u equal to u_i on the ith side is C^1 on a full disc D. Write the minimal surface equation for a graph in the ambient metric in divergence form, $\partial_i(a^i(x, u, Du)) + b(x, u, Du) = 0$, with a and b smooth and the equation uniformly elliptic on bounded gradients. Each u_i solves it classically on its own side; for a test function φ supported in D, integrating by parts on each side produces two interface integrals of $a^i(x, u, Du)\nu_i\varphi$ along the projection of e, and these cancel because Du is continuous across it and $\nu_1 = -\nu_2$. Hence u is a Lipschitz weak solution on all of D.

Interior regularity for quasilinear divergence-form elliptic equations now applies: difference quotients of u solve linear divergence-form equations with bounded measurable uniformly elliptic coefficients, so De Giorgi–Nash–Moser gives $Du \in C^{0,\alpha}$, after which the coefficients $a^i(x, u, Du)$ are $C^{0,\alpha}$ and Schauder bootstrapping gives $u \in C^\infty$; see [7, 2nd ed., Chs. 6 and 8]. So $P_1 \cup P_2$ is a smooth surface near q, and it is minimal because u solves the equation. □

Lemma 2.8 (Hopf at a boundary point)**.** *Let $p \in \partial M$, let $T \subset T_pM$ be a plane, let $U \subset T$ be a closed circular sector with vertex 0 and opening angle at least π, and let W be the graph over $U \cap B_\delta(0)$ of a C^1 map into the orthogonal complement of T which vanishes to first order at 0; so $p \in W$ and $T_pW = T$. Assume that $W \subset M$, that W is a minimal surface over the interior of U, and that W meets ∂M only over ∂U — over one edge, say, or over the vertex alone. Then*

$$dr_p|_T \neq 0, \qquad \textit{equivalently} \qquad T \neq T_p\partial M.$$

Proof. Put $u := r \circ \Phi$ on $U \cap B_\delta(0)$, where Φ is the graph map. Since $W \subset M = \{r \geq 0\}$ we have $u \geq 0$; u vanishes at 0 and, by hypothesis, nowhere on $\operatorname{Int} U$, so $u > 0$ there; and u is smooth over $\operatorname{Int} U$ and C^1 up to 0.

For a minimal surface W and a smooth function r,

$$\Delta_W u = \operatorname{tr}_W \operatorname{Hess} r + \langle \operatorname{grad} r, H\rangle = \operatorname{tr}_W \operatorname{Hess} r,$$

H vanishing. In the collar $\operatorname{Hess} r = \varphi\varphi'\, g_{\partial M} \leq 0$ on vectors tangent to the level sets of r, by (2), while $\operatorname{Hess} r(\partial_r, \cdot) = 0$ because r is a geodesic coordinate; so $\operatorname{Hess} r \leq 0$ as a quadratic form and $\Delta_W u \leq 0$. Read in the coordinates of U, the operator Δ_W is uniformly elliptic with continuous coefficients and no zeroth-order term, so $-u$ is a subsolution.

The opening angle of U is at least π, so there is an open disc $B \subset \operatorname{Int} U$ with $0 \in \partial B$. On B we have $-u < 0 = -u(0)$, the interior sphere condition holds at 0 trivially, and the outward normal derivative exists because u is C^1 up to 0. Hopf's boundary point lemma [7, 2nd ed., Lemma 3.4, p. 34] gives $\partial_\nu(-u)(0) > 0$, that is $\partial_\nu u(0) < 0$ for the outward ν. The differential of Φ at 0 is the inclusion $T \hookrightarrow T_pM$, so $\partial_\nu u(0) = dr_p(\nu)$, whence $dr_p|_T \neq 0$. □

The lemma consumes less convexity than the divergence-form comparison it replaces: only $\operatorname{Hess} r \leq 0$, that is $\varphi' \leq 0$ in (1), and not the strict positivity of the mean curvature of ∂M. It is used twice below, once inside Proposition 2.9 and once, in the simplest case U a half-plane, as the whole of Lemma 2.11.

Nothing else is needed. In particular the argument below never compares F' with a least area disc, never leaves the piecewise smooth category, and uses no result of [8] beyond the existence statement (E)(i) itself.

Proposition 2.9 (Minimisers with fixed boundary are smooth up to ∂M)**.** *Let $F, \mathcal{F}, I, F'$ be as at the start of this subsection. Then F' is a smooth embedded minimal surface, smooth up to ∂M, with $F' \cap \partial M = \partial F' = \lambda$; and F' is stable.*

Proof. By clause (i) of Lemma 2.6 every piece is a minimal surface, and by clause (ii) of that lemma together with Lemma 2.7 the surface F' is smooth and minimal near every interior point of every arc of Γ lying in $\operatorname{Int} F'$. Three things remain: the vertices of Γ in $\operatorname{Int} F'$, the points of λ, and stability.

Vertices in the interior. Let q be a vertex of Γ in $\operatorname{Int} F'$. Finitely many arcs of Γ emanate from q and cut a disc neighbourhood N of q in F' into closed sectors $S_1, \ldots, S_m$ in cyclic order, each contained in a piece, hence minimal and smoothly embedded up to its own boundary, in particular up to q. Along the interior of each shared arc the two adjacent sectors have the same tangent plane, by the previous paragraph; and the tangent plane of a sector extends continuously to q. So all the S_i have one and the same tangent plane T at q.

Take coordinates with $q = 0$ and $T = \{y = 0\}$, and shrink N so that each S_i is the graph of a C^1 function over a closed sector $\Sigma_i \subset T$ with vertex 0 and opening angle θ_i, the Σ_i consecutive. Here $\theta_i \in (0, 2\pi)$, the two boundary arcs of S_i having distinct tangent rays at q by the corner clause of Definition 2.1; no bound $\theta_i < \pi$ is available, and none is used. Let π denote the orthogonal projection onto T. Then π carries $N \setminus \{q\}$ onto a punctured disc $D_\varepsilon \setminus \{0\}$ as a covering map of some degree k, with $2\pi k = \sum_i \theta_i$. Over each point of $D_\varepsilon \setminus \{0\}$ the k preimages are distinct points of the embedded surface F', hence have pairwise distinct heights; the heights depend continuously on the point and never coincide, so they order the sheets consistently and the covering is trivial. But $N \setminus \{q\}$, a disc minus an interior point, is connected. Hence $k = 1$.

So N is the graph of a function u over D_ε. It is C^1, the sectors having matching tangent planes along the arcs; it is smooth off the arcs; and it is a weak solution of the minimal surface equation on $D_\varepsilon \setminus \{0\}$, the interface terms cancelling as in the proof of Lemma 2.7. It is then a weak solution on all of D_ε: cutting a test function off at radius δ about 0 changes the weak formulation by at most $C\delta^{-1} \cdot \pi\delta^2 \to 0$, u being Lipschitz. By the regularity quoted in Lemma 2.7, u is smooth. So F' is smooth near q, and F' is a smooth minimal surface on all of $\operatorname{Int} F'$.

Points of λ. Since Γ is properly embedded, a point $p \in \lambda$ either lies off Γ or is one of the finitely many endpoints of arcs of Γ on λ. In the first case some relative neighbourhood of p misses Γ altogether, so lies in $F' \setminus \Gamma$, which by Definition 2.1 is a smooth submanifold of M — smooth, that is, up to ∂M and including its boundary points — and F' is smooth up to ∂M at p. (The clause about the pieces would give only C^1 here, which is why it is not the clause used.) So only the finitely many endpoints remain; fix such a p.

Those arcs cut a half-disc neighbourhood N of p in F' into closed sectors $S_0, \dots, S_m$, where S_0 and S_m carry the two arcs ℓ_0, ℓ_m into which λ is divided at p. Exactly as in the interior case, all the S_i share a tangent plane T at p; take coordinates with $p = 0$, $T = \{y = 0\}$, write $\Sigma_i \subset T$ for the tangent sector of S_i, of angle $\theta_i \in (0, 2\pi)$ as before, and put $\Theta := \sum_i \theta_i$. The extreme edges of Σ_0 and Σ_m are the two tangent rays of λ at p, which are opposite; hence

$$\Theta \equiv \pi \pmod{2\pi}, \qquad \text{and in particular } \Theta \geq \pi. \tag{6}$$

What is wanted is a sector of opening angle in $[\pi, 2\pi)$ over which F' is a graph. Let j be least with $\Theta_j := \theta_0 + \cdots + \theta_j \geq \pi$; such a j exists by (6). If $\Theta_j < 2\pi$, put $U := \Sigma_0 \cup \cdots \cup \Sigma_j$ and $W := S_0 \cup \cdots \cup S_j$. If instead $\Theta_j \geq 2\pi$ then $j \geq 1$, since $\theta_0 < 2\pi$, and $\Theta_{j-1} < \pi$ by the minimality of j, so that the single angle θ_j already exceeds π; put $U := \Sigma_j$ and $W := S_j$.

Either way U is a sector of opening angle in $[\pi, 2\pi)$ and W is a C^1 graph over it; W lies in M; it is minimal over $\operatorname{Int} U$, by the previous step; and W meets ∂M only over ∂U, because $F' \cap \partial M = \lambda$ by properness while $\lambda \cap N = \ell_0 \cup \ell_m$, and ℓ_0, ℓ_m are the extreme edges, carried by S_0 and S_m. Lemma 2.8 applies and gives $dr_p|_T \neq 0$.

Now let v be a unit vector tangent at p to one of the sectors. That sector is C^1 at p and contained in $M = \{r \geq 0\}$, so its points at parameter t in the direction v have $r = t\, dr_p(v) + o(t) \geq 0$, forcing $dr_p(v) \geq 0$. The sectors are in cyclic order about p and consecutive ones share an edge, so the set of such v is the image in the unit circle of T of an interval of length Θ, swept monotonically from the tangent ray of ℓ_0; and it lies in the closed half-circle $\{dr_p \geq 0\}$, whose length is π because $dr_p|_T \neq 0$. Were $\Theta \geq 2\pi$ that sweep would cover the whole circle, which no half-circle contains; so $\Theta < 2\pi$, the sweep is injective, its image is an arc of angular length Θ, and $\Theta \leq \pi$. With (6), $\Theta = \pi$. In particular each $\theta_i < \pi$ after all; that bound is a conclusion here, not a hypothesis, which is why the selection of U above could not be allowed to use it.

So the Σ_i tile a half-disc $H \subset T$ and N is the graph of a function u over H. It is C^1 on $\overline{H}$ and, by the previous step and the cancellation of the interface terms as in Lemma 2.7, a weak solution of the minimal surface equation on $\operatorname{Int} H$. The curved side of ∂H is the projection of λ, a smooth curve, so near 0 the domain H has smooth boundary; and there u takes the boundary values given by λ itself, which are smooth. Boundary regularity for the Dirichlet problem now finishes: a $C^{0,1}$ weak solution of a quasilinear divergence-form elliptic equation with smooth data on a smooth boundary portion is $C^{1,\alpha}$ up to that portion by the boundary form of De Giorgi–Nash–Moser, and then C^∞ up to it by boundary Schauder estimates; see [7, 2nd ed., Chs. 6 and 8]. Hence F' is a smoothly embedded surface with boundary up to ∂M near p.

It is worth recording that this last step, and not a matching of jets, is what lets Definition 2.1 ask only C^1 of the pieces.

Globally and stability. The two steps give a smooth structure near every point of the compact set F', so F' is a smooth embedded minimal surface, smooth up to ∂M; and $F' \cap \partial M = \partial F' = \lambda$ is part of the hypothesis that F' is properly embedded, which is the membership condition of $\mathcal{F}$. Finally F' has least area among the piecewise smooth surfaces isotopic to it rel $\partial F'$, so all nearby normal deformations fixing the boundary increase area: F' is stable. □

Theorem 2.10 ((R1): boundary regularity of relative minimisers)**.** *Let $F \subset E(K)$ be properly embedded, incompressible and piecewise smooth with ∂F a leaf of J, and let*

$$A := \inf\{\operatorname{Area}(G) : G \textit{ piecewise smooth and properly embedded,} \quad \textit{properly isotopic to } F,\ \partial G \subset J\}.$$

Then there is a surface Σ properly isotopic to F, with $\partial\Sigma$ a leaf of J and $\operatorname{Area}(\Sigma) = A$, which is the image of a smooth proper embedding. In particular Σ is a stable minimal surface, for some $\rho > 0$ the set $\Sigma \cap \{0 \le r \le \rho\}$ is an embedded smooth surface with boundary up to $\{r = 0\}$, and

$$\Sigma \cap \partial E(K) = \partial\Sigma.$$

Proof. Let G_i be a minimising sequence for A. Each G_i is properly embedded, piecewise smooth, incompressible — incompressibility being invariant under isotopy — with ∂G_i a leaf of J. Apply (E)(i) to G_i to obtain F_i' of least area $I(G_i)$ in the class $\mathcal{F}(G_i)$ of piecewise smooth surfaces isotopic to G_i rel ∂G_i, and apply Proposition 2.9 to it. Then F_i' is a smooth embedded stable minimal surface, smooth up to $\partial E(K)$, with $F_i' \cap \partial E(K) = \partial F_i' = \partial G_i$ a leaf of J; and F_i' is properly isotopic to F, so that

$$A \ \le\ \operatorname{Area}(F_i') \ =\ I(G_i) \ \le\ \operatorname{Area}(G_i) \longrightarrow A.$$

In particular each F_i' is a smooth proper embedding of the compact connected surface S underlying F, whose preimage of $\partial E(K)$ is the whole of ∂S, whose image is incompressible, and whose boundary is a curve of J. In the notation of [16, p. 897], recalled in the proof of Lemma 4.2, this says exactly that $F_i' \in \mathcal{M}(S)$, and $\operatorname{Area}(F_i') \le a := A + 1$ for large i puts it in $\mathcal{M}_a$.

Membership in $\mathcal{M}(S)$ has two requirements, and it is worth naming them separately. The first is that F_i' be a smooth proper embedding, which is Proposition 2.9 and is the whole point of §2.2. The second is stability, and for Kapovich this means stability under variations fixing the boundary: it is the fixed-boundary least area surface of [8, Theorem 6.12] that he calls "necessarily a stable minimal surface" [16, p. 897], and the proof of [16, Proposition A.1] uses stability only through Schoen's interior estimates at finitely many interior points. Having least area rel $\partial F_i'$, F_i' is stable in that sense.

By [16, Corollary A.3, p. 898] the space $\mathcal{M}_a$ is the disjoint union of finitely many open and closed subsets, each consisting of mutually isotopic surfaces; so after passing to a subsequence all the F_i' lie in one such subset $\mathcal{N}$. By [16, Proposition A.1, p. 897] the space $\mathcal{M}_a$ is compact, so after a further subsequence $F_i' \to \Sigma$ in the C^1 topology with $\Sigma \in \mathcal{M}_a$; and $\mathcal{N}$ being closed, $\Sigma \in \mathcal{N}$, so Σ is isotopic to the F_i' and hence properly isotopic to F. Area is continuous in the C^1 topology, so $\operatorname{Area}(\Sigma) = \lim \operatorname{Area}(F_i') = A$. Membership in $\mathcal{M}(S)$ means that Σ is a stable minimal surface which is the image of a smooth proper embedding $f\colon (S, \partial S) \to (E(K), \partial E(K))$ with $f^{-1}(\partial E(K)) = \partial S$ and $f|_{\partial S}$ a parametrised curve of J. This is smoothness up to $\partial E(K)$ in a collar, together with $\Sigma \cap \partial E(K) = \partial\Sigma$. □

Nothing in the proof used that $\partial E(K)$ is connected, or that S has a single boundary circle: Kapovich's standing hypothesis [16, p. 897] is that $f^{-1}(\partial_i M)$ be a single component of ∂S for each boundary component $\partial_i M$ of M, which is what a spanning surface for an n-component link satisfies. Theorem 2.10 therefore holds verbatim in the n-component setting used in §8.4, and we refer to that case as the n-component form of (R1).

Lemma 2.11 ((R2): transversality along the boundary)**.** *Let Σ be as in Theorem 2.10. Then $T_p\Sigma \neq T_p\partial E(K)$ for every $p \in \partial\Sigma$; that is, Σ meets $\partial E(K)$ transversally along $\partial\Sigma$, and is therefore neatly embedded.*

Proof. This is Lemma 2.8 in its simplest case. By Theorem 2.10, Σ is a smooth surface with boundary, smooth up to $\partial E(K)$, with $\Sigma \cap \partial E(K) = \partial\Sigma$. Let $p \in \partial\Sigma$ and $T := T_p\Sigma$. In a boundary chart at p the surface Σ is the graph over a half-disc $U \subset T$, of opening angle π and with p on its straight edge, of a smooth map vanishing to first order at p; the graph lies in $E(K)$, is minimal, and meets $\partial E(K)$ exactly in the image of that straight edge. Lemma 2.8 gives $dr_p|_T \neq 0$, that is $T_p\Sigma \neq T_p\partial E(K)$. □

Remark 2.12*.* The proof consumes only $\operatorname{Hess} r \leq 0$, that is $\varphi' \leq 0$ in (1), and not the strict positivity of the mean curvature of $\partial E(K)$ that an earlier version of this argument used through a divergence-form comparison and an explicit annular barrier. With Lemma 2.8 in hand that machinery is not needed, and §2 no longer uses the theory of weak solutions of [7, Ch. 8] at all; Chapter 8 returns only in §5, where the operator is genuinely a linearisation with a zeroth-order term of uncontrolled sign.

Remark 2.13 (The route through least area *maps*)*.* One could instead minimise over the *homotopy* class rel ∂F, using [8, Theorem 6.13, p. 113], which asserts a least area *smooth immersion* in that class; make it an embedding by the results of [6]; and return to the isotopy class by the rigidity theorem of [18], whose hypothesis of ∂-incompressibility is Lemma 3.1′. That would dispense with §2.2 altogether. We have not taken it, because both analytic steps are weaker than the ones they replace. Theorem 6.13 is stated without proof, with the remark that "a version of" it "was proved by Lemaire", and its content at ∂F is precisely the boundary regularity that §2.2 establishes. The theorems of [6] are proved for *closed* surfaces; the bounded case is discussed in their §7, which opens by isolating an *Existence Statement* 7.3 of which the authors write, on [6, p. 636], "Presumably, the statement we give below can be proved . . . but we have not done this", and then that "on the assumption that this statement holds" the earlier sections extend. So the alternative would trade the single dependency of Remark 2.2 for an unproved regularity assertion and a result its authors state they did not write.

3. The Kakimizu complex

3.1. **Surfaces, vertices, simplices.** A *knot* is a smooth oriented embedding $K \hookrightarrow S^3$. Choose an open tubular neighbourhood $N(K)$ and write $E(K) = S^3 \setminus \operatorname{Int} N(K)$ for the compact *exterior*; its boundary is a torus. The *preferred longitude* is the oriented slope on $\partial E(K)$ whose linking number with K is zero. A *Seifert surface in the exterior* is a compact, connected, orientable, neatly embedded surface $S \subset E(K)$ whose oriented boundary is a preferred longitude; since S is connected with a single boundary component,

$$\chi(S) = 1 - 2g(S), \tag{7}$$

and $g(K) = \min\{g(S)\}$. A *compressing disc* for S is an embedded disc $D \subset E(K)$ with $D \cap S = \partial D$ such that ∂D is essential on S; if none exists, S is *incompressible*. *Boundary incompressibility is not required*, and no step below uses it; see §3.2.

An *ambient isotopy* is a smooth family of diffeomorphisms $H_t\colon E(K) \to E(K)$ with $H_0 = \mathrm{id}$; a *normal isotopy* is a smooth family $f_t\colon S \to E(K)$ of neat embeddings. Two surfaces determine the same vertex if and only if they are ambient isotopic in $E(K)$. The boundary longitude is allowed to move on the boundary torus: the isotopy is not required to fix any chosen boundary curve pointwise. That an ambient isotopy restricts to a normal isotopy is immediate; the converse is Lemma 3.9, so on Seifert surfaces the two equivalence relations agree, and the relative area minimisation of §2, which is performed in a normal isotopy class, computes an invariant of a vertex.

The exterior is irreducible [12, p. 228]; this is used in Lemma 3.7.

Definition 3.1. The vertices of $\mathrm{IS}(K)$ are the ambient isotopy classes of incompressible Seifert surfaces in $E(K)$. A finite set of pairwise distinct vertices spans a simplex if and only if its classes admit *simultaneously* pairwise disjoint representatives. For an integer $\ell \geq g(K)$,

$$\mathrm{IS}_\ell(K) = \mathrm{IS}(K)\big[\{[S] : g(S) \leq \ell\}\big]$$

is the *full* subcomplex on the vertices of genus at most ℓ; in particular $\mathrm{IS}_{g(K)}(K) = \mathrm{MS}(K)$ is the minimal-genus complex. We write $\mathrm{dist}(u, w)$ for the graph distance in the 1-skeleton. A complex is *flag* if every finite clique in its 1-skeleton spans a simplex.

Two notions of disjointness must be kept apart. An *edge* speaks only of a single pair of vertices, and different edges may use mutually incompatible representatives; a *simplex of dimension at least* 2 requires representatives chosen once and for all that are pairwise disjoint as a family. The passage from the former to the latter is Proposition 3.3, whose proof uses (E) and (U); it is never implicit.

Finally, dist is not merely a graph metric but Kakimizu's own distance: by [12, Proposition 3.1(1), p. 231] the minimum length of an edge path in $\mathrm{IS}(L)$ between two classes equals the distance d obtained by counting lifts to the infinite cyclic cover. We use this in §5, where the hypothesis of the exchange lemma is $\mathrm{dist}(u, w) = 2$ and what the construction is handed is a pair of surfaces at Kakimizu distance 2. The statement is strictly stronger than connectedness and does not appear in version 1 of this work.

3.2. **Which definitions these are.** Three conventions have to be pinned down before Kakimizu's theorems may be quoted, because the literature does not agree on them.

Vertices. We use Kakimizu's definition, not the paraphrase of it that has entered the literature. A spanning surface for a link L is, in [12, p. 225], a surface $S = \Sigma \cap E(L)$ with Σ oriented in S^3, $\partial\Sigma = L$, Σ having no closed component and *possibly disconnected*, and $\Sigma \cap N(L)$ a collar of $\partial\Sigma$; it is incompressible if *each component* is incompressible in $E(L)$. In particular ∂-incompressibility is not required. Przytycki and Schultens describe the same complex as having for vertices the classes of surfaces "incompressible and ∂-incompressible" [15]; that description adds a condition which is not in [12], and a manuscript which inherits it owes a lemma identifying the two vertex sets. We do not impose it, so the vertex set of Definition 3.1 is Kakimizu's on the nose; and the condition is in any case automatic, which we record now, both because it settles the discrepancy and because it is the hypothesis under which the rigidity theorems of [18] are stated.

Lemma 3.1′ (Incompressible spanning surfaces are ∂-incompressible)**.** *Let L be a link in S^3 and let $\Sigma \subset E(L)$ be an incompressible spanning surface, neatly embedded, with $\partial\Sigma$ the union of the longitudes. Then Σ is ∂-incompressible: there is no embedded disc $D \subset E(L)$ with $D \cap \Sigma = a$ and $D \cap \partial E(L) = b$, where $\partial D = a \cup b$ and a is an arc which is essential in Σ, that is, which does not cut a disc off Σ together with an arc of $\partial\Sigma$. For $n = 1$ this says that every incompressible Seifert surface in a knot exterior is ∂-incompressible.*

Proof. Suppose such a D exists. Since $D \cap \Sigma = a$ and $\lambda := \Sigma \cap \partial E(L) = \partial\Sigma$ lies in Σ, the interior of b misses λ; so b is an arc properly embedded in the surface obtained by cutting $\partial E(L)$ along λ, and both endpoints of b lie on the same torus T_i, on which the cut yields an annulus A_i with boundary circles λ_i^+, λ_i^-, the two sides of λ_i in T_i. A properly embedded arc in an annulus either has both endpoints on the same boundary circle, and is then ∂-parallel, or joins the two.

Both endpoints on the same circle. Then b together with an arc $\lambda_0 \subset \lambda_i$ bounds a disc $\Delta \subset A_i$, so $\operatorname{Int}\Delta \cap \lambda = \emptyset$. Now $D \cup \Delta$ is an embedded disc, the two meeting exactly in b, its boundary is the closed curve $a \cup \lambda_0 \subset \Sigma$, and its interior misses Σ; pushing Δ off $\partial E(L)$ makes it a disc meeting Σ exactly in its boundary. If $a \cup \lambda_0$ is essential in Σ this is a compressing disc, contradicting incompressibility. If it is inessential it bounds a disc in Σ, so a cuts a disc off Σ together with an arc of $\partial\Sigma$ and was not essential.

Endpoints on the two different circles. Near a point of λ_i the pair $(E(L), \Sigma)$ is the pair $(\{z \geq 0\}, \{x = 0, z \geq 0\})$, so the two sides of λ_i in T_i are the traces of the two sides of Σ. Perturb ∂D by pushing a off Σ to the side on which D lies. The interior of b misses Σ, since $b \subset \partial E(L)$ and $\Sigma \cap \partial E(L) = \lambda$, so the resulting closed curve meets Σ only near the two endpoints of b; and at such an endpoint it crosses or does not according as b leaves the endpoint on the far side of Σ from D or on the near one. In the case at hand b leaves its two endpoints on opposite sides, so exactly one of them is the far side and there is exactly one crossing. Hence the algebraic intersection number of ∂D with Σ is ± 1. But Σ is Poincaré dual to the total linking homomorphism $\varphi\colon H_1(E(L)) \to \mathbb{Z}$, $\gamma \mapsto \operatorname{lk}(\gamma, L)$, which is the homomorphism of §8.3 — nothing proved there is needed here, the duality holding for any spanning surface of any link by a count of meridians. So that number is $\varphi(\partial D)$, which vanishes because ∂D bounds the disc D in $E(L)$. □

The only remaining discrepancy, connectedness, is vacuous for a knot: a spanning surface for a knot has no closed component and exactly one boundary circle, hence is connected, so "connected and incompressible" and "incompressible componentwise" select the same surfaces.

Non-splitness. Kakimizu's Theorem A is stated for a non-split oriented link, and the hypothesis is used, not decorative: it is what makes $E(L)$ irreducible. For a knot it is vacuous. It returns in Theorem D, where it is assumed.

Edges. Kakimizu spans an edge on two classes with disjoint representatives [12, 1.3(b), p. 227]; Przytycki and Schultens instead require Kakimizu distance one, and point out that the two conditions differ for links, precisely because spanning surfaces there are allowed to be disconnected. We use the disjointness convention, as in Definition 3.1. For a knot the two agree, since spanning surfaces are then connected, so the discrepancy is vacuous here; for Theorem D it is neutralised by the linking condition, which forces every spanning surface to be connected.

3.3. **Connectedness, flagness, non-emptiness.**

Proposition 3.2 (Connectedness)**.** IS(unknot) *is a single point. For every non-trivial knot K, the complex* IS(K) *is connected.*

Proof. The exterior of the unknot is a solid torus V, the complementary handlebody of a genus-one Heegaard splitting of S^3, and the preferred longitude is a meridian of V. If $S \subset V$ is an incompressible Seifert surface then $\pi_1(S) \to \pi_1(V) \cong \mathbb{Z}$ is injective, since an essential circle of S that died in V would, after innermost-circle and outermost-arc surgery on a null-homotopy in general position, bound a compressing disc. A free group of rank $2g(S) \geq 2$ does not embed in $\mathbb{Z}$, so $g(S) = 0$ and S is a meridian disc; two meridian discs may be isotoped to share their boundary, made disjoint by innermost-circle surgery, and then the sphere they assemble bounds a ball by Alexander–Schoenflies, which isotopes one to the other. For non-trivial K the statement is [12, Theorem A, p. 226], quoted in §1. $\square$

The engine behind that theorem is [12, Theorem 2.1, p. 228], whose proof splits on p. 229 into a minimal-genus case and an incompressible case; it is the second that we need. Nothing is claimed here about the connectedness of a truncation $\mathrm{IS}_\ell(K)$. That is proved in §8, as part of Theorem C and by way of Corollary 6.7; the implication runs from the global statement to the truncations and never in the reverse direction, and an independent source is recorded in Remark 8.1.

Proposition 3.3 (Flagness)**.** $\mathrm{IS}(K)$ *is flag, and hence so is every full subcomplex* $\mathrm{IS}_\ell(K)$*.*

Proof. Let $\{v_1, \ldots, v_n\}$ be a clique in the 1-skeleton. Its vertices are distinct, so the corresponding surfaces are pairwise non-isotopic and all incompressible. By (E) fix, once and for all and one class at a time, a relative area minimiser A_i in the class v_i. Let $i \neq j$. Since $\{v_i, v_j\}$ is an edge, the two classes contain disjoint representatives, so Theorem 2.3(i) applies to the pair A_i, A_j and yields that they are disjoint or coincide; and $v_i \neq v_j$ excludes coincidence. The A_i were fixed before any pair was examined and do not depend on (i, j), so $\{A_1, \ldots, A_n\}$ is a family of *simultaneously* pairwise disjoint representatives, which is the simplex required. Fullness makes the truncations retain the same simplices. $\square$

The proof consumes only the two-surface clause of (U); the n-surface clause is not needed, because the choice made in advance already converts pairwise information into simultaneous information. Flagness is therefore conditional on two citations and on no hypothesis. The same mechanism, applied to a vertex and two of its neighbours, is what the exchange lemma is handed, and we isolate it here because it is the exact point at which the argument for links breaks.

Proposition 3.4 (Disjointness from two neighbours at once)**.** *Let u, w be vertices of* $\mathrm{IS}(K)$ *with* $\mathrm{dist}(u, w) = 2$*, let $x \in \mathcal{I}(u, w)$ be a common neighbour, and let $S_+ \in u$, $S_- \in w$ and $S \in x$ be relative area minimisers supplied by* (E)*, chosen one class at a time. Then*

$$S \cap (S_+ \cup S_-) = \emptyset.$$

Proof. Since x and u are adjacent, their classes contain disjoint representatives, so Theorem 2.3(i) applies to S and S_+: they are disjoint or they coincide. A Seifert surface for a knot is connected, so "coincide" means that S and S_+ are equal as sets and hence that $x = u$, which is excluded. Therefore $S \cap S_+ = \emptyset$, and the same argument on the edge $\{x, w\}$ gives $S \cap S_- = \emptyset$. The three minimisers were fixed before any pair was examined, so the two conclusions hold for one and the same S. $\square$

The step that fails for a general link is the sentence beginning "A Seifert surface for a knot is connected". For a link, Kakimizu's spanning surfaces may be disconnected, Theorem 2.3(i) then yields only that each component of S is disjoint from or equal to a component of S_+, and two distinct vertices may share a component; the intersection $S \cap S_+$ need not be empty. This is the obstruction analysed in §8.4, and the linking condition of Theorem D removes it by forcing every spanning surface to be connected.

Proposition 3.5 (Non-emptiness)**.** *$\mathrm{IS}(K) \neq \emptyset$ for every knot K, and $\mathrm{IS}_\ell(K) \neq \emptyset$ for every integer $\ell \geq g(K)$.*

Proof. Seifert's algorithm, applied to a regular diagram of K, produces a compact connected oriented surface $S_0 \subset S^3$ with $\partial S_0 = K$; made transverse to $N(K)$, it meets it in an annulus, and $S = S_0 \cap E(K)$ is a Seifert surface in the exterior; connectedness of S_0 is part of the definition over which the existence statement quantifies [11, Definition 2.1 and Theorem 2.2, pp. 15–16]. If S is incompressible then $[S]$ is a vertex. Otherwise compress and keep the component carrying the longitudinal boundary; by Lemma 3.6 each such step strictly lowers the genus, so after finitely many steps the process stops at an incompressible Seifert surface. For the truncations, a Seifert surface of least genus is incompressible, since a compression would produce one of smaller genus, so it is a vertex of genus $g(K)$ and lies in every $\mathrm{IS}_\ell(K)$ with $\ell \geq g(K)$. ☐

3.4. **Compression and the robustness of neighbours.** Throughout this subsection A and B are transverse, neatly embedded compact surfaces. Cutting both along $A \cap B$, the resulting components are called *patches* and the intersection curves and arcs duplicated on their boundaries are called *exchange seams*; a *disc patch* is a disc component of the result, never a component of a set-theoretic symmetric difference. This is the vocabulary of §5.

Lemma 3.6 (Euler characteristic and compression)**.** *Let A, B be compact and transverse. Cut along $A \cap B$ and reglue by the orientation-compatible normal exchange, and let P_{tot} be the union of the output components. Then $\chi(P_{\mathrm{tot}}) = \chi(A) + \chi(B)$, so that if $P_{\mathrm{tot}} = P \sqcup C_1 \sqcup \cdots \sqcup C_m$ then*

$$\chi(P) = \chi(A) + \chi(B) - \sum_j \chi(C_j). \tag{8}$$

Moreover, compressing a connected Seifert surface along an essential circle and keeping the component that carries the longitudinal boundary strictly lowers the genus; iterating therefore terminates at an incompressible Seifert surface.

Proof. Triangulate $A \cup B$ so that each intersection curve is a subcomplex. Cutting duplicates exactly the cells of the intersection trace and the normal regluing identifies the same cells again, so the Euler correction is the same before and after the exchange; additivity over components gives (8). For the second statement: compressing along a non-separating circle removes a handle and keeps the unique boundary component, so the genus drops by one; compressing along a separating essential circle discards a closed side, which has positive genus since otherwise the circle would bound a disc, and this again lowers the genus of the component with boundary. The genus is a non-negative integer, so the process is finite, and it stops only where no compression is available. ☐

Lemma 3.7 (Prescribed compressions performed away from a fixed neighbour)**.** *Let M be an irreducible orientable 3-manifold, let $P \subset M$ be connected, neatly embedded, with non-empty boundary, and let $Q \subset M$ be incompressible, neatly embedded and disjoint from P. Given any finite compression sequence for P, each step keeping the component with boundary, there is a compression sequence whose intermediate and final ambient isotopy classes agree with the given ones one by one and all of whose discs are disjoint from Q. In particular the final surface has a representative disjoint from Q.*

Proof. It suffices to clean a single prescribed compressing disc D and then induct along the sequence, transporting the next prescribed disc to the current representative at each step. Put D transverse to Q. As $\partial D \subset P$ and $P \cap Q = \emptyset$, the intersection $D \cap Q$ consists of circles. Each is null-homotopic in M, being the boundary of a subdisc of D, hence null-homotopic in Q by incompressibility, hence bounds a disc in Q; choose such a disc $E \subset Q$ innermost, so

$\operatorname{Int} E \cap D = \emptyset$. Its boundary circle bounds in D a unique subdisc F not containing ∂D, and $E \cup F$ is a sphere with a corner along that circle; rounding the corner by the qualitative part of Lemma 5.1, at a width small enough that the tube misses P, gives a smoothly embedded sphere, and by irreducibility it bounds a ball B. By clause (5) of that lemma the rounding does not change the region bounded, up to the tube; so B agrees off the tube with the region bounded by $E \cup F$, and we use it in that form below. That sphere is disjoint from P, since B lies in the interior of M while P is connected with its boundary on ∂M. Pushing F across B to a parallel copy of E is an isotopy of D rel ∂D through compressing discs, keeps the disc interiors off P, and removes that circle together with every intersection circle inside F. Repeating produces D' disjoint from Q and isotopic to D rel boundary, so compressing along D' gives a surface ambient isotopic to the one obtained by compressing along D. The component with boundary remains disjoint from Q and furnishes the input for the next step; the discarded closed components play no role. □

Corollary 3.8 (Robustness of neighbours)**.** *Let P be connected, neatly embedded, with non-empty boundary and disjoint from an incompressible surface Q. Then* every *ambient isotopy class obtained by compressing P down to an incompressible descendant with boundary is, in* $\mathrm{IS}(K)$*, equal or adjacent to* $[Q]$.

Proof. Apply Lemma 3.7 to the chosen compression sequence; its final representative is disjoint from Q. If the two classes coincide they are equal, and otherwise their disjoint representatives span an edge. □

3.5. **Normal isotopies are ambient.**

Lemma 3.9 (Isotopy extension; [19, Theorem 2.4.6, p. 52])**.** *Let M and F be compact smooth manifolds with boundary and let $f\colon F \times [0,1] \to M$ be a smooth isotopy through neat embeddings, that is, $f_t^{-1}(\partial M) = \partial F$ and $f_t(F)$ meets ∂M transversally along its boundary. Then there is a smooth ambient isotopy $H_t\colon M \to M$ with $H_0 = \mathrm{id}_M$ and $H_t \circ f_0 = f_t$ for all t.*

Wall's hypotheses are compactness of F and neatness of the embeddings; there is no orientability hypothesis and no dimension condition relating F to M, and the ambient isotopy is covered by a diffeotopy of M, which is what we use.

Taking $M = E(K)$, which is compact with boundary, and F a Seifert surface, every normal isotopy is a smooth family of neat embeddings, so the lemma applies: a normal isotopy class, as used in relative area minimisation, is exactly an ambient isotopy vertex.

Remark 3.10 (No silent strengthening)*.* The ambient isotopy produced above preserves $\partial E(K)$ *setwise*, and the longitudinal boundary curve is allowed to move. Strengthening this to "fixing the boundary pointwise" would make the statement false: on $M = S^1 \times [0,1]$ the rotational isotopy of the boundary is realised by no ambient isotopy fixing the boundary pointwise. In citations, and in formalisation, this must be written setwise.

4. Complexity and the frontier construction

Two things are assembled here. The first is the complexity function that will be fed to Definition 6.1, together with the one fact about it that is not formal, namely that the set of values it attains is well-ordered. The second is the cyclic-cover construction which, given a vertex and two of its neighbours, manufactures the pair of candidate surfaces on which §5 operates.

4.1. **The complexity.**

Definition 4.1 (Complexity)**.** Let v be a vertex of $\mathrm{IS}(K)$. Write $g(v)$ for the genus of any representative, which by (7) is determined by v, and

$$A(v) \;:=\; \inf\{\mathrm{Area}(F) \;:\; F \in v,\ \partial F \subset J\},$$

the infimum of area over the surfaces in the class with boundary on a leaf of the longitudinal foliation. Set

$$c(v) \;:=\; \big(g(v), A(v)\big) \in \mathbb{N} \times \mathbb{R}_{>0},$$

and order $\mathbb{N} \times \mathbb{R}_{>0}$ lexicographically. Finally put

$$W \;:=\; c\big(\mathrm{IS}(K)^{(0)}\big),$$

the set of *attained* values, with the induced order.

The infimum is attained, by (E)(ii) of Theorem 2.4 applied to the normal isotopy class of a representative — which is the vertex, by Lemma 3.9 — so $A(v)$ is a minimum and is realised by a stable minimal surface, smooth and embedded in $\mathrm{Int}\, E(K)$. Both coordinates are therefore invariants of v and not of a representative.

It is W, and not $\mathbb{N} \times \mathbb{R}_{>0}$, that is required to be well-ordered in Definition 6.1, and the distinction is not cosmetic: $\mathbb{N} \times \mathbb{R}_{>0}$ is not well-ordered, since $\mathbb{R}_{>0}$ is not. What has to be proved is that the values actually taken by c are sparse. That is the content of the next lemma, and it is the only place in this paper where a statement about the totality of vertices of $\mathrm{IS}(K)$ is needed.

Lemma 4.2 (Well-foundedness of the complexity)**.** *Let K be a non-trivial knot, with the relative metric and the foliation J of §2. Then:*

- (i) *for every $h \in \mathbb{N}$ and every $a > 0$ the set $\{v \in \mathrm{IS}(K)^{(0)} : g(v) = h,\ A(v) \le a\}$ is finite;*
- (ii) *consequently, for each h the set $W_h := \{A(v) : g(v) = h\}$ is a closed discrete subset of $\mathbb{R}_{>0}$ and is well-ordered of order type at most ω;*
- (iii) *W is well-ordered of order type at most ω^2; in particular there is no infinite strictly descending chain in W, and every non-empty subset of W has a least element;*
- (iv) $\mathrm{IS}(K)$ *has countably many vertices.*

Proof. Fix h and let S be the compact orientable surface of genus h with one boundary circle. We use the compactness theorem of Kapovich's appendix to [16] in the form printed there, and we transcribe its membership conditions rather than paraphrasing them. Its setting [16, p. 897] is a P^2-irreducible compact Riemannian 3-manifold M with smooth strictly convex boundary, together with a compact family $\mathcal{F}$ of smooth curves on ∂M, the case of interest being that in which ∂M is a single torus and $\mathcal{F}$ is a smooth foliation of it by closed curves. For a proper embedding $f\colon (S, \partial S) \to (M, \partial M)$ whose class is admissible — meaning that $f^{-1}(\partial M)$ is a single component of ∂S and $f(S)$ is incompressible — one writes $M([f])$ for the set of stable minimal surfaces in the proper isotopy class $[f]$ whose boundary is a parametrised curve of $\mathcal{F}$, modulo reparametrisation, and

$$\mathcal{M}(S) \;=\; \bigcup_{[f]} M([f]),$$

the union being over all such classes, topologised by C^1 convergence; $\mathcal{M}_a$ denotes the subset of surfaces of area at most a. Two features of this definition are used below and are worth naming. The union runs over *all* admissible proper isotopy classes at once, and the boundary of a member is required only to be a curve of $\mathcal{F}$, not a prescribed one.

These hypotheses hold for $M = E(K)$, $\mathcal{F} = J$: the exterior is compact and orientable and is irreducible [12, p. 228], and an orientable 3-manifold contains no two-sided projective plane, so $E(K)$ is P^2-irreducible; $\partial E(K)$ is a torus, made strictly convex by (1); and J is a smooth — indeed real-analytic — foliation of it by closed curves. A Seifert surface in the exterior is properly embedded, incompressible, and meets $\partial E(K)$ in its single boundary circle, so its class is admissible.

Now let v be a vertex with $g(v) = h$ and $A(v) \leq a$, and let Σ_v be a minimiser supplied by (E)(ii). It is a stable minimal surface with boundary a leaf of J and area $A(v) \leq a$, so $\Sigma_v \in \mathcal{M}_a$. By [16, Corollary A.3, p. 898] the set $\mathcal{M}_a$ is the disjoint union of finitely many open and closed subsets $\mathcal{M}_a([g])$, each consisting of surfaces isotopic to one another. Distinct vertices are distinct ambient isotopy classes, hence, by Lemma 3.9 and the convention fixed with it, distinct proper isotopy classes; so $v \mapsto$ the piece containing Σ_v is injective on the set in (i), which is therefore finite.

For (ii): $W_h \cap (0, a]$ is the image of the finite set of (i), hence finite, for every $a > 0$; a subset of $\mathbb{R}_{>0}$ meeting every interval $(0, a]$ in a finite set is closed and discrete in $\mathbb{R}_{>0}$ and is order-isomorphic to an initial segment of $\mathbb{N}$. For (iii): lexicographically, W is the ordered sum $\sum_{h\in\mathbb{N}} W_h$ of well-orders of type at most ω indexed by $\mathbb{N}$, hence a well-order of type at most $\omega \cdot \omega = \omega^2$. For (iv): each set in (i) is finite and $\mathrm{IS}(K)^{(0)} = \bigcup_{h\in\mathbb{N}} \bigcup_{n\in\mathbb{N}} \{v : g(v) = h,\ A(v) \leq n\}$. □

Remark 4.3 (Least elements without choice). Remark 6.5 records that in general the passage from "no infinite descending chain" to "every non-empty subset has a least element" uses dependent choice. For the complexity of Definition 4.1 it does not. Let $\emptyset \neq T \subseteq W$. The set $\{h : (h, \alpha) \in T \text{ for some } \alpha\}$ is a non-empty subset of $\mathbb{N}$ and so has a least element h_0. Pick any α_1 with $(h_0, \alpha_1) \in T$ — a single instance, not a choice function. By Lemma 4.2(i) the set $\{\alpha : (h_0, \alpha) \in T,\ \alpha \leq \alpha_1\}$ is finite and non-empty, so it has a least element α_0, and $(h_0, \alpha_0) = \min T$. No form of choice is used. The use of choice flagged in Remark 6.5 is elsewhere: in the well-ordering of the fibres in Lemma 6.6.

Remark 4.4 (Two remarks on the hypotheses of Lemma 4.2). First, the appendix [16, p. 897] carries two standing hypotheses on the surface: that it is connected, and that its preimage of each boundary component of M is a single component of its own boundary. For a knot both are automatic, a spanning surface in the sense of §3.2 having no closed component and exactly one boundary circle; for links they are supplied by Lemma 8.3.

Second, on this route the hypothesis that K is non-trivial is not consumed. Every knot exterior, the unknot's included, satisfies those hypotheses, so the finiteness statement (i) holds for the unknot as well — where, by Proposition 3.2, it is anyway visible. The hypothesis is kept in the statement because it is the standing hypothesis of Theorem A and of every result of §5, and because a version of this lemma proved instead by normalising surfaces in the infinite cyclic cover *does* consume it. The exponent in ω^2 is an upper bound only: whether W has order type ω is exactly the question (F) of §7.4, which is open.

4.2. **The cyclic-cover frontier construction.** The exchange lemma is handed a vertex x and two vertices u, w adjacent to it and at distance 2 from one another. What Proposition 3.4 supplies is one surface disjoint from two others which meet each other; the construction below turns that configuration into two new Seifert surfaces. It is stated for surfaces, not for classes, and no minimality is assumed or concluded.

Lemma 4.5 (Frontier construction). *Let S, S_+, S_- be Seifert surfaces in $E(K)$ with*

$$S \cap (S_+ \cup S_-) = \emptyset, \tag{9}$$

with S_+ meeting S_-, the pair S_+, S_- transverse, $\partial S_+ \cap \partial S_- = \emptyset$, and with no disc patch produced by cutting S_+ and S_- along $S_+ \cap S_-$. Then there are Seifert surfaces $P_\uparrow, P_\downarrow$ such that

(1) *each is disjoint from S, S_+, S_- and from the other;*
(2) *every compact surface Z disjoint from $S_+ \cup S_-$ is disjoint from some choice of representatives of $P_\uparrow, P_\downarrow$;*
(3) *writing $T_\uparrow, T_\downarrow$ for the possibly disconnected frontier outputs containing $P_\uparrow, P_\downarrow$,*

$$\chi(T_\uparrow) + \chi(T_\downarrow) = \chi(S_+) + \chi(S_-), \tag{10}$$

and every component discarded from either output is a closed surface of Euler characteristic at most 0.

Proof. Let $p\colon \widetilde{E} \to E(K)$ be the infinite cyclic cover determined by the linking number homomorphism $\pi_1(E(K)) \twoheadrightarrow \mathbb{Z}$ and let τ generate its positive deck group. A Seifert surface carries the primitive class dual to p, hence is non-separating, and cutting along S produces the fundamental domain C between a lift S_0 and $S_1 = \tau S_0$, with $p|_C\colon C \to E(K) \setminus S$ a homeomorphism. Hypothesis (9) is what makes S_+ and S_- lie in $E(K) \setminus S$, so the inverse of that homeomorphism lifts them to $S_0^+, S_0^- \subset C$; this is the only place (9) is used, and it is used for both surfaces at once. Each of $S_0^\pm$ is a wall in the corresponding stacking of C-translates and its complement in $\widetilde{E}$ has two components; write $M_\uparrow^\pm, M_\downarrow^\pm$ for them, the upper one being the one containing S_1. Put

$$\widetilde{B}_\downarrow = \mathrm{fr}\big(M_\downarrow^- \cap M_\downarrow^+\big), \qquad \widetilde{B}_\uparrow = \mathrm{fr}\big(M_\uparrow^- \cap M_\uparrow^+\big). \tag{11}$$

These are assembled from complementary subsurfaces of $S_0^+ \cup S_0^-$ and are piecewise smooth with corners exactly along $\Gamma := S_0^+ \cap S_0^-$, which by transversality and $\partial S_+ \cap \partial S_- = \emptyset$ is a finite family of disjoint smooth closed curves in the interior. Round the corners by Lemma 5.1 to get smooth neatly embedded $\widehat{B}_\uparrow, \widehat{B}_\downarrow$, then push each into its own open region at scale σ by Lemma 5.2. The two regions are opposite, so the results are disjoint from the lifted inputs and from each other; they lie in C, on which p is injective, and project to embedded surfaces $T_\uparrow(\sigma), T_\downarrow(\sigma)$ disjoint from S, S_+, S_-. By clause (3) of Lemma 5.2 all $\sigma \in (0, \sigma_0]$ give isotopic outputs, so σ labels a representative and not a class, and is suppressed.

Each frontier is the frontier of a region in an orientable cover, hence two-sided and consistently oriented across the resolved seams, and the algebraic intersection number of a lifted meridian with its boundary is $+1$.

We count the boundary curves, since the conclusion depends on it. The preimage of $\partial E(K)$ in $\widetilde{E}$ is an annulus, the longitude lifting and the meridian not, and ∂S_0^+ and ∂S_0^- are two disjoint core circles of it; so one of them separates the other from the positive end. Say ∂S_0^- lies below ∂S_0^+. Points of S_0^+ near ∂S_0^+ then lie above S_0^-, that is in $M_\uparrow^-$, and S_0^+ separates $M_\uparrow^+$ from $M_\downarrow^+$; so they are frontier points of $M_\uparrow^- \cap M_\uparrow^+$ and $\partial S_0^+ \subset \widetilde{B}_\uparrow$. Symmetrically, points of S_0^- near ∂S_0^- lie below S_0^+, so $\partial S_0^- \subset \widetilde{B}_\downarrow$. Since Γ is a family of closed curves in the interior, cutting along it creates no new boundary, and $\partial \widetilde{B}_\uparrow = \partial S_0^+$, $\partial \widetilde{B}_\downarrow = \partial S_0^-$: each frontier has exactly one boundary circle. Hence each output has exactly one component with boundary, a connected oriented Seifert surface whose boundary is a longitude, and the others are closed. These are $P_\uparrow, P_\downarrow$, and clause (1) holds.

For (2): if Z is compact and disjoint from the compact set $S_+ \cup S_-$ their distance is positive, so the push-off may be taken small enough to miss Z, and two sufficiently small push-offs on the same side are isotopic through the collar between them, so only the representative changes.

For (3): the two frontiers exhaust the cut-open $S_0^+ \cup S_0^-$, and Lemma 3.6 applied to this exchange gives (10). A discarded component is closed, and being cut from the orientable $S_+ \cup S_-$ it is orientable, so it has $\chi \leq 0$ unless it is a sphere; suppose one were. If it carried no exchange seam it would be a whole component of one of the inputs with the intersection curves deleted, and no such component is closed, the inputs being connected with non-empty boundary. So it carries a seam, and an innermost seam on it bounds a seam-free disc on the sphere: a disc patch, which the hypothesis excludes. □

Remark 4.6. The corresponding construction in the literature assumes minimal genus in its hypotheses and asserts it in its conclusion. Nothing of the kind is used or claimed here: Lemma 4.5 says nothing about the genus of either output beyond what (10) forces, and that is all §5 consumes. The hypothesis (9) is, by contrast, indispensable, and it is the exact sentence that fails for a general link; see §8.4.

5. The exchange lemma

This section proves axiom (N2) for IS(K). Given two vertices at distance 2 and a common neighbour, it produces two new vertices, each an apex of the two-interval, whose complexities cannot both fail to drop. The mechanism is Lemma 4.5 applied to relative area minimisers, and the difficulty is quantitative: rounding the corners of a frontier gains area, pushing the result off the inputs costs area, and the gain must be pinned down before the cost — and, in §§5.3–5.5, before the further parameter introduced by making two tangent minimisers transverse.

Throughout, K is a non-trivial knot, and by Theorem 2.10 and Lemma 2.11 every minimiser supplied by (E) may be taken smooth up to $\partial E(K)$ and is then neatly embedded.

5.1. **Rounding and pushing off.** The frontier of Lemma 4.5 is piecewise smooth with corners, whereas $A(\cdot)$ is an infimum over smooth neatly embedded surfaces. Two steps separate the two, and their order is the pivot of the quantifier structure below. Lemma 5.1 is stated pointwise — varying angle, varying width, gain an integral — because in §5.4 the angle degenerates near a tangency point and a single global width would collapse with it.

Lemma 5.1 (Corner rounding, pointwise form)**.** *Let Γ be a finite disjoint union of compact C^2 curves without boundary in a compact Riemannian 3-manifold, let $\widetilde{B}$ be an embedded surface with a corner along Γ, smooth off Γ and with its two sheets smooth up to Γ — every constant below depends on them only through their C^2 bounds — and let $\Gamma \cap \partial\widetilde{B} = \emptyset$. Assume that in the normal slice at each $x \in \Gamma$ the trace of $\widetilde{B}$ consists of the two edges of a wedge with vertex x and opening angle $\alpha(x) \in (0, \pi)$ — so that α, and with it the slope $m := \cot(\alpha/2)$, is C^1 along Γ — and put*

$$\theta(x) := \min\{\alpha(x), \pi - \alpha(x)\} > 0. \tag{12}$$

Then there are continuous $\eta_{\max}, \kappa \colon \Gamma \to (0, \infty)$, a continuous non-decreasing $g_0 \colon (0, \pi/2] \to (0, \infty)$, and non-decreasing $\underline{\eta}, \underline{\kappa} \colon (0, \pi/2] \times (0, \infty)^2 \to (0, \infty)$ determined by the ambient metric, by the C^2 bounds of the two sheets on the normal slice at x, by the local reach

$$\begin{aligned} \varrho(x) := \sup\Big\{\varrho > 0 :\ \exp^{\perp} \ &\textit{is injective on} \\ &\{(z, \xi) : z \in \Gamma,\ |\xi| < \varrho,\ d_\Gamma(z, x) < \varrho\}, \\ &\textit{and } B(x, \varrho) \cap \Gamma \textit{ is connected}\Big\} \end{aligned} \tag{13}$$

and by $\varrho_\partial(x) := d(x, \partial\widetilde{B})$, such that

$$\eta_{\max}(x) \ \geq \ \underline{\eta}(\theta(x), \varrho(x), \varrho_\partial(x)), \qquad \kappa(x) \ \geq \ \underline{\kappa}(\theta(x), \varrho(x), \varrho_\partial(x)), \tag{14}$$

and such that for every C^1 width function η subject to

$$0 < \eta(x) \leq \eta_{\max}(x) \qquad \textit{and} \qquad |\eta'(x)| \leq \kappa(x) \tag{15}$$

there is a compact C^∞ neatly embedded surface $\widehat{B} = \widehat{B}(\eta)$ with:

(1) $\widehat{B}$ *agrees pointwise with* $\widetilde{B}$ *outside the tubular neighbourhood* $\{d(\cdot,x) < \eta(x)\}$ *of* Γ*; in particular* $\partial\widehat{B} = \partial\widetilde{B}$*;*
(2) *inside that neighbourhood* $\widehat{B}$ *lies on the closed side of* $\widetilde{B}$ *into which the wedge of angle* α *opens; consequently, if* $\widetilde{B}$ *is the frontier of an open* $U \subset E(K)$ *and the wedge opens into* U *at every point of* Γ*, then* $\widehat{B} \subset \overline{U}$ *and there is a one-sided collar* $\widehat{B} \times [0, \epsilon_1) \to E(K)$ *into* U*;*
(3) $\widehat{B}$ *is carried to* $\widetilde{B}$ *by an* ambient *isotopy of the ambient manifold supported in that neighbourhood;*
(4) *the area gain is non-negative slice by slice, and*

$$\operatorname{Area}(\widehat{B}) \ \le\ \operatorname{Area}(\widetilde{B}) - \int_\Gamma g_0(\theta(s))\,\eta(s)\,ds; \tag{16}$$

(5) (The bounded region is unchanged.) *Write* $T := \{d(\cdot,x) < \eta(x)\}$ *for the tubular neighbourhood of (1). If* $\widetilde{B}$ *is a closed surface separating the ambient manifold, then so is* $\widehat{B}$*, and the complementary regions correspond under the ambient isotopy of (3): if* R *is a complementary region of* $\widetilde{B}$ *and* $\widehat{R}$ *the corresponding one of* $\widehat{B}$*, then* $\widehat{R} \setminus T = R \setminus T$*.*

In particular the integral in (4) may be restricted to any measurable subset of Γ *and the rest discarded.*

Proof. Take an odd $\chi \in C^\infty([-1,1],[-1,1])$ with $0 \le \chi \le 1$ on $[0,1]$, $\chi \equiv 1$ near 1 and $\chi(0) = 0$, and put $\Theta(t) = c_0 + \int_0^t \chi$ with $c_0 := 1 - \int_0^1 \chi \ge 0$. Then Θ is even, $\Theta(\pm 1) = 1$, $\Theta'(\pm 1) = \pm 1$ and $\Theta^{(k)}(\pm 1) = 0$ for $k \ge 2$, so Θ meets $|\cdot|$ to infinite order at $|t| = 1$; and $t \mapsto \Theta(t) - t$ is non-increasing on $[0,1]$ with value 0 at 1, so $\Theta \ge |\cdot|$; finally $\Theta'(0) = 0$ and $|\Theta'| \le 1$. For $m > 0$ put

$$g(m) := \int_{-1}^{1} \Big(\sqrt{1+m^2} - \sqrt{1+m^2\Theta'(\tau)^2}\Big)d\tau,$$

which is positive because $|\Theta'| \le 1$ everywhere and $\Theta'(0) = 0$, and continuous in m. Moreover g is non-decreasing: differentiating under the integral sign,

$$\partial_m g(m) = \int_{-1}^{1} m\Big[(1+m^2)^{-1/2} - \Theta'(\tau)^2\big(1+m^2\Theta'(\tau)^2\big)^{-1/2}\Big]d\tau \ \ge\ 0,$$

since $u \mapsto u(1+m^2u)^{-1/2}$ has derivative $(1+m^2u)^{-3/2}(1+\frac{1}{2}m^2u) > 0$ and $\Theta'(\tau)^2 \le 1$. Set $g_0(\theta) := \frac{1}{2}\min\{g(m) : m \in [\tan(\theta/2), \cot(\theta/2)]\} > 0$; since $\alpha(x)$ lies in $[\theta(x), \pi - \theta(x)]$, the slope $m(x) = \cot(\alpha(x)/2)$ lies in that interval. The minimum is therefore attained at the left endpoint, $g_0(\theta) = \frac{1}{2}g(\tan(\theta/2))$, and g_0 is *non-decreasing*: as $\theta \downarrow 0$ the wedge degenerates to a slit, the two sheets are nearly parallel, and smoothing them saves almost nothing, so $g_0(\theta) \to 0$.

Straighten first. Each sheet is smooth up to Γ, and the two meet there at the angle $\alpha \in (0,\pi)$, so they are transverse along Γ. Extend both across Γ to smooth surfaces $\widehat{A}_\pm$ meeting transversally in Γ. Straighten $\widehat{A}_+$ to $\{y = 0\}$ with Γ the z-axis; transversality makes $\widehat{A}_-$ a graph $\{x = G(y,z)\}$ with $G(0,z) = 0$, and $(x,y,z) \mapsto (x - G(y,z), y, z)$ is a diffeomorphism carrying it to $\{x = 0\}$ and fixing $\{y = 0\}$. So there is a smooth chart, fixing Γ pointwise, in which the two sheets are half planes meeting along the Γ-axis.

The metric in that chart is arbitrary, and correcting it is where the construction has to be done with care. Both sheets are planes *containing* the Γ-axis, so both are preserved not only by linear maps of the normal plane but also by shears along Γ, $(z,x,y) \mapsto (z + ax + by,\ x,\ y)$; and it is the shears that are needed, because a linear map of the normal plane alone cannot alter the cross terms $h(\partial_z, \partial_x)$, $h(\partial_z, \partial_y)$, which the straightening step — a purely smooth normal form, carrying no metric information — leaves arbitrary. With both at hand, apply Gram–Schmidt in the order forced by the sheets: normalise ∂_z; project ∂_x and ∂_y h-orthogonally to it, which is a shear and so keeps each sheet planar; then map the two resulting rays to the rays at angles $\pm\alpha(s)/2$ from $+\xi_1$, which is a linear map of the now h-orthogonal normal plane. Doing this smoothly in s leaves s arclength on Γ, the slices $\{s = \text{const}\}$ h-orthogonal to Γ, and $h = \delta$ at every point of Γ — cross terms included — while the wedge is the standard one of opening angle $\alpha(s)$, that angle being measured in h. Writing (s, ξ_1, ξ_2) for this chart, with the bisector along $+\xi_1$,

$$\widetilde{B} = \{\xi_1 = m(s)|\xi_2|\} \ \text{ near } \Gamma, \qquad h = \delta + O(|\xi|). \tag{17}$$

What matters is that the first of these is an equality. A replacement profile matched to the model wedge $m|\xi_2|$ while the true sheets are only $m|\xi_2| + O(|\xi|^2)$ would miss them by $O(\eta^2)$ at the seam, and the surface obtained by gluing would fail to be continuous there, let alone smooth; the chart removes that mismatch rather than estimating it. The price is that C_{met} below now depends on the two sheets, through the distortion of the straightening, and not on the ambient metric alone.

On $|\xi_2| \le \eta(s)$ replace the two sheets by $\xi_1 = m(s)\,\eta(s)\,\Theta(\xi_2/\eta(s))$. At $|\xi_2| = \eta(s)$ this equals $m(s)\eta(s) = m(s)|\xi_2|$; and the difference of the two profiles is $m(s)\eta(s)\Psi(\xi_2/\eta(s))$ with $\Psi := \Theta - |\cdot|$, every derivative of which is a finite sum of terms carrying a factor $\Psi^{(k)}(t)$, so all of them vanish at $|t| = 1$. The glued surface is therefore C^∞ across the seam and coincides with $\widetilde{B}$ outside the tube. Since $\Theta \ge |\cdot|$ the new sheet lies in the closed wedge side $\{\xi_1 \ge m(s)|\xi_2|\}$, and in its interior where $|\xi_2| < \eta$ — which is the local half of (2), the global half following from it under the additional hypothesis stated there. Taking $\eta_{\max}$ less than half the normal injectivity radius keeps the tubes disjoint and embedded, and less than half ϱ_∂ keeps them clear of $\partial\widetilde{B}$. This is (1)–(2), the collar in (2) coming from compactness.

For (3): inside the tube the old and the new sheets are, in each normal slice, two graphs over the same segment $|\xi_2| \le \eta(s)$, namely $\xi_1 = m|\xi_2|$ and $\xi_1 = m\eta\Theta(\xi_2/\eta)$, agreeing to infinite order at the two ends. The linear interpolation between them is therefore a family of embedded surfaces, constant outside the tube, and pushing across the region they cobound gives an ambient isotopy supported in the tube. Clause (5) is then immediate, an ambient isotopy carrying complementary regions to complementary regions, together with the fact that the isotopy is the identity outside T; separation of the ambient manifold is preserved because $\widehat{B}$ is the image of $\widetilde{B}$ under an ambient homeomorphism. This is the statement that "rounding does not change the region bounded", used three times below and in §3, and it is proved here once rather than asserted there.

A varying width does not spoil the matching. Writing $t := \xi_2/\eta(s)$, the new sheet is the graph $\xi_1 = F(s, \xi_2)$ with $F = m\eta\Theta(t)$, so

$$F_{\xi_2} = m(s)\Theta'(t), \qquad F_s = \eta(s)m'(s)\Theta(t) + m(s)\eta'(s)\big(\Theta(t) - t\Theta'(t)\big), \tag{18}$$

and at the seam $|t| = 1$ we have $\Theta(1) = 1$ and $\Theta'(\pm 1) = \pm 1$, so $\Theta - t\Theta'$ vanishes there and $F_s = \eta m'$, which by (17) is exactly the corresponding derivative of the old sheet: that sheet is $m(s)|\xi_2|$, whose s-derivative at $|\xi_2| = \eta$ is $m'\eta$. The η' term switches itself off exactly at the seam, and the two pieces still meet to infinite order along the boundary curves $|\xi_2| = \eta(s)$, which are now non-parallel.

That same η' term is why (15) constrains η' at all. It is $O(\eta')$ and not $O(\eta)$, so no shrinking of $\eta_{\max}$ controls it: for $\eta(s) = \epsilon\big(2 + \sin(s/\epsilon^2)\big)$ one has $\eta \le 3\epsilon \to 0$ while $|\eta'| \to \infty$. Since $c_0 \le \Theta \le 1$ and $|\Theta'| \le 1$ on $[-1, 1]$ we have $|\Theta - t\Theta'| \le 2$, whence

$$F_s^2 \ \le\ 2\eta^2 m'^2 + 8m^2\eta'^2. \tag{19}$$

One slice. Fix s. Over $|\xi_2| \le \eta(s)$ the old surface is the two graphs $\xi_1 = \pm m\xi_2$ of (17), so in the Euclidean model of the chart their total length is $2\eta\sqrt{1 + m^2} + E_1$ with $E_1 = 0$. The term is carried through the estimate below under the weaker bound $|E_1| \le C_1\eta^2$, C_1 depending only on the C^2 bounds of the two sheets, so that nothing depends on the straightening having been exact; in the chart just built the third entry of (22) may be read as $+\infty$, as the second already is where $m' = 0$. The new profile has length

$$\eta \int_{-1}^{1} \sqrt{1 + m^2\Theta'^2 + F_s^2}\, dt \;\le\; \eta \int_{-1}^{1} \sqrt{1 + m^2\Theta'^2}\, dt + \eta \max_t F_s^2,$$

by $\sqrt{a+b} \le \sqrt{a} + b/(2\sqrt{a})$ with $a \ge 1$ over an interval of length 2. So the Euclidean decrease in the slice is at least $\eta\bigl[g(m) - 2\eta^2 m'^2 - 8m^2\eta'^2 - C_1\eta\bigr]$, using (19).

Passing to the true metric. By (17) $h = \delta + O(|\xi|)$ in the chart built above, and inside the tube $|\xi| \le \eta\sqrt{1+m^2}$. Let a graph $\xi_1 = u$ have Euclidean first fundamental form Q and true one $Q + \mathcal{E}$. Every tangent vector v satisfies $|\mathcal{E}(v,v)| \le \|h - \delta\|\,|v|^2 = \|h-\delta\|\,Q(v,v)$, so the eigenvalues of $Q^{-1}\mathcal{E}$ are bounded by $\|h-\delta\|$ *irrespective of* $|\nabla u|$, and

$$\Bigl|\sqrt{\det(Q+\mathcal{E})} - \sqrt{\det Q}\Bigr| \;\le\; C_{\mathrm{met}}\,\eta\sqrt{1+m^2}\,\sqrt{\det Q}. \tag{20}$$

The uniformity in the gradient is what lets the estimate survive $m \to \infty$. Both profiles have Euclidean length $O(\eta\sqrt{1+m^2})$, so (20) costs at most $6C_{\mathrm{met}}\eta^2(1+m^2)$, and

$$\begin{aligned}\operatorname{Area}(\widetilde{B}) - \operatorname{Area}(\widehat{B}) \;\ge\; \int_\Gamma \eta\Bigl[&g(m) - 2\eta^2 m'^2 - 8m^2\eta'^2 \\ &- C_1\eta - 6C_{\mathrm{met}}\eta(1+m^2)\Bigr]\,ds.\end{aligned} \tag{21}$$

Choosing $\eta_{\max}$ and κ. Since $g(m(s)) \ge 2g_0(\theta(s))$ by the definition of g_0, put

$$\begin{aligned}\eta_{\max} &:= \min\Bigl\{\frac{\varrho}{2\sqrt{1+m^2}},\ \sqrt{\frac{g_0(\theta)}{8m'^2}},\ \frac{g_0(\theta)}{4C_1}, \\ &\qquad\quad \frac{g_0(\theta)}{24C_{\mathrm{met}}(1+m^2)},\ \frac{\varrho_\partial}{2}\Bigr\}, \\ \kappa &:= \frac{\sqrt{g_0(\theta)}}{6m},\end{aligned} \tag{22}$$

all five entries being read pointwise at $x \in \Gamma$, with ϱ the local reach (13), $\varrho_\partial(x) = d(x, \partial\widetilde{B}) > 0$ by the hypothesis $\Gamma \cap \partial\widetilde{B} = \emptyset$, and the second entry read as $+\infty$ where $m' = 0$. The fifth entry is what makes the words “in particular $\partial\widehat{B} = \partial\widetilde{B}$” in (1) legitimate: the conclusion that $\widehat{B}$ agrees with $\widetilde{B}$ outside the tube says nothing about the boundary unless the tube is known to miss it, and none of the other four entries mentions $\partial\widetilde{B}$. Both ϱ and ϱ_∂ are positive and lower semicontinuous on Γ, and it is only their values on the subset over which the gain is collected that enter the estimates below; this is the reason for stating them pointwise rather than as the global normal injectivity radius and the global distance $d(\Gamma, \partial\widetilde{B})$, both of which degenerate in the applications of §§5.4–5.5. The second, third and fourth entries, together with κ, make the four error terms in (21) at most $\frac14 g_0(\theta)$ each: the second bounds $2\eta^2 m'^2$, the third $C_1\eta$, the fourth $6C_{\mathrm{met}}\eta(1+m^2)$, while $8m^2\eta'^2 \le 8m^2\kappa^2 = \frac29 g_0(\theta) < \frac14 g_0(\theta)$. The first and fifth entries do not enter (21) at all: the first secures the legitimacy of the Fermi chart and the disjointness and embeddedness of the tubes, the fifth their avoidance of $\partial\widetilde{B}$. So the bracket is at least $2g_0(\theta) - \frac{35}{36}g_0(\theta) > g_0(\theta) \ge 0$ for each s separately. That is (16) together with the slicewise non-negativity asserted in (4), and it is what allows the integral to be restricted to a measurable subset.

Finally (14). Where $\theta(x) \ge \theta_\bullet$, $\varrho(x) \ge \varrho_\bullet$ and $\varrho_\partial(x) \ge \varrho_{\partial,\bullet}$, the slope $m = \cot(\alpha/2)$ lies in $[\tan(\theta_\bullet/2), \cot(\theta_\bullet/2)]$ and $g_0(\theta) \ge g_0(\theta_\bullet)$, so (22) is bounded below by the quantity obtained from it by replacing m with $\cot(\theta_\bullet/2)$, m' with $M' := \sup_\Gamma |m'| < \infty$, $g_0(\theta)$ with $g_0(\theta_\bullet)$, ϱ with

$\varrho_\bullet$ and ϱ_∂ with $\varrho_{\partial,\bullet}$; call these $\underline{\eta}(\theta_\bullet, \varrho_\bullet, \varrho_{\partial,\bullet})$ and $\underline{\kappa}(\theta_\bullet, \varrho_\bullet, \varrho_{\partial,\bullet})$. Both are non-decreasing in each argument, since g_0 is non-decreasing and $\cot(\theta/2)$ is decreasing. The dependence on θ cannot be removed: the first entry of (22) forces $\eta_{\max} \lesssim \varrho/m \to 0$ as $\alpha \to 0$, and $g(m)/(1+m^2) \to 0$ at *both* ends $\alpha \to 0$ and $\alpha \to \pi$. So no bound in terms of the reach and the metric alone is available, and Corollary 5.9 below must supply an angle bound before it can supply a width bound. □

Lemma 5.2 (Pushing off). *Let $\widehat{B}$ be a compact C^∞ neatly embedded surface satisfying clause (2) of Lemma 5.1 for an open region U, with $\partial\widehat{B}$ a leaf of the longitudinal foliation. Then there are $\sigma_0 > 0$ and $C < \infty$, depending only on $\widehat{B}$ and on the ambient metric near it, such that for every $\sigma \in (0, \sigma_0]$ there is a compact C^∞ neatly embedded $T(\sigma)$ with $T(\sigma) \subset U$; $\partial T(\sigma)$ a leaf; $T(\sigma)$ isotopic to $\widehat{B}$ and any two of them isotopic to each other; and $|\mathrm{Area}(T(\sigma)) - \mathrm{Area}(\widehat{B})| \le C\sigma$.*

Proof. Let ν be the unit normal of $\widehat{B}$ pointing into U and n the outward normal of $\partial E(K)$, extended through the boundary collar; put $X := \nu - \psi(r)\langle \nu, n\rangle n$ with $\psi \equiv 1$ near $r = 0$ and supported in the collar. Then X is tangent to $\partial E(K)$, and neatness with compactness gives $\langle \nu, n\rangle^2 \le 1 - c$, so $\langle X, \nu\rangle \ge c > 0$. On $\partial E(K)$ replace X by a field transverse to J whose flow carries leaves to leaves, oriented towards U, and interpolate in r, shrinking the collar so that $\langle X, \nu\rangle \ge c/2$ persists. Its flow Ψ_s exists for $s \in [0, \sigma_0]$, preserves $\partial E(K)$ and carries leaves to leaves; put $T(\sigma) := \Psi_\sigma(\widehat{B})$. Transversality makes $(x, s) \mapsto \Psi_s(x)$ an embedding into the one-sided collar for σ_0 small, which gives the first three clauses, the last of them because Ψ is an ambient isotopy and $\Psi_{\sigma'}\Psi_\sigma^{-1}$ joins two push-offs. With $K_1 := \sup|\nabla X|$ near $\widehat{B}$, the Jacobian of $\Psi_s|_{\widehat{B}}$ obeys $\partial_s \log J = \operatorname{div} X \le 2K_1$ in absolute value, so $|\,\mathrm{Area}(T(\sigma)) - \mathrm{Area}(\widehat{B})| \le (e^{2K_1\sigma} - 1)\,\mathrm{Area}(\widehat{B}) \le C\sigma$ with $C := 2K_1 e^{2K_1\sigma_0}\,\mathrm{Area}(\widehat{B})$, independent of σ. □

The order of the two steps is forced: a normal push-off of a cornered surface is not embedded, while the frontier of an inner parallel set is again cornered. Hence C is determined by $\widehat{B}$, that is, by the cornered frontier and the once-chosen width η: *the rounding gain is fixed before σ is.*

The next lemma excludes a disc in the symmetric difference. It carries an error term because in §§5.4–5.5 one of the two surfaces is a perturbation of a minimiser and not a minimiser.

Lemma 5.3 (No disc patch). *Let A_0, B be relative area minimisers in two distinct vertices of $\mathrm{IS}(K)$, let A be neatly embedded, incompressible, normally isotopic to A_0 with ∂A a leaf of J, and put $e := \mathrm{Area}(A) - \mathrm{Area}(A_0) \ge 0$. Suppose A and B are transverse with $\partial A \cap \partial B = \emptyset$, and suppose that cutting A and B along $\Gamma := A \cap B$ produces a disc patch. For a closed curve $\gamma \subset \Gamma$ and a width function admissible along it, let $\mathcal{G}(\gamma)$ denote the rounding gain (16) along γ; by (12) it is one and the same number for either of the two pairs of opposite sectors at γ. Then there are closed curves $a, b \subset \Gamma$, not necessarily distinct, with*

$$\mathcal{G}(a) + \mathcal{G}(b) \ \le\ e. \tag{23}$$

One point about $\mathcal{G}$ has to be made before the proof, because the lemma is applied with a width function chosen for a *different* surface. In §§5.4–5.5 the gain along γ is computed for the cornered frontiers of Lemma 4.5, whereas the proof below rounds the cornered surfaces produced by a disc exchange. A single η serves both. Indeed, by (22) the bounds $\eta_{\max}$ and κ at $x \in \gamma$ depend only on: the opening angle $\theta(x)$, which by (12) is the same for all four sectors at x and hence for both surfaces; the local reach (13) of γ, which depends on γ alone; the C^2 bounds of the two sheets meeting along γ, which in both cases are those of A and of B, the exchange surface differing from one of them by a push-off whose scale we let tend to 0 and in which all the estimates below are continuous; and $\varrho_\partial(x)$. For the last, every surface occurring here — the two frontiers, A, B, and the exchanged surfaces — has its boundary on $\partial E(K)$, so $\varrho_\partial(x) \ge d(x, \partial E(K))$ in every case; replacing ϱ_∂ by $d(\cdot, \partial E(K))$ in (22) only decreases $\eta_{\max}$, and we do so throughout. With that reading a width admissible along γ for one of these surfaces is admissible for all of them, and $\mathcal{G}(\gamma)$ is unambiguous.

Proof. No patch has a boundary arc on its frontier, since the endpoints of such an arc would lie in $\partial A \cap \partial B = \emptyset$; so Γ is a finite disjoint union of closed curves in $\operatorname{Int} E(K)$ and the frontier of a disc patch is a single one of them.

Step 0*: the discs* $D_A(\gamma)$*.* A simple closed curve γ in the interior of A bounds at most one disc in A: the two sides of a separating γ cannot both be discs, or A would be a sphere, and the disc side is the one missing ∂A. Write $D_A(\gamma)$ for that disc when it exists and A_γ for the closure of $A \setminus D_A(\gamma)$; since A is connected, a component of A_γ missing γ would be a component of A, so A_γ is connected, and so is $A_\gamma \setminus \gamma$. Use the same notation for B. Let $\mathcal{D} \subset \Gamma$ be the set of curves bounding a disc in A or in B. For $\gamma \in \mathcal{D}$ *both* discs exist: γ bounds a disc, hence is null-homotopic in $E(K)$; a two-sided incompressible surface is π_1-injective, so γ is null-homotopic in the other surface as well; and a null-homotopic simple closed curve on a surface bounds a disc there. A disc patch is $D_A(\gamma)$ or $D_B(\gamma)$ for γ its frontier, by the uniqueness just proved, so $\mathcal{D} \neq \emptyset$. Since $D_A(\gamma) \subset A$ and $A \cap B = \Gamma$, the sets

$$\mathcal{A} := \{\gamma \in \mathcal{D} : \operatorname{Int} D_A(\gamma) \cap B = \emptyset\}, \qquad \mathcal{B} := \{\gamma \in \mathcal{D} : \operatorname{Int} D_B(\gamma) \cap A = \emptyset\}$$

are the sets of curves whose disc on the respective side is a disc patch.

Step 1*:* $\mathcal{A}$ *and* $\mathcal{B}$ *are non-empty.* Γ is finite, hence so is $\mathcal{D}$. Let $\gamma \in \mathcal{D}$ have $D_A(\gamma)$ minimal under inclusion among $\{D_A(\gamma') : \gamma' \in \mathcal{D}\}$. If a curve γ' of Γ met $\operatorname{Int} D_A(\gamma)$ it would lie in it, and would cut off a subdisc of $D_A(\gamma)$, which is $D_A(\gamma')$ by Step 0; then $D_A(\gamma') \subsetneq D_A(\gamma)$ contradicts minimality. So $\gamma \in \mathcal{A}$, and symmetrically $\mathcal{B} \neq \emptyset$.

Step 2*: one-sided exchange.* Let $\gamma \in \mathcal{A}$ and write $D := D_A(\gamma)$, $D' := D_B(\gamma)$. Then $\operatorname{Int} D \cap B = \emptyset$ forces $D \cap B = \gamma$, so $D \cup D'$ is an embedded sphere, and it lies in $\operatorname{Int} E(K)$ because $A \cap \partial E(K) = \partial A$ is not met by D, and likewise for D'. It has a corner along γ; round it by the qualitative part of Lemma 5.1, at a width $\eta_\sharp$ small enough that the tube T about γ misses $\partial E(K)$. By irreducibility the resulting smooth sphere bounds a ball Q, and by clause (5) of that lemma Q agrees off T with the region bounded by $D \cup D'$; moreover $Q \cap \partial E(K) = \emptyset$, since a point of $\partial E(K)$ in Q would have to lie on $\partial Q \subset \operatorname{Int} E(K)$. Now

$$B \cap \operatorname{Int} Q = \emptyset. \tag{24}$$

Indeed, $B_\gamma \setminus \gamma$ is connected by Step 0 and is disjoint from $D \cup D'$ — from D' because $B_\gamma \cap D' = \gamma$, and from D because $D \cap B = \gamma$. It is therefore disjoint from ∂Q: outside T the two coincide, and inside T the sphere ∂Q lies, by clause (2) of Lemma 5.1, in the closed sector of the normal slice bounded by D and D', meeting the boundary of that sector only in $D \cup D'$, whereas the two sheets of B at γ bound the two *other* sectors and so meet that closed sector only along γ, which is not on ∂Q. Hence $B_\gamma \setminus \gamma$ lies in a single complementary region of ∂Q, and that region is not Q because it contains ∂B, which lies on $\partial E(K)$ while Q does not meet $\partial E(K)$. As for D': outside T it lies on ∂Q, and inside T it lies on the boundary of the sector while ∂Q lies in the sector's interior, so it is disjoint from $\operatorname{Int} Q$ as well. Since $B = D' \cup B_\gamma$, this is (24).

By (24), pushing D' across Q to a parallel copy of D is an ambient isotopy of $E(K)$, supported in a neighbourhood of Q; this is the only point at which the position of the second surface matters, and it is what a simultaneous replacement of both discs would not supply. The isotopy carries B to a neatly embedded B' with $\partial B' = \partial B$ and with $\operatorname{Area}(B') \to \operatorname{Area}(B) - \operatorname{Area}(D') + \operatorname{Area}(D)$ as the push-off tends to 0. Along the corner circle B' has a genuine corner, since $A \pitchfork B$; rounding by Lemma 5.1 gives a smooth neatly embedded B'', isotopic to B, with boundary still the same leaf and $\mathrm{Area}(B'') \leq \mathrm{Area}(B) - \mathrm{Area}(D') + \mathrm{Area}(D) - \mathcal{G}(\gamma)$. As B minimises in its class, $\mathrm{Area}(B'') \geq \mathrm{Area}(B)$, whence

$$\mathrm{Area}(D_B(\gamma)) + \mathcal{G}(\gamma) \;\leq\; \mathrm{Area}(D_A(\gamma)), \qquad \gamma \in \mathcal{A}. \tag{25}$$

The argument used only that the disc in the *first* surface is a patch, so exchanging the roles of A and B it applies to $\gamma \in \mathcal{B}$ and produces a smooth neatly embedded A'', isotopic to A and hence to A_0, with the same boundary leaf and $\mathrm{Area}(A'') \leq \mathrm{Area}(A) - \mathrm{Area}(D_A(\gamma)) + \mathrm{Area}(D_B(\gamma)) - \mathcal{G}(\gamma)$. Here $\mathrm{Area}(A'') \geq \mathrm{Area}(A_0) = \mathrm{Area}(A) - e$, whence

$$\mathrm{Area}(D_A(\gamma)) + \mathcal{G}(\gamma) \;\leq\; \mathrm{Area}(D_B(\gamma)) + e, \qquad \gamma \in \mathcal{B}. \tag{26}$$

Step 3*: the two curves.* Put $\varphi(\gamma) := \mathrm{Area}(D_A(\gamma)) - \mathrm{Area}(D_B(\gamma))$ on $\mathcal{D}$, so that (25) reads $\varphi \geq \mathcal{G}$ on $\mathcal{A}$ and (26) reads $\varphi \leq e - \mathcal{G}$ on $\mathcal{B}$. Choose $a \in \mathcal{A}$ with $\mathrm{Area}(D_A(a))$ least and $b \in \mathcal{B}$ with $\mathrm{Area}(D_B(b))$ least. Then $\mathrm{Area}(D_A(a)) \leq \mathrm{Area}(D_A(b))$: if $b \in \mathcal{A}$ this is the choice of a, and otherwise $\mathrm{Int}\, D_A(b)$ meets Γ, and choosing among the curves of Γ inside it one, c, whose disc $D_A(c)$ — a subdisc of $D_A(b)$, by Step 0 — is minimal under inclusion, the argument of Step 1 gives $c \in \mathcal{A}$ and hence $\mathrm{Area}(D_A(a)) \leq \mathrm{Area}(D_A(c)) < \mathrm{Area}(D_A(b))$. Symmetrically $\mathrm{Area}(D_B(b)) \leq \mathrm{Area}(D_B(a))$. Adding the two, $\varphi(a) \leq \varphi(b)$, so $\mathcal{G}(a) \leq \varphi(a) \leq \varphi(b) \leq e - \mathcal{G}(b)$, which is (23). □

The exchange is performed one surface at a time because (24) is available only on that side: a simultaneous replacement of both discs would require $D_A(\gamma)$ and $D_B(\gamma)$ to be patches for one and the same γ, which no innermost choice supplies. What replaces it is the pair (25), (26) at two possibly distinct curves, glued by the monotonicity of φ. Note also that the proof uses only that A_0 and B separately minimise, not that A does; the no-product-region statements in the literature assume both surfaces minimal, which the perturbed A of §§5.4–5.5 is not.

5.2. **The exchange theorem.**

Theorem 5.4 (Incompressible exchange lemma)**.** *Let K be a non-trivial knot and let x, u, w be vertices of* $\mathrm{IS}(K)$ *with*

$$\mathrm{dist}(x,u) = \mathrm{dist}(x,w) = 1, \qquad \mathrm{dist}(u,w) = 2.$$

Then there are vertices $w_\uparrow, w_\downarrow$, determined before any further vertex is considered, such that

- (a) *each of $w_\uparrow, w_\downarrow$ is equal or adjacent to each of x, u, w;*
- (b) for these two fixed vertices*, every vertex z equal or adjacent to both u and w is equal or adjacent to both $w_\uparrow$ and $w_\downarrow$;*
- (c) *$g(w_\uparrow) + g(w_\downarrow) \leq g(u) + g(w)$;*
- (d) *if* (c) *is an equality then $A(w_\uparrow) + A(w_\downarrow) < A(u) + A(w)$.*

Clause (a) *asserts adjacency pair by pair and does* not *assert that the five vertices carry simultaneously disjoint representatives; they cannot, since* $\mathrm{dist}(u,w) = 2$ *forces the two minimisers to meet.*

Proof when the minimisers are transverse. By (E) fix, one class at a time, relative area minimisers $S \in x$, $S_+ \in u$, $S_- \in w$; by Proposition 3.4 the single surface S is disjoint from both others, and $S_+ \cap S_- \neq \emptyset$ since $\mathrm{dist}(u,w) = 2$. Assume in this proof that S_+ and S_- are transverse with $\partial S_+ \cap \partial S_- = \emptyset$.

Write $\Gamma := S_+ \cap S_-$, which by neatness and $\partial S_+ \cap \partial S_- = \emptyset$ is a finite family of disjoint smooth closed curves in the interior, of total length $\ell_0 > 0$. In the normal slice at $y \in \Gamma$ the two tangent planes cut the plane into four sectors of angles $\alpha, \pi-\alpha, \alpha, \pi-\alpha$ with $\alpha(y) \in (0,\pi)$, and transversality together with the compactness of Γ makes

$$\theta_0 := \min_{\Gamma} \min\{\alpha, \pi-\alpha\} > 0. \tag{27}$$

Choose a constant width $\eta \leq \min_\Gamma \eta_{\max}$ once and for all: the minimum is positive because Γ is compact and $\theta \geq \theta_0 > 0$ on it, and a constant width satisfies the constraint on η' in (15) vacuously. With this width the gain (16) along a closed curve $\gamma \subset \Gamma$ is at least $g_0(\theta_0)\,\eta\,\mathrm{length}(\gamma) > 0$, uniformly in γ. Now S_- is itself a relative area minimiser, so Lemma 5.3 applies with $e = 0$: were there a disc patch it would supply closed curves a, b with $\mathcal{G}(a) + \mathcal{G}(b) \leq 0$, whereas both gains have just been bounded below by a positive number. So cutting S_+ and S_- along Γ produces no disc patch, Lemma 4.5 applies, and it yields $P_\uparrow, P_\downarrow$.

By (10) and $\chi \leq 0$ for the discarded closed components, $\chi(P_\uparrow) + \chi(P_\downarrow) \geq \chi(S_+) + \chi(S_-)$; substituting (7) gives

$$g(P_\uparrow) + g(P_\downarrow) \;\leq\; g(u) + g(w). \tag{28}$$

Compress each P along essential circles, always keeping the component with the longitudinal boundary, until it is incompressible; by Lemma 3.6 the genus does not increase and the process terminates. Call the resulting vertices $w_\uparrow, w_\downarrow$; with (28) this is (c). Each P is disjoint from each of S, S_+, S_-, so Corollary 3.8, applied separately to the three, gives (a). For (b): if $z \notin \{u, w\}$ is equal or adjacent to both, take a minimiser $Z \in z$; by Theorem 2.3(i) it is disjoint from S_+ and from S_-, so by clause (2) of Lemma 4.5 the coarse outputs have representatives disjoint from Z, and Corollary 3.8 applies again. That the push-off scale depends on Z is harmless: all small push-offs represent the same coarse classes and the chosen compression sequence is transported along the isotopy between them, so the output *vertices* do not move. If $z \in \{u, w\}$, (a) already gives the conclusion.

Suppose now (c) is an equality. Then (28) and every compression inequality is an equality, so no compression occurred and each P is already incompressible and represents the corresponding w. The two frontiers (11) pick out, in the normal slice at $y \in \Gamma$, a pair of opposite sectors, so both have the same opening angle $\alpha(y)$ and θ_0 of (27) bounds the angle for both. The two cornered frontiers partition $S_+ \cup S_-$ up to the measure-zero set Γ, so

$$\mathrm{Area}(\widetilde{B}_\uparrow) + \mathrm{Area}(\widetilde{B}_\downarrow) = \mathrm{Area}(S_+) + \mathrm{Area}(S_-). \tag{29}$$

Round both frontiers with the width η already fixed; by (16) the total gain is at least $2\varepsilon_0$ with $\varepsilon_0 := g_0(\theta_0)\,\eta\,\ell_0 > 0$, which depends on $S_\pm$ and on this one choice and on nothing later. Apply Lemma 5.2 to the two rounded surfaces, obtaining σ_0 and C fixed before σ, and choose $\sigma \in (0, \sigma_0]$ with $C\sigma < \varepsilon_0$. Combining, and discarding the closed components, which have positive area,

$$\begin{aligned} \mathrm{Area}(P_\uparrow) + \mathrm{Area}(P_\downarrow) \;&\leq\; \mathrm{Area}(S_+) + \mathrm{Area}(S_-) - 2\varepsilon_0 + C\sigma \\ &<\; A(u) + A(w) - \varepsilon_0. \end{aligned}$$

These are smooth neatly embedded competitors with boundary a leaf of J, so $A(w_\uparrow) + A(w_\downarrow)$ is at most the left-hand side, which is (d). □

5.3. **Non-transverse minimisers.** The proof just given assumed two things, and both may fail. We record first that the failures are of exactly two kinds.

Both ∂S_+ and ∂S_- are leaves of the foliation J, and two leaves of a foliation are equal or disjoint. So either $\partial S_+ \cap \partial S_- = \emptyset$ or $\partial S_+ = \partial S_-$, and in each case the two surfaces may or may not be transverse in the interior. Three configurations therefore remain to be treated beyond the one just settled:

(T0) $\partial S_+ \cap \partial S_- = \emptyset$ and $S_+ \pitchfork S_-$; this is the case proved above;

(T1) $\partial S_+ \cap \partial S_- = \emptyset$, with S_+ and S_- tangent at one or more interior points;
(T2) $\partial S_+ = \partial S_-$, so that the two surfaces meet along a whole curve of common boundary and the argument above cannot even start.

The surfaces of (T2) need not be tangent along the shared leaf, and in general are not: by (34) in clause (2) of Proposition 5.5 they are transverse at all but finitely many of its points. What fails in (T2) is not transversality but the disjointness of the boundaries, on which the cutting and pasting of §5.2 depends throughout. Configuration (T1) is not a curiosity of the write-up either: two distinct minimal surfaces with disjoint boundaries may perfectly well touch at an interior point, and there it is transversality that fails. Freedman, Hass and Scott record on [6, p. 636] that non-transverse behaviour occurs and give no structure theorem for it; Proposition 5.5 is that structure theorem, its first clause covering the interior tangencies of both (T1) and (T2), its second the collar of (T2). Both configurations are then disposed of by perturbing S_- to a transverse competitor and running the transverse argument on the perturbation; what has to be controlled in each case is that the area lost to the perturbation stays below the area gained by rounding, and it is Lemma 5.3, with its error term e, that makes the two comparable. We take them in the order (T2), §5.4, then (T1), §5.5, and not in the order of their labels: the argument for (T1) is the argument for (T2) with the collar deleted, so writing (T2) first is what makes (T1) short. The labels themselves group by boundary behaviour, (T0) and (T1) being the cases with $\partial S_+ \cap \partial S_- = \emptyset$.

Throughout §§5.3–5.5 write $A := S_-$ and $B := S_+$, keeping S fixed and disjoint from both.

The interior picture. Let p be a point of $\operatorname{Int} A \cap \operatorname{Int} B$ at which $T_pA = T_pB$. In normal coordinates (x_1, x_2, y) at p with the common tangent plane $\{y = 0\}$, both surfaces are graphs over a disc D about the origin, say $B = \{y = \beta\}$ and $A = \{y = \hat{f}\}$ with $\beta(0) = \hat{f}(0) = 0$ and $D\beta(0) = D\hat{f}(0) = 0$; both are smooth by Theorem 2.10. Both graphs are minimal, so subtracting the two minimal graph equations and applying the mean value theorem along the segment joining the two gradients gives, classically on D,

$$L_p f_p := \partial_i(a^{ij}\partial_j f_p) + b^i\partial_i f_p + c f_p = 0, \qquad f_p := \hat{f} - \beta, \tag{30}$$

with a^{ij} smooth and uniformly elliptic and b^i, c smooth. The tangency points of A and B in D are exactly the points of $\{f_p = 0,\ df_p = 0\}$.

The collar picture, in configuration (T2) *only.* Suppose the two boundary leaves coincide along a longitude L. Take a vector field on $\partial E(K)$ transverse to J whose flow carries leaves to leaves, extend it into the collar, and use its direction as a coordinate y, the inward distance $r \geq 0$, and arclength s along L. The resulting chart (s, r, y) is bi-Lipschitz onto its image with a constant $C'_{\mathrm{met}} < \infty$ depending only on the ambient metric near L and on the chosen vector field. By Lemma 2.11 both surfaces are neat, so the extension may be taken transverse along L to both tangent planes, and both are graphs over one half-cylinder:

$$\partial E(K) = \{r = 0\}, \quad B = \{y = 0\}, \quad A = \{y = f(s,r)\}, \quad f(s,0) = 0. \tag{31}$$

By Theorem 2.10 the function f is smooth up to $\{r = 0\}$. Both graphs are minimal on $\{r > 0\}$, so subtracting the two minimal graph equations and applying the mean value theorem along the segment joining the two gradients gives, classically on the closed half-disc,

$$Lf := \partial_i(a^{ij}\partial_j f) + b^i\partial_i f + cf = 0, \tag{32}$$

with a^{ij} smooth and uniformly elliptic and b^i, c smooth, all up to $\{r = 0\}$.

Proposition 5.5 (Structure of the tangency set)**.** *Let A and B be as above: distinct relative area minimisers in two vertices at distance* 2*, meeting each other. Then:*

(1) (Interior; no hypothesis on the boundaries.) *Write $\mathcal{T} \subset \operatorname{Int} A \cap \operatorname{Int} B$ for the set of interior tangency points. Every $p \in \mathcal{T}$ is isolated in $\operatorname{Int} A \cap \operatorname{Int} B$ and has an integer order $m_p \geq 1$ and constants $c_1, c_2 > 0$ with $|df_p| \geq c_1\rho^{m_p} - c_2\rho^{m_p+1} > 0$ at distance ρ from p on a punctured neighbourhood, f_p being the local difference of (30); so p is the only critical point of f_p in a disc D_p. Moreover*

$$\text{for each } p \in \mathcal{T},\ f_p \text{ takes both positive and negative values in every neighbourhood of } p. \tag{33}$$

(2) (Collar; in configuration (T2).) *If $\partial A = \partial B = L$, then with f as in (31) there is $r_0 > 0$ with $\mathcal{T} \subset \{r \geq r_0\}$ and*

$$\begin{aligned}&\text{in the open collar } \{0 < r < r_0\} \text{ one has } df \neq 0 \text{ everywhere,}\\ &\text{while the critical points of } f \text{ on } \{r = 0\} \text{ are finite in}\\ &\text{number and all lie on } \{f = 0\}.\end{aligned} \tag{34}$$

(3) *In both configurations $\mathcal{T}$ is finite; put $N := \#\mathcal{T}$.*

We refer to (34) and (33) as the collar conditions, though strictly (34) concerns the collar and (33) the interior tangency set. In configuration (T2) they are used together. In configuration (T1) only (33) is available, and only it is needed, because there is no collar to control: the whole intersection is then at a positive distance from $\partial E(K)$.

Proof. Both clauses rest on the same two devices, applied on different domains: the zeroth-order term is removed by dividing by a positive solution, and the resulting divergence-form equation without zeroth-order term is factorised in the sense of Vekua. We set the first up once, for a domain Δ which is either a disc $D_R(p)$ about an interior tangency point, carrying the operator L_p of (30), or a half-disc D_R^+ about a point of L, carrying the operator L of (32). In both cases we write L for the operator and f for its solution.

Removing the zeroth-order term. Solve $Lh = 0$ on Δ with $h = 1$ on $\partial\Delta$. We do *not* assume a sign condition on the zeroth-order coefficient, so the existence theorems of [7, 2nd ed., Ch. 8], which all carry hypothesis (8.8), are not available as stated; smallness of R replaces the sign condition. Indeed for $u \in W_0^{1,2}(\Delta)$ the Poincaré inequality gives $\|u\|_{L^2} \leq CR\|Du\|_{L^2}$, whence the associated bilinear form satisfies

$$B[u,u] \ \geq \ \left(\lambda - C\|b\|_\infty R - C^2\|c\|_\infty R^2\right)\|Du\|_{L^2}^2,$$

which is positive definite for R small; B is bounded on $W_0^{1,2}(\Delta)$ because the coefficients are, so the Lax–Milgram theorem [7, 2nd ed., Theorem 5.8, p. 83] yields a unique weak solution. Writing $h = 1 + \varpi$, the function ϖ solves $L\varpi = -c$ with zero boundary data. Rescale $x \mapsto x/R$ to the unit disc or half-disc, under which the drift becomes $O(R)$ and the zeroth-order coefficient $O(R^2)$; testing the equation against ϖ and using the same Poincaré inequality gives $\|\varpi\|_{L^2} = O(R^2)$. The global maximum principle [7, 2nd ed., Theorem 8.15, p. 189], applied to ϖ and to $-\varpi$, both of which lie in $W_0^{1,2}$ and so are ≤ 0 on the boundary in the generalised sense, then gives $\|\varpi\|_\infty \leq C(\|\varpi\|_{L^2} + \nu^{-1}\|c\|_{L^{q/2}}) = O(R^2)$, so $h \geq \frac{1}{2}$ for R small. We use Theorem 8.15 rather than the sharper [7, 2nd ed., Theorem 8.16, p. 191] precisely because the latter assumes hypothesis (8.8): in Theorem 8.15 the zeroth-order coefficient enters only through $\lambda^{-1}|d|$ in (8.6), so its sign is unconstrained, and the $\|\varpi\|_{L^2}$ term that this costs is supplied by the energy estimate just made. Put $v := f/h$; expanding $L(hv) = 0$ with $Lh = 0$ gives, weakly,

$$\partial_i(h\, a^{ij}\partial_j v) + (a^{ij}\partial_j h + h\, b^i)\partial_i v = 0, \tag{35}$$

with Lipschitz leading coefficients, bounded drift and *no zeroth-order term*; in the half-disc case one has in addition $v = 0$ on L, since f does. Multiplying by the reciprocal square root of the determinant of the leading matrix normalises that determinant to 1 and changes only the drift. The zeroth-order term has to be removed afresh on each domain on which the factorisation or the maximum principle is used: with a zeroth-order term of uncontrolled sign both fail.

Proof of (1). Let $p \in \mathcal{T}$, run the previous paragraph on $\Delta = D_R(p)$ with the operator L_p of (30), and write $v_p := f_p/h_p$ for the resulting solution of (35); note $v_p(p) = 0$. Choose a smooth Jordan subdomain $\Omega \Subset D_R(p)$ containing p. As in the paragraph "An isothermal chart up to L" below — the argument is the same and shorter, no boundary arc being distinguished — Lichtenstein's theorem [10, Theorem 4.1, case $m = 1$] supplies a $C^{1,\alpha}$ diffeomorphism of $\overline{\mathbb{D}}$ onto $\overline{\Omega}$, conformal for the metric (36), and by the density law recorded there the equation for $\hat{v}_p$ becomes (37) on the disc, with no boundary condition and no reflection needed. The factorisation of the paragraph "Finite order" below then applies verbatim: either $\mathfrak{w} := 2\partial_z \hat{v}_p$ has isolated zeros of finite integer order m with $c_1'|z - z_0|^m \le |\mathfrak{w}| \le c_2'|z - z_0|^m$ nearby, or $\mathfrak{w} \equiv 0$. In the second case $\nabla \hat{v}_p \equiv 0$, so $\hat{v}_p$ is constant, and the constant is 0 because $\hat{v}_p$ vanishes at the image of p; then $f_p \equiv 0$ near p, the set on which the two sheets agree is open by this argument applied at each of its points and closed by continuity, so the two surfaces coincide, contradicting $\operatorname{dist}(u, w) = 2$. Transporting through the bi-Lipschitz chart gives $|d\hat{v}_p| \asymp \rho^m$ and $|\hat{v}_p| = O(\rho^{m+1})$ at distance ρ from p, and $f_p = h_p v_p$ with $h_p \ge \frac{1}{2}$ gives $|df_p| \ge c_1\rho^m - c_2\rho^{m+1} > 0$ on a punctured neighbourhood of p. In particular p is the only critical point of f_p in a disc D_p, and, the tangency points in D_p being critical points of f_p, p is isolated in $\operatorname{Int} A \cap \operatorname{Int} B$.

For (33), suppose $f_p \ge 0$ near p, the other sign being identical. Then $v_p \ge 0$ on the disc on which it is defined and $v_p(p) = 0$, so by the strong maximum principle at this regularity [7, 2nd ed., Theorem 8.19, p. 198], whose hypotheses are met since the operator is in divergence form with no zeroth-order term and $v_p \in W^{1,2}$, we get $v_p \equiv 0$, hence $f_p \equiv 0$, and the same contradiction as above.

Proof of (2). Assume $\partial A = \partial B = L$ and run the removal of the zeroth-order term on half-discs $\Delta = D_R^+$ against L, writing $v := f/h$ as there.

An isothermal chart up to L. Choose a smooth Jordan subdomain Ω of the half-disc whose boundary contains a subarc of L around the point in question. Write $a = (a^{ij})$ for the leading matrix of (35) after the normalisation $\det a \equiv 1$, b for its drift, and

$$g := a^{-1}, \tag{36}$$

a Riemannian metric with $\det g = 1$; it is the *inverse* of the leading matrix that carries the relevant conformal structure, as the computation below shows. Being the inverse of a Lipschitz uniformly elliptic matrix, g is Lipschitz on $\overline{\Omega}$, hence $C^{0,\alpha}$, and $\partial\Omega$ is $C^{1,\alpha}$; so by the global form of Lichtenstein's theorem [10, Theorem 4.1, case $m = 1$] there is a diffeomorphism $\tau\colon \overline{\mathbb{D}} \to \overline{\Omega}$ of class $C^{1,\alpha}$, conformal from the flat metric to g. Let φ be a Möbius map of $\mathbb{D}$ onto the upper half-plane carrying the arc $\tau^{-1}(L \cap \partial\Omega)$ into $\mathbb{R}$ and τ^{-1} of the base point to 0, and put $F := \varphi \circ \tau^{-1}$. Then F is $C^{1,\alpha}$ up to L, sends L into $\mathbb{R}$ and the interior into the upper half-plane, and is bi-Lipschitz, DF being continuous and invertible on a compact set.

It remains to see that in the coordinate $z = F(\cdot)$ the operator becomes the flat Laplacian with a bounded drift. A divergence-form operator transforms under a $C^{1,\alpha}$ diffeomorphism by the density law: testing $\int (a\,\nabla v) \cdot \nabla\phi\, dx$ against ϕ and substituting $x = F^{-1}(z)$ turns the leading matrix into $DF\,a\,DF^{\mathsf{T}}/|\det DF|$ and the drift into $DF\,b/|\det DF|$, no derivative of DF appearing anywhere. Three consequences. First, the transformed drift is again in L^∞, since DF and $(\det DF)^{-1}$ are bounded. Second, the transformed leading matrix again has determinant 1, the factors $(\det DF)^2$ cancelling. Third, that matrix is a positive multiple of the identity exactly when $a = \lambda\,(DF^{\mathsf{T}}DF)^{-1}$ for a positive function λ, that is, exactly when $g = a^{-1}$ is a conformal multiple of the pullback $F^*\delta$ of the flat metric — which is what the choice of τ arranged, φ being conformal. A positive multiple of the identity of determinant 1 is the identity. Hence (35) becomes

$$\Delta\hat{v} + \hat{b}\cdot\nabla\hat{v} = 0, \qquad \hat{b}\in L^\infty, \qquad \hat{v} = 0 \ \text{ on } \ \mathbb{R}. \tag{37}$$

Odd reflection. Extend $\hat{v}$ to the full disc by $\hat{v}(z) := -\hat{v}(\bar{z})$ for $\operatorname{Im} z < 0$, and extend $\hat{b}$ by $\hat{b}_1(\bar{z}) = \hat{b}_1(z)$, $\hat{b}_2(\bar{z}) = -\hat{b}_2(z)$. For a test function ϕ supported in the full disc, split $\int\nabla\hat{v}\cdot\nabla\phi - \int(\hat{b}\cdot\nabla\hat{v})\phi$ over the two half-discs and substitute $z\mapsto\bar{z}$ in the lower one: the two interior terms cancel by (37), and the two conormal boundary terms along $\mathbb{R}$ cancel because the leading matrix is the identity, so its off-diagonal entry vanishes on $\mathbb{R}$ and the conormal derivative is $\partial_2\hat{v}$, which is even under the reflection while the outward normals are opposite. Hence the extension solves (37) weakly on the full disc. This is the one point at which the vanishing of the off-diagonal coefficient on L is used, and it is why the isothermal chart is taken before the reflection and not after.

Finite order. Put $\mathfrak{w} := 2\partial_z\hat{v}$. Since $\partial_{\bar{z}}\partial_z = \frac{1}{4}\Delta$ and $\hat{b}\cdot\nabla\hat{v} = \frac{1}{2}[(\hat{b}_1 + i\hat{b}_2)\mathfrak{w} + (\hat{b}_1 - i\hat{b}_2)\overline{\mathfrak{w}}]$ for real $\hat{v}$, equation (37) reads

$$\partial_{\bar{z}}\mathfrak{w} = -\tfrac{1}{4}(\hat{b}_1 + i\hat{b}_2)\,\mathfrak{w} - \tfrac{1}{4}(\hat{b}_1 - i\hat{b}_2)\,\overline{\mathfrak{w}},$$

a generalized analytic function with coefficients in $L^\infty\subset L^p$ for every $p > 2$; this is exactly Vekua's standing hypothesis [17, Ch. III, §1, condition (1.10), p. 137], the equation already being in the canonical form [17, Ch. III, (1.5), p. 131], whose leading coefficient is identically 1. By the similarity principle [17, Ch. III, §4.1, Basic Lemma, p. 144, and formula (4.3), p. 146], $\mathfrak{w} = \Phi e^{\omega}$ with Φ holomorphic and ω bounded and Hölder; so unless $\mathfrak{w}\equiv 0$ its zeros are isolated of finite integer order m, with $c_1'|z - z_0|^m \le |\mathfrak{w}| \le c_2'|z - z_0|^m$ nearby [17, Ch. III, §4.2, p. 145, Theorem 3.5, pp. 146–147, and the inequalities (4.15), p. 152]. The alternative $\mathfrak{w}\equiv 0$ forces $\nabla\hat{v}\equiv 0$, hence $\hat{v}\equiv 0$ since $\hat{v} = 0$ on $\mathbb{R}$, hence $f\equiv 0$ on an open set; the set where the two minimal sheets agree is then open by this argument applied at each of its points and closed by continuity, so the sheets coincide, contradicting $\operatorname{dist}(u, w) = 2$. Transporting through the bi-Lipschitz F gives $|d\hat{v}|\asymp\rho^m$ and $|\hat{v}| = O(\rho^{m+1})$ at distance ρ, and $f = hv$ with $h\ge\frac{1}{2}$ gives $|df|\ge c_1\rho^m - c_2\rho^{m+1} > 0$ on a punctured half-neighbourhood.

Every critical point of f on L is therefore isolated in the closed half-neighbourhood, and at a non-critical boundary point continuity gives a neighbourhood free of critical points; L is compact, so finitely many critical points remain and a uniform r_0 exists with no critical point in $\{0 < r < r_0\}$. They lie on $\{f = 0\}$ because f vanishes on L. Since a tangency point in the collar would be a critical point of f, this also gives $\mathcal{T}\subset\{r\ge r_0\}$, and clause (2) is proved.

Proof of (3). In configuration (T1) one has $A\cap B\cap\partial E(K) = \partial A\cap\partial B = \emptyset$, so $A\cap B$ is a compact subset of $\operatorname{Int} E(K)$; in configuration (T2), clause (2) confines $\mathcal{T}$ to the compact set $A\cap B\cap\{r\ge r_0\}$, which again misses $\partial E(K)$ — a point of $A\cap B$ on $\partial E(K)$ lies on L, where $r = 0$. In both cases $\mathcal{T}$ is a subset of a compact subset C of $\operatorname{Int} A\cap\operatorname{Int} B$. Tangency is a closed condition, the tangent planes varying continuously, so $\mathcal{T}$ is closed in $\operatorname{Int} A\cap\operatorname{Int} B$ and hence closed in C; being also discrete by clause (1), it is finite. □

5.4. **Configuration (T2): a common boundary leaf.**

Lemma 5.6 (No small closed intersection curves)**.** *Let f_t be the perturbation of f described in* (38) *below. There are $\rho_0 > 0$ and $\delta_1 > 0$ such that for every $t\in(0,\delta_1]$ every closed component γ of $A_t\cap B$ has* $\operatorname{diam}(\gamma)\ge\rho_0$.

We first describe the perturbation, since the lemma depends on its exact form. Its sign is a free parameter, and fixing it correctly is what keeps the intersection curves away from L; so we begin with that choice.

Lemma 5.7 (A sign, and an arc of L swept clear)**.** *Put $a(s) := \partial_r f(s,0)$ and $\Lambda := \sup|\partial_r^2 f| < \infty$. Then $a \not\equiv 0$, and there are $\varsigma \in \{\pm 1\}$, a closed arc $I \subset L$, and constants $a_0 > 0$, $r_1 > 0$, all depending only on A and B, such that*

$$\varsigma f(s,r) \;\geq\; \tfrac{1}{2}a_0\, r \;>\; 0 \qquad \textit{for } s \in I,\; 0 < r \leq r_1.$$

Proof. Since $f(s,0) \equiv 0$ we have $\partial_s f(s,0) \equiv 0$, so $df(s,0) = (0, a(s))$ and the critical points of f on $\{r = 0\}$ are exactly the zeros of a. By (34) there are finitely many, so $a \not\equiv 0$. Pick s_0 with $a(s_0) \neq 0$, set $\varsigma := \operatorname{sign} a(s_0)$ and $a_0 := \frac{1}{2}|a(s_0)|$, and use the continuity of a to choose a closed arc $I \ni s_0$ with $\varsigma a \geq a_0$ on I. Put $r_1 := \min\{a_0/(2\Lambda),\, r_0/4\}$. For $s \in I$ and $0 \leq \rho \leq r_1$,

$$\varsigma\, \partial_r f(s,\rho) \;\geq\; \varsigma a(s) - \Lambda\rho \;\geq\; a_0 - \Lambda r_1 \;\geq\; \tfrac{1}{2}a_0,$$

and integrating in r gives the claim. Nothing in the construction refers to the perturbation parameter. □

Take a cut-off $\chi(r)$ equal to 1 near $r = 0$, supported in the collar $\{r < r_0\}$, with its transition band placed where $\{u = 0\}$ is regular, and pairwise disjoint discs $D_p \Supset D'_p$ around the points of $\mathcal{T}$ with bumps ψ_p equal to 1 on D'_p; with ς as in Lemma 5.7 set

$$A_t := \Big\{\, y = f + \varsigma t\chi(r) - \varsigma t \sum_{p\in\mathcal{T}} \psi_p \,\Big\}. \tag{38}$$

Near the boundary $A_t \cap B$ is the level set $\{f = -\varsigma t\}$, regular for small $t \neq 0$ by (34); on the transition band, if transversality failed along $t_n \to 0$ the bad points would converge to a point with $f = 0$ and $df = 0$, excluded by the placement of the band. On D'_p one has $A_t \cap B = \{f = \varsigma t\}$, and by the last clause of Proposition 5.5 the only critical point of f in D_p is p, with critical value $f(p) = 0$, so every $t \neq 0$ is a regular value of either sign; on the annulus $D_p \setminus D'_p$ one has $|du| \geq c > 0$, hence $|d(f - \varsigma t\psi_p)| \geq c/2$ for t small. Elsewhere $A_t = A$ and transversality is open. Thus there is $\delta^{(1)} > 0$ such that for $t \in (0, \delta^{(1)}]$ the surface A_t is neat, incompressible, transverse to B, with ∂A_t the leaf $\{y = \varsigma t\}$ disjoint from ∂B, disjoint from S, and $A_t \to A$ in C^1. *The interior perturbation is of subtract-a-constant type, and this cannot be weakened*: a family guaranteeing only transversality would lose the fact that f_t and f differ by a constant near p, and the proof below would fail.

Proof of Lemma 5.6. Otherwise there are $t_n \to 0$ and closed components γ_n with $\operatorname{diam}(\gamma_n) \to 0$; by C^0 convergence, points $x_n \in \gamma_n$ subconverge to some $x \in A \cap B$. If x is a transverse point outside the collar, then in a fixed ball at x the intersection is a single arc crossing it, and C^1 stability keeps that true for small t, so γ_n cannot lie in the ball. If $x = p \in \mathcal{T}$, take D_p as above; for large n the curve γ_n bounds a disc $D_n \subset D_p$, and f_t vanishes on γ_n. If $f_t \equiv 0$ on D_n then $df_t \equiv 0$ there, impossible since f_t has at most the one critical point p in D_p; otherwise f_t attains a strict interior extremum on D_n, at a critical point, hence at p — but on D'_p the functions f_t and f differ by the constant t, and by (33) p is not a local extremum of f, hence not of f_t. If x lies in the collar, then D_n does not meet ∂B, so $D_n \subset \{r > 0\}$, where by (34) $df \neq 0$ and hence $|df_t| \geq c/2$ for small t; but f_t vanishes on ∂D_n, so both its maximum and its minimum on $\overline{D_n}$ are 0, forcing $f_t \equiv 0$ and $df_t \equiv 0$. Each case is a contradiction. □

Choice of ρ_2. The next two results share a scale, and we fix it here rather than inside either of them. Choose $\rho_2 > 0$ with the balls $B_{2\rho_2}(p)$, $p \in \mathcal{T}$, pairwise disjoint,

$$4N\rho_2 < \rho_0/8, \qquad \rho_2 \le r_0/8, \qquad \rho_2 \le r_1/(2C'_{\mathrm{met}}), \tag{39}$$

and put $P := \bigcup_p B_{2\rho_2}(p)$, $G := E(K) \setminus P$ and $G' := E(K) \setminus \bigcup_p B_{\rho_2}(p)$, so that $G \subset G'$. Only ρ_0, r_0, r_1 and $\mathcal{T}$ enter this choice. Lemma 5.8 below quotes clause (1) of Corollary 5.9, which is stated after it; there is no circle, as the proof of Corollary 5.9 uses only Lemma 5.6, Proposition 5.5 and Lemma 5.1.

Lemma 5.8 (Every closed component has uniform length away from L)**.** *Let ς, I, a_0, r_1 be as in Lemma 5.7 and write $\mathcal{C} := L \times (0, r_0)$ for the open collar. Then*

(i) *for every $t > 0$ the set $\Gamma_t := A_t \cap B$ is disjoint from $I \times (0, r_1]$;*
(ii) *no closed component of Γ_t contained in $\mathcal{C}$ is null-homotopic in $\mathcal{C}$;*
(iii) *there is $c_\star > 0$, depending only on A, B and the perturbation scheme, such that for every $t \in (0, \min\{\delta_1, \delta^{(1)}\}]$ every closed component γ of Γ_t satisfies*
$$\mathrm{length}(\gamma \cap G \cap \{r \ge r_1\}) \ \ge\ c_\star.$$

Proof. (i) In the collar Γ_t is the level set $\{f = -\varsigma t\}$, that is $\varsigma f = -t < 0$, whereas $\varsigma f \ge \frac{1}{2} a_0 r > 0$ on $I \times (0, r_1]$ by Lemma 5.7.

(ii) Suppose a closed component $\gamma \subset \mathcal{C}$ bounds a closed disc $D \subset \mathcal{C}$. Then $f \equiv -\varsigma t$ on ∂D. If $\max_{\overline{D}} f > -\varsigma t$ the maximum is interior and f has a critical point there, contradicting (34); likewise for the minimum. So $f \equiv -\varsigma t$ on D and $df \equiv 0$ there, contradicting (34) again.

(iii) Note first that $\gamma \cap \partial E(K) = \emptyset$, since $A_t \cap \partial E(K) = \partial A_t$ is the leaf $\{y = \varsigma t\}$ while $B \cap \partial E(K) = \partial B = L = \{y = 0\}$. Next, $\mathcal{T}$ lies in $\{r \ge r_0\}$ by Proposition 5.5(2), and $\rho_2 \le r_0/8$ by (39), so $P = \bigcup_p B_{2\rho_2}(p) \subset \{r > 3r_0/4\}$ and hence

$$\{r_1 \le r \le r_0/2\} \ \subset\ G \cap \{r \ge r_1\} =: Z. \tag{40}$$

Let γ be a closed component of Γ_t. Three cases exhaust the possibilities.

Case A: $\min_\gamma r \ge r_1$. Then $\gamma \subset \{r \ge r_1\}$, so $\gamma \cap G \subset Z$ and clause (1) of Corollary 5.9 gives $\mathrm{length}(\gamma \cap Z) \ge 3\rho_0/8$.

Case B: $\min_\gamma r < r_1$ *and* $\max_\gamma r > r_0/2$. The function r is continuous on the circle γ and takes a value below r_1 and a value above $r_0/2$, so γ contains two disjoint subarcs each joining $\{r = r_1\}$ to $\{r = r_0/2\}$ inside $\{r_1 \le r \le r_0/2\}$. In the collar chart r is 1-Lipschitz, so each has length at least $(r_0/2 - r_1) \ge r_0/4$ up to the bi-Lipschitz constant C'_{met}; by (40) both lie in Z, and together they give $\mathrm{length}(\gamma \cap Z) \ge r_0/(2C'_{\mathrm{met}})$.

Case C: $\min_\gamma r < r_1$ *and* $\max_\gamma r \le r_0/2$. Then $\gamma \subset \{0 < r \le r_0/2\} \subset \mathcal{C}$, so by (ii) γ is not null-homotopic in the annulus $\mathcal{C}$ and its projection to L is onto. In particular γ has points over every $s \in I$, and by (i) each of them has $r > r_1$; by hypothesis each also has $r \le r_0/2$, so the part of γ lying over I is contained in Z by (40). The projection is 1-Lipschitz in the collar chart, so that part has length at least the arclength $\ell(I)$ of I, up to C'_{met}.

Take $c_\star := \min\{3\rho_0/8,\ r_0/2,\ \ell(I)\}/C'_{\mathrm{met}}$. ☐

Corollary 5.9 (Bounds independent of t)**.** *With ρ_2, P, G, G' as in the choice of ρ_2 above, there are $\theta_1 \in (0, \pi/2]$, $\eta_1^{\mathrm{geo}} > 0$, $\kappa_1 > 0$ and $\delta_2 > 0$ such that for every $t \in (0, \delta_2]$ and every closed component γ of $A_t \cap B$:*

(1) $\mathrm{length}(\gamma \cap G) \ge 3\rho_0/8$*;*
(2) *the opening angle α along $\gamma \cap G' \cap \{r \ge r_1/2\}$ satisfies the* two-sided *bound* $\min\{\alpha, \pi - \alpha\} \ge \theta_1$*;*

(3) *along* $\gamma \cap G' \cap \{r \geq r_1/2\}$ *the functions of Lemma 5.1, read pointwise, satisfy* $\eta_{\max} \geq \eta_1^{\mathrm{geo}}$ *and* $\kappa \geq \kappa_1$.

Proof. (1) By Lemma 5.6 there are $a, b \in \gamma$ with $d(a,b) \geq \rho_0/2$. Collapse each of the N components of P to a point and let d' be the resulting quotient pseudometric. Collapsing N sets of diameter at most $4\rho_2$ decreases distances by at most $4N\rho_2$, so $d(a,b) \leq d'(a,b) + 4N\rho_2$; and the image of an arc of γ from a to b is a path of length exactly $\mathrm{length}(\gamma \cap G) =: \Lambda$. Hence $\rho_0/2 \leq \Lambda + 4N\rho_2 < \Lambda + \rho_0/8$, so $\Lambda > 3\rho_0/8$. The quotient formulation is needed, not a decoration: a curve may enter and leave the same ball an unbounded number of times, so the count "N balls, each costing $4\rho_2$" is not valid, whereas each component is collapsed only once.

(2) Put $G'' := E(K) \setminus \bigcup_p B_{\rho_2/2}(p)$, so $G' \subset G''$, and put

$$\mathcal{K} \;:=\; A \cap B \cap G'' \cap \{r \geq r_1/4\},$$

with the convention that r is extended by $r \equiv r_0$ outside the collar. The set $\mathcal{K}$ is compact, and A meets B transversally at every point of it. Indeed, at a point of $\mathcal{K}$ inside the collar the two sheets are the graphs $\{y = f\}$ and $\{y = 0\}$ of §5.3, and transversality along $\{f = 0\}$ is exactly the condition $df \neq 0$, which holds by (34); the tangency points of L itself lie on $\{r = 0\}$ and are excluded by the constraint $r \geq r_1/4$. At a point of $\mathcal{K}$ outside the collar Proposition 5.5 gives transversality away from $\mathcal{T}$, and $\mathcal{T}$ has been excised by G''. Hence $\min\{\alpha, \pi - \alpha\}$, a continuous positive function on $\mathcal{K}$, has a positive infimum $2\theta_1$; the upper bound is included because the two frontiers occupy opposite sectors and Lemma 5.1 needs α bounded away from π as well as from 0. For small t the curve $A_t \cap B$ lies in the $\min\{\rho_2/4, r_1/4\}$-neighbourhood of $A \cap B$ and on G'' the C^1 distance from A_t to A is $O(t)$, so the two-sided bound persists with θ_1 along $\gamma \cap G' \cap \{r \geq r_1/2\}$.

(3) By the last paragraph of the proof of Lemma 5.1, the lower bound $\underline{\eta}$ of (14) is obtained from (22) by replacing m with $\cot(\theta/2)$, m' with $\sup|m'|$ and $g_0(\theta)$ with $g_0(\theta_\bullet)$; besides the angle it involves the local reach ϱ of (13), the distance ϱ_∂ to $\partial\widetilde{B}$, and a constant C_1 bounding the C^2 norms of the two sheets. Each of the three is bounded below along $\gamma \cap G' \cap \{r \geq r_1/2\}$, uniformly in $t \leq \delta_2$.

For ϱ_∂: on $\{r \geq r_1/2\}$ the distance to $\partial E(K)$ is at least $r_1/(2C'_{\mathrm{met}})$, and $\widetilde{B}$ may be taken to contain the $r_1/(4C'_{\mathrm{met}})$-neighbourhood of that set, so $\varrho_\partial \geq r_1/(4C'_{\mathrm{met}})$.

For C_1 and $\sup|m'|$: by the construction of A_t the graph functions of A_t over A have C^3 norms bounded by a constant independent of $t \leq \delta^{(1)}$, and B is fixed; together with the two-sided angle bound of (2) this bounds the second fundamental forms of the two sheets and the derivative of the angle function along γ, uniformly in t.

For ϱ: the curvature of γ is bounded by a constant K_1 independent of t, by the previous paragraph, so the normal exponential map of γ is injective on a tube of radius $\varrho' > 0$ around any arc of γ of length at most ϱ', with ϱ' depending only on K_1 and the ambient injectivity radius. It remains to rule out that two arcs of γ far apart in γ come close in $E(K)$. Suppose not: there are $t_n \downarrow 0$ and points x_n, y_n on components γ_n of $A_{t_n} \cap B$, both in $G' \cap \{r \geq r_1/2\}$, with $d_{\gamma_n}(x_n, y_n) \geq \varrho'$ and $d(x_n, y_n) \to 0$. Passing to a subsequence, $x_n, y_n \to x \in \overline{G'} \cap \{r \geq r_1/2\}$, and $x \in A \cap B$ by C^1 convergence, so $x \in \mathcal{K}$ and A meets B transversally at x. Transversality gives a ball $B(x, \zeta)$ in which $A \cap B$ is a single embedded arc, and by C^1 convergence the same holds for $A_{t_n} \cap B$ for large n; but then x_n and y_n lie on one arc of $A_{t_n} \cap B$ of diameter at most 2ζ, whose length is at most $2\zeta \cdot (1 + o(1)) < \varrho'$ once ζ is small, contradicting $d_{\gamma_n}(x_n, y_n) \geq \varrho'$. Hence $\varrho \geq \varrho_1 > 0$ along $\gamma \cap G' \cap \{r \geq r_1/2\}$, uniformly in t.

Take $\eta_1^{\mathrm{geo}} := \underline{\eta}(\theta_1, \varrho_1, r_1/(4C'_{\mathrm{met}}))$ and $\kappa_1 := \underline{\kappa}(\theta_1, \varrho_1, r_1/(4C'_{\mathrm{met}}))$, both positive since $\theta_1 > 0$.

The angle bound has to come first: by the last paragraph of the proof of Lemma 5.1 the admissible width degenerates as the angle does, so there is no bound on $\eta_{\max}$ in terms of the reach and the metric alone. $\square$

Proof of Theorem 5.4, configuration (T2). Put $\eta_1 := \min\{\eta_1^{\text{geo}},\ \kappa_1\rho_2/4\}$ and

$$\varepsilon_* := g_0(\theta_1) \cdot \eta_1 \cdot c_\star > 0, \tag{41}$$

with $c_\star$ the constant of Lemma 5.8, which depends on A, B and on the once-chosen perturbation scheme and on nothing later.

No disc patch, uniformly in t. Suppose cutting A_t and B along $A_t \cap B$ produces a disc patch. Every closed component γ of $A_t \cap B$ satisfies, by Corollary 5.9, Lemma 5.8(iii) and the slicewise non-negativity in Lemma 5.1,

$$\mathcal{G}(\gamma) \ \geq \ g_0(\theta_1)\, \eta_1 \operatorname{length}\big(\gamma \cap G \cap \{r \geq r_1\}\big) \ \geq \ \varepsilon_*,$$

the gain being computed with the width function (43) constructed below, which equals η_1 on $G \cap \{r \geq r_1\}$, and the rest of the curve being discarded. So (23) gives $2\varepsilon_* \leq \mathcal{G}(a) + \mathcal{G}(b) \leq e(t) := \operatorname{Area}(A_t) - \operatorname{Area}(A)$, and $e(t) \to 0$ by C^1 convergence. So there is $\delta_3 > 0$ with $e(t) < \varepsilon_*$, hence no disc patch, *for every* $t \in (0, \delta_3]$. The quantifier order matters: ε_* is pinned down before t, since the patch exists only after t and both the length of its frontier and the angle along it vary with t.

The output vertices do not depend on t. For t in a compact subinterval of $(0, \delta^{(1)}]$ the surfaces A_t are transverse to B with disjoint boundaries and disjoint from S, so there is an ambient isotopy H_t with $H_{t_0} = \mathrm{id}$, $H_t(A_{t_0}) = A_t$, $H_t(B) = B$ and $H_t|_S = \mathrm{id}$, preserving $\partial E(K)$: take $X_t = \partial_t j_t$ for embeddings j_t with image A_t, correct it along the compact intersection track by a tangential term so that it becomes tangent to B there — possible because transversality makes the relevant map onto $T_pE(K)/T_pB$ surjective and the metric supplies a smooth right inverse — and extend by Hadamard's lemma and a partition of unity to a time-dependent field tangent to B and to $\partial E(K)$ and vanishing on S, whose flow is the required isotopy. Lift H_t to the infinite cyclic cover starting at the identity. It fixes S_0 and S_1 and the fundamental domain between them, commutes with the deck generator, and cannot interchange the two sides of a wall, since the upper side is the one containing S_1; so it carries the frontiers (11) at t_0 to those at t and small push-offs to admissible push-offs. Hence the two coarse classes are independent of $t > 0$. Fix once and for all, at one value of t, a compression sequence from each coarse surface to an incompressible descendant with boundary, giving $w_\uparrow, w_\downarrow$; transporting the coarse surfaces and the discs along the isotopy gives the same two vertices at every $t > 0$. *This is what makes (b) hold in the order stated*: given z with minimiser Z disjoint from $A \cup B$, one may choose $t_z > 0$ after Z is known so that $Z \cap A_{t_z} = \emptyset$, and take the push-offs at t_z small enough to avoid Z; the classes, and therefore the vertices, are the ones already fixed, so t_z changes only a representative.

The area estimate. Since $\partial A_t \cap \partial B = \emptyset$, the set $\Gamma_t := A_t \cap B$ is a finite union of closed curves, and it is non-empty, for otherwise A_t and B would be disjoint representatives of w and u. Corollary 5.9 and Lemma 5.8 apply to every component, and the slicewise gain is non-negative everywhere, so rounding the two cornered frontiers along Γ_t gains at least ε_* in total. The order of choices is

$$\begin{aligned} A, B &\Rightarrow (\text{perturbation scheme, including } \varsigma) \Rightarrow \rho_0 \Rightarrow r_1 \Rightarrow \rho_2 \\ &\Rightarrow (\theta_1, \varrho_1) \Rightarrow (\eta_1^{\text{geo}}, \kappa_1) \Rightarrow \eta_1 \Rightarrow c_\star \Rightarrow \varepsilon_* \\ &\Rightarrow \delta \Rightarrow t \Rightarrow \eta_2 \Rightarrow (\widehat{B}_\uparrow, \widehat{B}_\downarrow) \Rightarrow C \Rightarrow \sigma, \end{aligned}$$

each step using only what precedes it; in particular t appears in the chain, after δ and before the rounded surfaces. Choose

$$\begin{gathered}\delta \le \min\{\delta^{(1)}, \delta_1, \delta_2, \delta_3\} \quad \text{and small enough that} \\ \operatorname{Area}(A_t) < \operatorname{Area}(A) + \varepsilon_*/2 \ \ \text{for } t \in (0, \delta],\end{gathered} \tag{42}$$

and fix such a t. It remains to name the width function, and this is the one place where the constraint $|\eta'| \le \kappa$ in (15) is doing work: the width must be η_1 out on $G \cap \{r \ge r_1\}$, where the gain is collected, and must fall to something admissible inside the balls $B_{\rho_2}(p)$, where the angle degenerates and $\eta_{\max}$ with it, and inside the collar $\{r \le r_1/2\}$, where the two sheets meet $\partial E(K)$. Put

$$\begin{gathered} w(x) := \min\{\rho_2,\ d(x, \textstyle\bigcup_p B_{\rho_2}(p)),\ d(x, \{r \le r_1/2\})\}, \\ \eta_2 := \min\{\eta_1,\ \displaystyle\min_{\Gamma_t} \eta_{\max},\ \tfrac{1}{2} \min_{\Gamma_t} \varrho\} > 0, \end{gathered}$$

so that w is 1-Lipschitz on $E(K)$ and hence along Γ_t, and set

$$\eta(s) := \min\{\eta_1,\ \eta_2 + \tfrac{\kappa_1}{2} w(x(s))\}, \tag{43}$$

smoothed at a scale $\ll \rho_2$ so as to be C^1 with $|\eta'| \le \kappa_1$. On $\Gamma_t \cap G \cap \{r \ge r_1\}$ we have $w = \rho_2$: the second entry is ρ_2 because G omits the balls $B_{2\rho_2}(p)$, and the third is at least $r_1/(2C'_{\text{met}}) \ge \rho_2$ because r is C'_{met} -Lipschitz in the collar chart. Since $\eta_2 + \kappa_1\rho_2/2 \ge 2\eta_1$ by the choice $\eta_1 \le \kappa_1\rho_2/4$, we get $\eta \equiv \eta_1$ there. Where $w = 0$ we have $\eta \equiv \eta_2 \le \eta_{\max}$ pointwise, by the definition of η_2. In between, $w > 0$ forces the point to lie outside $\bigcup_p B_{\rho_2}(p)$, hence in G', and outside $\{r \le r_1/2\}$, hence in $\{r > r_1/2\}$, so Corollary 5.9(3) gives $\eta \le \eta_1 \le \eta_1^{\text{geo}} \le \eta_{\max}$ and $|\eta'| \le \kappa_1 \le \kappa$. So (15) holds along all of Γ_t, and by slicewise non-negativity the gain over $\Gamma_t \cap G \cap \{r \ge r_1\}$ alone already accounts for ε_*.

An absolute bound on η', rather than a relative one such as $|\eta'| \le \Lambda\eta$, is what makes this fit: under a relative bound the descent from η_1 to η_2 needs arclength $\Lambda^{-1}\log(\eta_1/\eta_2)$, unbounded as $\eta_2 \to 0$, whereas the transition length available is the fixed $\rho_2/2$; under the absolute bound it needs $(\eta_1 - \eta_2)/\kappa_1 \le \eta_1/\kappa_1$, independent of η_2. That an absolute bound is the right form is visible in (21), where the loss $m^2\eta'^2 \cdot \eta$ and the gain $g(m) \cdot \eta$ are of the same order in η.

Now obtain C from Lemma 5.2, and choose $C\sigma < \varepsilon_*/4$. Running steps (a)–(d) of the transverse case with (A_t, B) in place of (S_-, S_+) and ε_0 replaced by ε_* gives

$$\begin{aligned} \operatorname{Area}(P_\uparrow) + \operatorname{Area}(P_\downarrow) &\le \operatorname{Area}(A_t) + \operatorname{Area}(B) - \varepsilon_* + C\sigma \\ &< A(w) + A(u) - \varepsilon_*/4, \end{aligned}$$

whence (d). The genus computation is topological and is transported by the isotopy, so (c) is unchanged, and (a) follows as before. □

5.5. **Configuration (T1): disjoint boundaries, interior tangencies.** This is the easier of the two non-transverse configurations, and it is treated by the same device: perturb A to a transverse competitor and bound the area lost against the area gained. What makes it easier is that the intersection stays away from $\partial E(K)$ altogether, so that the collar apparatus of §5.4 — the sign ς and the arc I, the length estimate inside the collar, the third scale of the width ramp — is not needed and is replaced by a single positive number.

Assume $\partial A \cap \partial B = \emptyset$ and $\mathcal{T} \neq \emptyset$. Since $A \cap \partial E(K) = \partial A$ and $B \cap \partial E(K) = \partial B$ we have $A \cap B \cap \partial E(K) = \partial A \cap \partial B = \emptyset$, and $A \cap B$ is compact, so

$$d_0 := \operatorname{dist}(A \cap B,\ \partial E(K)) > 0. \tag{44}$$

The perturbation. By Proposition 5.5(1) each of the finitely many $p \in \mathcal{T}$ carries a disc D_p, inside the graph chart of (30), on which p is the only critical point of f_p and $f_p(p) = 0$. Shrinking, take the D_p pairwise disjoint and contained in $\{d(\cdot, \partial E(K)) > d_0/2\}$, choose concentric smaller discs $D_p' \Subset D_p$, and take bumps ψ_p supported in D_p and equal to 1 on D_p'. Let A_t be obtained from A by replacing, over each D_p, the graph $\{y = f_p\}$ by

$$\{y = f_p - t\psi_p\}. \tag{45}$$

Then $\partial A_t = \partial A$, still a leaf of J disjoint from ∂B; A_t is smooth, neatly embedded and normally isotopic to A, hence incompressible and a representative of w; and $A_t = A$ outside a compact subset of $\operatorname{Int} E(K)$, so $A_t \cap S = \emptyset$ for small t and $A_t \to A$ in C^∞. As in (38), the perturbation is of subtract-a-constant type on D_p', and this is what the second case of Lemma 5.10(1) consumes.

Transversality holds for all small $t > 0$: on D_p' one has $A_t \cap B = \{f_p = t\}$, and $t \neq 0$ is a regular value because the only critical point of f_p in D_p is p, with critical value 0; on $D_p \setminus D_p'$ one has $|df_p| \geq c > 0$, hence $|d(f_p - t\psi_p)| \geq c/2$ for t small; elsewhere $A_t = A$, which meets B transversally away from $\mathcal{T}$. Fix $\delta^{(1)} > 0$ accordingly. For $t \in (0, \delta^{(1)}]$ the set $\Gamma_t := A_t \cap B$ is therefore a finite union of disjoint smooth closed curves — closed, because an arc would have its endpoints in $\partial A_t \cap \partial B = \emptyset$ — lying in $\{d(\cdot, \partial E(K)) \geq d_0/2\}$; and it is non-empty, for otherwise A_t and B would be disjoint representatives of w and u, contradicting $\operatorname{dist}(u, w) = 2$. Finally

$$e(t) := \operatorname{Area}(A_t) - \operatorname{Area}(A) \longrightarrow 0 \qquad (t \downarrow 0), \tag{46}$$

and $\operatorname{Area}(A) = A(w)$, A being a minimiser.

Lemma 5.10 (Bounds independent of t, tangency case)**.** *There are $\rho_0 > 0$, $\rho_2 > 0$, $\theta_1 \in (0, \pi/2]$, $\eta_1^{\mathrm{geo}} > 0$, $\kappa_1 > 0$ and $\delta_2 > 0$, depending only on A, B and the perturbation scheme, such that, writing $P := \bigcup_p B_{2\rho_2}(p)$, $G := E(K) \setminus P$ and $G' := E(K) \setminus \bigcup_p B_{\rho_2}(p)$, the following hold for every $t \in (0, \delta_2]$ and every closed component γ of Γ_t:*

1. $\operatorname{diam}(\gamma) \geq \rho_0$ *and* $\operatorname{length}(\gamma \cap G) \geq 3\rho_0/8$*;*
2. *the opening angle α satisfies the two-sided bound* $\min\{\alpha, \pi - \alpha\} \geq \theta_1$ *along $\gamma \cap G'$;*
3. *along $\gamma \cap G'$ the functions of Lemma 5.1, read pointwise, satisfy $\eta_{\max} \geq \eta_1^{\mathrm{geo}}$ and $\kappa \geq \kappa_1$.*

Proof. Each clause is the corresponding statement of §5.4 with the collar deleted, and we say what the deletion is.

Diameter. Run the proof of Lemma 5.6. Suppose $t_n \to 0$ and closed components γ_n with $\operatorname{diam}(\gamma_n) \to 0$; points $x_n \in \gamma_n$ subconverge to some $x \in A \cap B$. Two cases now exhaust the possibilities, the third case of that proof being vacuous because Γ_t misses the collar by (44). If x is a transverse point, a fixed ball at x meets $A \cap B$ in a single arc crossing it, and C^1 stability keeps that true for small t, so γ_n cannot lie in the ball. If $x = p \in \mathcal{T}$, then for large n the curve γ_n bounds a disc $D_n \subset D_p$ on which $f_p - t\psi_p$ vanishes on the boundary; it cannot vanish identically, since then its differential would vanish on D_n while p is its only critical point in D_p; so it attains a strict interior extremum at a critical point, necessarily p — but on D_p' it differs from f_p by the constant t, and by (33) p is not a local extremum of f_p. This gives $\rho_0 > 0$, valid for all sufficiently small t. Now choose $\rho_2 > 0$ with the balls $B_{2\rho_2}(p)$ pairwise disjoint, $4N\rho_2 < \rho_0/8$ and $2\rho_2 < d_0/2$; the collapsing argument of Corollary 5.9(1), which uses nothing but ρ_0 and this choice of ρ_2, gives $\operatorname{length}(\gamma \cap G) \geq 3\rho_0/8$.

Angle. Put $G'' := E(K) \setminus \bigcup_p B_{\rho_2/2}(p)$ and $\mathcal{K} := A \cap B \cap G''$, a compact set on which A meets B transversally, $\mathcal{T}$ having been excised. The restriction $\{r \geq r_1/4\}$ of Corollary 5.9(2) is not needed and has no analogue here: there it removed the tangencies along L, and here $\partial A \cap \partial B = \emptyset$. So $\min\{\alpha, \pi - \alpha\}$ has a positive infimum $2\theta_1$ on $\mathcal{K}$, and for t small the two-sided bound persists with θ_1 along $\gamma \cap G'$.

Widths. As in Corollary 5.9(3), $\underline{\eta}$ and $\underline{\kappa}$ are controlled by the angle, by bounds on the C^2 norms of the two sheets and on $\sup |m'|$, by the local reach ϱ and by ϱ_∂. The first three are bounded exactly as there. For ϱ_∂ the collar estimate is replaced by (44): the boundary of the cornered frontier lies on $\partial E(K)$, and Γ_t is at distance at least $d_0/2$ from it. For ϱ, the compactness argument of Corollary 5.9(3) applies verbatim with $\mathcal{K}$ as above. □

Proof of Theorem 5.4, configuration (T1). Put $\eta_1 := \min\{\eta_1^{\mathrm{geo}},\ \kappa_1\rho_2/4\}$ and

$$\varepsilon_* := g_0(\theta_1) \cdot \eta_1 \cdot \tfrac{3}{8}\rho_0 > 0, \tag{47}$$

fixed before t, in the order $A, B \Rightarrow$ perturbation scheme $\Rightarrow \rho_0 \Rightarrow \rho_2 \Rightarrow \theta_1 \Rightarrow (\eta_1^{\mathrm{geo}}, \kappa_1) \Rightarrow \eta_1 \Rightarrow \varepsilon_* \Rightarrow \delta \Rightarrow t$.

The width function. Put $w(x) := \min\{\rho_2,\ d(x, \bigcup_p B_{\rho_2}(p))\}$, which is 1-Lipschitz, and, after t is chosen, $\eta_2 := \min\{\eta_1,\ \min_{\Gamma_t} \eta_{\max},\ \frac{1}{2}\min_{\Gamma_t} \varrho\} > 0$; set $\eta(s) := \min\{\eta_1,\ \eta_2 + \frac{\kappa_1}{2} w(x(s))\}$, smoothed at a scale $\ll \rho_2$ so as to be C^1 with $|\eta'| \le \kappa_1$. On $\Gamma_t \cap G$ we have $w = \rho_2$, so $\eta \equiv \eta_1$ by $\eta_1 \le \kappa_1\rho_2/4$; where $w = 0$ we have $\eta \equiv \eta_2 \le \eta_{\max}$ pointwise; and where $w > 0$ the point lies in G', so Lemma 5.10(3) gives $\eta \le \eta_1 \le \eta_1^{\mathrm{geo}} \le \eta_{\max}$ and $|\eta'| \le \kappa_1 \le \kappa$. So (15) holds along all of Γ_t, and by the slicewise non-negativity of Lemma 5.1(4) the gain along a closed component γ satisfies

$$\mathcal{G}(\gamma) \ \ge\ g_0(\theta_1)\, \eta_1\, \mathrm{length}(\gamma \cap G) \ \ge\ \varepsilon_*,$$

the rest of the curve being discarded.

No disc patch, uniformly in t. If cutting A_t and B along Γ_t produced a disc patch, Lemma 5.3 — applicable because A is a minimiser, A_t is normally isotopic to it with ∂A_t a leaf, and $A_t \pitchfork B$ with $\partial A_t \cap \partial B = \emptyset$ — would give closed curves a, b of Γ_t with $2\varepsilon_* \le \mathcal{G}(a) + \mathcal{G}(b) \le e(t)$. By (46) there is $\delta_3 > 0$ with $e(t) < \varepsilon_*$, hence no disc patch, for every $t \in (0, \delta_3]$.

Conclusion. Choose $\delta \le \min\{\delta^{(1)}, \delta_2, \delta_3\}$, small enough in addition that $\mathrm{Area}(A_t) < \mathrm{Area}(A) + \varepsilon_*/2$ for $t \in (0, \delta]$, and fix such a t. Lemma 4.5 applies to the pair (A_t, B) and yields $P_\uparrow, P_\downarrow$; rounding the two cornered frontiers along Γ_t with the width η gains at least ε_* in total, and Lemma 5.2 then costs $C\sigma$, which we take below $\varepsilon_*/4$. Exactly as in configuration (T2), an ambient isotopy carrying A_{t_0} to A_t, fixing B and S and preserving $\partial E(K)$, shows that the two coarse classes, and hence the two output vertices $w_\uparrow, w_\downarrow$, do not depend on t; the construction of that isotopy used only transversality, the disjointness of the boundaries and the disjointness from S, all of which hold here. Then

$$\begin{aligned} \mathrm{Area}(P_\uparrow) + \mathrm{Area}(P_\downarrow) &\le \mathrm{Area}(A_t) + \mathrm{Area}(B) - \varepsilon_* + C\sigma \\ &< A(w) + A(u) - \varepsilon_*/4, \end{aligned}$$

which is (d); the genus computation is topological and transported by the isotopy, giving (c); and (a) follows as in the transverse case. For (b) the t-independence just established is what is needed, exactly as in configuration (T2): given z with minimiser Z disjoint from $A \cup B$, choose $t_z > 0$ after Z is known so that $Z \cap A_{t_z} = \emptyset$ and take the push-offs at t_z small enough to avoid Z; the coarse classes, and hence the vertices, are the ones already fixed, so t_z changes only a representative. □

The three configurations (T0), (T1), (T2) exhaust the possibilities, so Theorem 5.4 is proved.

5.6. Axiom (N2) for $\mathrm{IS}(K)$.

Proposition 5.11 (Apices of a two-interval)**.** *Let K be a non-trivial knot. Let u, w be vertices of* $\mathrm{IS}(K)$ *with* $\mathrm{dist}(u, w) = 2$*. Then*

1. $\mathcal{I}(u, w)$ *has an apex; indeed both outputs of Theorem 5.4, applied to u, w and any common neighbour, are apices;*
2. *some apex x of $\mathcal{I}(u, w)$ satisfies $c(x) < \max\{c(u), c(w)\}$;*

(3) *if* $g(u), g(w) \leq \ell$ *then the apex in* (2) *has* $g(x) \leq \ell$.

Consequently $(\mathrm{IS}(K), c)$ *is an exchange complex in the sense of Definition 6.1.*

Proof. (1) Let $x_0 \in \mathcal{I}(u,w)$, which is non-empty since $\mathrm{dist}(u,w) = 2$, and let y be either output of Theorem 5.4 applied to $(x_0; u, w)$. By (a), y is equal or adjacent to u and to w; it cannot equal u, since it would then be adjacent to w, contradicting $\mathrm{dist}(u,w) = 2$, and likewise it cannot equal w; so $y \in \mathcal{I}(u,w)$. By (b), every z equal or adjacent to both u and w — in particular every member of $\mathcal{I}(u,w)$ — is equal or adjacent to y. So y is an apex.

Local descent. Suppose $x \in \mathcal{I}(u,w)$ satisfies $c(x) \geq c(u)$ and $c(x) \geq c(w)$, and let $y_\uparrow, y_\downarrow$ be the outputs of Theorem 5.4 applied to $(x; u, w)$. Lexicographic comparison gives $g(x) \geq g(u)$ and $g(x) \geq g(w)$, so by (c) $g(y_\uparrow) + g(y_\downarrow) \leq g(u) + g(w) \leq 2g(x)$ and the minimum of the two genera is at most $g(x)$. If it is strictly smaller, the corresponding output y has $c(y) < c(x)$ and $g(y) \leq g(x)$. If it equals $g(x)$, then both genera equal $g(x)$, the sum is $2g(x)$, and (c) is an equality with $g(u) = g(w) = g(x)$; then $A(x) \geq A(u)$ and $A(x) \geq A(w)$, so (d) gives $A(y_\uparrow) + A(y_\downarrow) < A(u) + A(w) \leq 2A(x)$ and the output of smaller area has $g(y) = g(x)$, $A(y) < A(x)$, hence again $c(y) < c(x)$ and $g(y) \leq g(x)$.

(2) The set of complexities of apices of $\mathcal{I}(u,w)$ is a non-empty subset of W, so by Lemma 4.2(iii) it has a least element; let x_* be an apex attaining it. Suppose $c(x_*) \geq \max\{c(u), c(w)\}$. An apex is a common neighbour, so local descent applies to $(x_*; u, w)$ and produces an output y with $c(y) < c(x_*)$. By the argument in (1), y is itself an apex of $\mathcal{I}(u,w)$, contradicting the minimality of $c(x_*)$. Hence $c(x_*) < \max\{c(u), c(w)\}$, which is (2).

(3) From $c(x_*) < \max\{c(u), c(w)\}$ and the lexicographic order, $g(x_*) \leq \max\{g(u), g(w)\} \leq \ell$.

For the last sentence: $\mathrm{IS}(K)$ is non-empty by Proposition 3.5, connected by Proposition 3.2, and flag by Proposition 3.3; W is well-ordered by Lemma 4.2(iii); (N1) is clause (1) and (N2) is clause (2). ☐

Remark 5.12 (What is and is not chosen)*.* Theorem 5.4 produces its two outputs from minimisers chosen one class at a time, and different common neighbours may produce different apices; no map $(u,w) \mapsto x$ is defined anywhere above, and none is needed, because Definition 6.1 asks only for existence. This is exactly the point at which the obstruction identified in [15] is bypassed rather than solved. The passage in (2) to an apex of least complexity is a single instantiation inside a proof by contradiction, legitimate in ZF by Remark 4.3.

6. Exchange complexes

From this section to the end of §7 no three-dimensional topology is used. Nothing is quoted from §§2–5 except Definition 6.1 itself.

6.1. **The axioms.** Let X be a connected flag simplicial complex with vertex set $X^{(0)}$, and write dist for the edge-path metric on $X^{(0)}$. For u, w with $\mathrm{dist}(u,w) = 2$ the *two-interval* is

$$\mathcal{I}(u,w) \;=\; N(u) \cap N(w),$$

the set of common neighbours; it is non-empty by definition of the distance. A vertex $x \in \mathcal{I}(u,w)$ is an *apex* of the interval if x is equal or adjacent to every member of $\mathcal{I}(u,w)$; equivalently, since X is flag, the full subcomplex on $\mathcal{I}(u,w)$ is a cone with apex x.[1]

[1]The word "apex" is used in a different sense by Chepoi, Chalopin, Hirai and Osajda [3] for a vertex of a graph adjacent to all others. We keep it here because it is the natural word for a cone point, but the two usages should not be conflated.

Definition 6.1 (Exchange structure)**.** Let W be a well-ordered set — one in which every non-empty subset has a least element. An *exchange structure* on X is a map $c\colon X^{(0)} \to W$ such that

(N1) every two-interval has an apex; and
(N2) whenever $\operatorname{dist}(u, w) = 2$, *some* apex x of $\mathcal{I}(u, w)$ satisfies $c(x) < \max\{c(u), c(w)\}$.

A non-empty connected flag complex carrying such a c is an *exchange complex.*

Remark 6.2 ((N1) is implied by (N2)). Axiom (N2) asserts that *some apex* of $\mathcal{I}(u, w)$ has a certain property, and in particular that an apex exists; so (N1) follows from it. We keep both because the two are used for different purposes — (N1) alone in the manufacture of the cone structure, (N2) in the descent — and because in §8 they are verified by different arguments.

Three further comments on the definition, each of which is load-bearing.

Remark 6.3 (No apex is selected). The axioms assert that an apex *exists*; they never assert that one can be *chosen*, and no map $(u, w) \mapsto x$ is part of the data. This does not solve the difficulty identified in [15] — that a projection cannot be made well defined on isotopy classes — it avoids it. Everything below is written so that no such choice is ever needed.

Remark 6.4 (Non-emptiness cannot be dropped). The induction of Theorem 7.1 is based at "order type 1", and the empty complex, of order type 0, is not contractible.

Remark 6.5 (The choice-theoretic dependence, stated once). For a total order, "every non-empty subset has a least element" implies "no infinite strictly descending sequence" in ZF; the converse needs dependent choice. We use the well-ordering theorem, hence the axiom of choice, in Lemma 6.6, to well-order the fibres of the complexity. That is the only use in this paper, and we flag it here rather than repeatedly. In particular §7 does not add one: it treats $|X|$ as a CW complex with one open cell per simplex, and never needs the total ordering of the vertices that a Δ-complex structure would require [9, p. 107]. For the complexity actually used in §8 the descending-chain form is what Lemma 4.2 proves directly, and the passage to least elements is available there in ZF; see the remark following that lemma.

6.2. **Breaking ties.**

Lemma 6.6 (Ties may be broken)**.** *If X carries an exchange structure, then it carries one whose complexity is injective and well-orders $X^{(0)}$.*

Proof. The set of attained values is well-ordered. Well-order each fibre $c^{-1}(a)$ and order $X^{(0)}$ lexicographically, first by the old value and then within the fibre. A lexicographic sum of well-orders indexed by a well-order is a well-order, and the new complexity is injective. Axiom (N1) does not mention c. If the old (N2) had a witness x with $c(x) < \max\{c(u), c(w)\}$, then the value block of x strictly precedes the block of the larger endpoint, so the strict inequality survives the refinement. □

From here on the complexity is assumed injective.

6.3. **The ordering theorem.** An injective exchange structure well-orders the vertices, and axiom (N2) then says exactly that any two vertices at distance 2 have a common neighbour strictly below their maximum. That hypothesis — without the word apex, without flagness, and without (N1) — is already enough for the following.

Corollary 6.7. *Let (X, c) be an exchange complex with c injective and let $D \subseteq X^{(0)}$ be downward closed for c. Then the full subcomplex X_D on D is isometrically embedded in X, and is connected.*

In particular, for any two vertices p, q there is a geodesic from p to q all of whose vertices z satisfy $c(z) \leq \max\{c(p), c(q)\}$.

Proof. Let $u, w \in D$ and let $n := \mathrm{dist}_X(u, w)$, finite because X is connected. If $n \leq 1$ there is nothing to prove, so assume $n \geq 2$. Among all paths $v_0 = u, v_1, \ldots, v_n = w$ of length n in X — a non-empty set — consider

$$M(v_\bullet) := \max\{c(v_i) : 0 < i < n\} \ \in W.$$

The order on W is a well-order, so the set of attained values has a least element; fix a path attaining it.

Suppose some interior vertex has $c(v_i) > \max\{c(u), c(w)\}$, and let v_i be the interior vertex attaining M, unique because c is injective. Then $c(v_{i-1}) < c(v_i)$ and $c(v_{i+1}) < c(v_i)$: each of $v_{i\pm1}$ is either an endpoint, and then its complexity is below $c(v_i)$ by assumption, or an interior vertex, and then its complexity is at most the maximum, with equality excluded because c is injective and $v_{i\pm1} \neq v_i$. The vertices v_{i-1} and v_{i+1} are neither equal nor adjacent, since a path of length n between vertices at distance n admits no shortcut; hence $\mathrm{dist}(v_{i-1}, v_{i+1}) = 2$, and (N2) supplies $x \in \mathcal{I}(v_{i-1}, v_{i+1})$ with

$$c(x) < \max\{c(v_{i-1}), c(v_{i+1})\} < c(v_i).$$

Replace v_i by x. The result is a walk of length n from u to w; it is a path, for a repeated vertex would allow a loop to be excised and produce a walk of length $< n$, contradicting $\mathrm{dist}_X(u, w) = n$. Every interior vertex of it other than x had complexity strictly below $c(v_i)$, by injectivity, and $c(x) < c(v_i)$; so M has strictly dropped, contradicting minimality.

So every vertex of the chosen path satisfies $c(v_i) \leq \max\{c(u), c(w)\}$, and D being downward closed with $u, w \in D$, the whole path lies in X_D. Hence $\mathrm{dist}_{X_D}(u, w) \leq n = \mathrm{dist}_X(u, w)$, and the reverse inequality is automatic; X_D is isometrically embedded, and connected because these distances are finite. The last sentence of the statement is the displayed property of the chosen path, with $D = X^{(0)}$. □

Remark 6.8 (Where this sits in the literature). A considerably more general statement is [3, Lemma 9.13, p. 150]: for a well-order $\preceq$ on the vertex set of a graph such that any u, w at distance 2 admit $v \in \mathcal{I}(u, w) \setminus \{u, w\}$ with $v \prec \max\{u, w\}$, every level set $L_v = \{u : u \preceq v\}$ induces an isometric subgraph. Their hypotheses are strictly weaker than ours — no apex, no flagness, no local finiteness — and the argument above is theirs: they too take a shortest path whose largest vertex is least in the well-order and exchange at that vertex, which is what is done here with c in place of $\preceq$. We have nevertheless given it, so that §§6–7 are self-contained and no citation stands between the axioms and Theorem 7.1.

6.4. **Heredity.**

Lemma 6.9 (Heredity). *Let (X, c) be an exchange complex with c injective. For a vertex v let $X_{<v}$ be the full subcomplex on $\{u : c(u) < c(v)\}$ and let $L(v)$ be the full subcomplex on the* descending link $L_c(v) = \{u \in N(v) : c(u) < c(v)\}$. *Then*

(1) *if non-empty, $X_{<v}$ is an exchange complex;*
(2) *if non-empty, $L(v)$ is an exchange complex, and it is non-empty unless v is the least vertex.*

Proof. A full subcomplex of a flag complex is flag. Let u, w be non-adjacent vertices, either at distance 2 in $X_{<v}$ or both in $L_c(v)$. In the first case a common neighbour inside the sublevel set

witnesses $\mathrm{dist}_X(u, w) = 2$; in the second v itself is a common neighbour, so again $\mathrm{dist}_X(u, w) = 2$. By (N2) there is an apex x of $\mathcal{I}(u, w)$ with

$$c(x) \ < \ \max\{c(u), c(w)\} \ < \ c(v). \tag{48}$$

For $X_{<v}$ this places x in the sublevel set at once. For $L(v)$, the apex is equal or adjacent to every member of $\mathcal{I}(u, w)$, and $v \in \mathcal{I}(u, w)$; the strict inequality (48) excludes $x = v$, so x is adjacent to v and lies in $L_c(v)$. In both cases x lies in the smaller interval and is an apex of it, which is (N1) there, and (48) is (N2).

Connectedness of $L(v)$: two non-adjacent members have the common neighbour x just produced, so the graph diameter is at most 2. Connectedness of $X_{<v}$ is Corollary 6.7 with $D = \{u : c(u) < c(v)\}$.

Finally, if v is not the least vertex v_0, apply Corollary 6.7 to v and v_0: the vertex adjacent to v on the resulting geodesic is a neighbour $z \neq v$ with $c(z) \leq c(v)$, and injectivity gives $c(z) < c(v)$, so $z \in L_c(v)$. □

Remark 6.10 (The strict inequality is not decoration). The only point at which (48) is used with its full strength is the exclusion of $x = v$ in the descending-link case. Everything in §7.4 that fails, fails there.

6.5. **A machine-checked fragment.**

Remark 6.11. Four statements of this section and the next have been formalised in Lean 4 against `mathlib`: the ordering statement quoted in Remark 6.8, in the form used here, for an arbitrary linearly ordered well-founded W, an injective complexity, the weak form of (N2) and a possibly infinite vertex set; the reduction to an injective complexity of Lemma 6.6 without any finiteness assumption; a combinatorial reformulation of dismantlability; and the seven-vertex example of §7.3, whose four defining properties are settled by kernel evaluation. The development is 913 lines, contains no `sorry`, no `axiom` declaration and no appeal to `native_decide`, and every headline declaration depends only on `propext`, `Classical.choice` and `Quot.sound`. *Theorem 7.1 itself has not been formalised and no such claim is made.*

7. Contractibility

7.1. **Two facts about CW complexes.**

The geometric realisation $|X|$ of a simplicial complex carries the weak topology: a set is closed exactly when its intersection with every closed simplex is closed. For a complex that is not locally finite this is *strictly finer* than the metric topology, and $\mathrm{IS}(K)$ is not locally finite [12, p. 231]. All topological statements below refer to the weak topology, under which $|X|$ is a CW complex with one open cell per simplex. We use two standard facts, both cited and neither reproved.

(A) A compact subspace of a CW complex is contained in a finite subcomplex [9, Proposition A.1, p. 520].

(B) If (Y, A) is a CW pair and A is contractible, then the quotient map $Y \to Y/A$ is a homotopy equivalence. This is [9, Proposition 0.17, p. 15], whose hypothesis is the homotopy extension property, combined with [9, Proposition 0.16, p. 15], which supplies that property for a CW pair.

Fact (B) is what replaces, in this version, the appeal to "a cofibration which is a homotopy equivalence is the inclusion of a strong deformation retract". That implication is true but was quoted in version 1 without proof and without a checked reference; the argument below does not need it.

7.2. **The main combinatorial theorem.**

Theorem 7.1. *Every exchange complex is contractible.*

Proof. By Lemma 6.6 we may assume the complexity c is injective, so that it well-orders the vertex set. For an exchange complex (Z, c) with c injective, the *order type* of (Z, c) is the order type of $(Z^{(0)}, c)$; if it is α we write $Z^{(0)} = \{z_\beta\}_{\beta<\alpha}$ in increasing order and let Z_γ be the full subcomplex on $\{z_\beta : \beta < \gamma\}$, so that $Z_0 = \emptyset$ and $Z_\alpha = Z$. By Lemma 6.9(1), Z_γ with $1 \leq \gamma \leq \alpha$ is again an exchange complex with injective complexity, of order type γ.

We prove by transfinite induction on α the statement

> $(*_\alpha)$ *Every* exchange complex with injective complexity and order type at most α is contractible.

The universal form is not a convenience but a necessity: the successor step below applies the inductive hypothesis to a descending link, which is not one of the subcomplexes Z_γ of the complex being treated. Lemma 6.9(2) is what makes that application legitimate — the descending link of the largest vertex has all its vertices below that vertex, hence order type at most α, which is strictly less than $\alpha + 1$.

Base. Order type 1: a single vertex.

Successor. Let $\alpha \geq 1$, assume $(*_\alpha)$, and let Z be an exchange complex with injective complexity of order type $\alpha + 1$. Since $\alpha + 1$ is a successor, Z has a c-largest vertex v; put $Y = Z$, $A = Z_\alpha$ (the full subcomplex on the remaining vertices) and $L = L(v)$, the descending link of v. Since Y is the full subcomplex on an initial segment containing v, the neighbours of v in Y are exactly the vertices of L. A simplex of Y either omits v, and then has all its vertices below v and so lies in A; or it is $\{v\} \cup \tau$ with τ a simplex of Y contained in $L_c(v)$, hence a simplex of L because L is a full subcomplex. Conversely $L \subseteq A$, again by fullness. Hence

$$Y = A \cup (v * L), \qquad A \cap (v * L) = L,$$

the second equality because a simplex of $v * L$ lies in A exactly when it omits v. The paragraph above gives the inclusion $Y \subseteq A \cup (v * L)$; the reverse inclusion $v * L \subseteq Y$ is where flagness is used, and it should be said, because it is easy to read the displayed equality as pure bookkeeping. A simplex of $v * L$ is $\{v\} \cup \tau$ with τ a simplex of L, and $\tau \subseteq L_c(v) \subseteq N(v)$, so $\{v\} \cup \tau$ is a clique of the 1-skeleton; that it is a *simplex* of X is exactly flagness. This costs nothing — flagness is part of Definition 6.1, so $(*_\alpha)$ and Theorem 7.1 carry it — but the hypothesis is used here and not only in Lemma 6.9. Both A and L are exchange complexes with injective complexity of order type at most α (Lemma 6.9, using $\alpha \geq 1$ for the non-emptiness of L), so both are contractible by $(*_\alpha)$.

Now apply fact (B) twice. First to the CW pair $(v * L,\, L)$: the cone $v * L$ is contractible — the straight-line homotopy towards the cone point is continuous for the weak topology because $|v * L| \times I$ is again a CW complex, I being locally compact — and L is contractible, so $(v * L)/L$ is contractible. Second to the CW pair (Y, A): A is contractible, so $Y \to Y/A$ is a homotopy equivalence. Collapsing A collapses $L \subseteq A$, and no cell of $v * L$ outside L lies in A, so

$$Y/A \;=\; (v * L)/L,$$

which we have just seen to be contractible. Therefore Y is contractible.

Limit. Let λ be a limit ordinal, assume $(*_\alpha)$ for every $\alpha < \lambda$, and let Z be an exchange complex with injective complexity of order type λ. Then $|Z| = \bigcup_{1 \leq \alpha < \lambda} |Z_\alpha|$, an increasing union of non-empty subcomplexes, each contractible by $(*_\alpha)$, and Z is connected by Corollary 6.7. Let $n \geq 0$ and let $f \colon S^n \to |Z|$ be continuous. Its image is compact, hence by fact (A) meets only finitely many simplices; those simplices have finitely many vertices in total, and each has index $< \lambda$, so there is $\alpha < \lambda$ with $f(S^n) \subseteq |Z_\alpha|$ — here λ being a limit is used to find a single α strictly above the finitely many indices. Since Z_α is contractible, f is null-homotopic in $|Z_\alpha|$, hence in $|Z|$. Thus $\pi_n(|Z|) = 0$ for all $n \geq 0$, and $|Z|$ is a CW complex, so it is contractible by Whitehead's theorem [9, Theorem 4.5, p. 346, and the remark on p. 348].

Applying $(*_\kappa)$, where κ is the order type of (X, c), to X itself gives the theorem. □

Remark 7.2. No step of the proof selects an apex. Axiom (N2) is used only inside Lemma 6.9, and there only to assert that *some* vertex with the stated properties exists; the induction never transports a choice from one stage to the next. This is the whole reason the argument applies to $\mathrm{IS}(K)$ and not merely to $\mathrm{MS}(K)$.

7.3. **The exchange axioms do not give a dismantling.** A vertex u of a graph is *dominated* by w if $N[u] \subseteq N[w]$, and a finite graph is *dismantlable* if repeated deletion of dominated vertices reduces it to a single vertex [15, Definition 7.1, p. 1503]. Deleting a dominated vertex is exactly deleting a vertex whose link is a cone, and it is by this mechanism that dismantlability yields contractibility [2].

Axioms (N1) and (N2) do *not* deliver that mechanism. What they deliver, through Lemma 6.9(2), is a descending link of diameter at most 2 — contractible, by Theorem 7.1 applied to it — and a complex of diameter 2 need not be a cone.

Proposition 7.3. *There is a finite exchange complex X on seven vertices, with complexity the identity on $\{0, 1, \ldots, 6\}$, whose descending link at the top vertex is contractible but is not a cone.*

Proof. Let X be the flag complex on $\{0, \ldots, 6\}$ generated by the hexagon 0–1–2–3–4–5–0, the three chords 0–2, 2–4, 4–0, and the six edges joining 6 to each of $0, \ldots, 5$. Take $c = \mathrm{id}$.

The descending link of 6 is the full subcomplex L on $\{0, \ldots, 5\}$. Its maximal simplices are the triangles $\{0, 1, 2\}$, $\{2, 3, 4\}$, $\{4, 5, 0\}$ and $\{0, 2, 4\}$: a triangulated hexagonal disc, hence contractible. It is not a cone, because a cone point would have to be adjacent to all five other vertices of L, and none is: inside L the vertices $1, 3, 5$ have exactly two neighbours each, while 0 misses 3, 2 misses 5 and 4 misses 1.

That X satisfies (N1) and (N2) for $c = \mathrm{id}$ is a finite verification over the pairs at distance 2; it has been carried out by kernel evaluation in Lean 4 (Remark 6.11), as has the failure of the cone property. □

Remark 7.4 (What Proposition 7.3 does and does not say)*.* It says that the exchange structure c is not a dismantling order: one cannot run the induction of Theorem 7.1 by deleting dominated vertices in the order given by c. It does *not* say that this particular X is undismantlable — it is dismantlable, since $N[1] = \{0, 1, 2, 6\} \subseteq N[0]$ — and no such claim is needed. The point is that the axioms hand one a contractible descending link and nothing stronger, so Theorem 7.1 is not a corollary of the dismantlability theory even in the finite case; and in the infinite case, which is the case of $\mathrm{IS}(K)$, there is no dismantlability theory to be a corollary of.

7.4. **The relation to the descending-link criterion.** The remaining comparison is with Bestvina–Brady Morse theory in the form given by Zaremsky, which applies to arbitrary simplicial complexes.

Definition 7.5 ([20, Definition 2.1])**.** A map $h \colon K \to \mathbb{R}$ on a simplicial complex K is a *Morse function* if the image $h(K^{(0)})$ is a closed discrete subset of $\mathbb{R}$ and the restriction of h to any simplex takes distinct values on the vertices of that simplex. For $t \in \mathbb{R}$ let $K^{h \leq t}$ denote the full subcomplex on $\{v : h(v) \leq t\}$, and let $\mathrm{lk}^{\downarrow}_h(v)$ denote the full subcomplex of $\mathrm{lk}(v)$ on $\{u \in \mathrm{lk}(v) : h(u) < h(v)\}$.

When h is injective on vertices, this $\mathrm{lk}_h^{\downarrow}(v)$ agrees with the descending link of [20, Definition 2.2], which is defined through the descending star: a simplex of $\mathrm{lk}(v)$ all of whose vertices are h-below v spans, together with v, a simplex having v as its maximum.

Theorem 7.6 ([20, Lemma 2.3])**.** *Let $h\colon K \to \mathbb{R}$ be a Morse function and let $s < t$ in $\mathbb{R}$. If $\mathrm{lk}_h^{\downarrow}(v)$ is contractible for every vertex v with $s < h(v) \leq t$, then the inclusion $K^{h \leq s} \hookrightarrow K^{h \leq t}$ is a homotopy equivalence. If $\mathrm{lk}_h^{\downarrow}(v)$ is contractible for every vertex v with $h(v) > t$, then the inclusion $K^{h \leq t} \hookrightarrow K$ is a homotopy equivalence.*

The statement is relative, and necessarily so. It concludes that an inclusion is a homotopy equivalence, never that K is contractible, and it could not conclude the latter: for K a disjoint union of two copies of the bi-infinite line with h the vertex index, every descending link is a single point while K is not even connected. To obtain contractibility from Theorem 7.6 one must exhibit a level t at which $K^{h \leq t}$ is already contractible — in practice, a level below which nothing lies.

There is no local finiteness hypothesis here. Consequently the fact that $\mathrm{IS}(K)$ fails to be locally finite is by itself no obstruction to applying Theorem 7.6, and we make no novelty claim on that basis.

What does obstruct a direct application is the order type. Zaremsky's h takes values in a closed discrete subset of $\mathbb{R}$, and such a subset is order-isomorphic to a subset of $\mathbb{Z}$; in particular it contains no increasing ω-sequence with a supremum belonging to it. The complexity of §4 is not of this kind: by Lemma 4.2 the set of values attained by $c = (g, A)$ has order type at most ω^2, and we do not know that it is at most ω. So c cannot simply be substituted for h, and the question is whether it can be replaced. The honest comparison runs through the following condition.

Definition 7.7. An exchange complex (X, c) satisfies (F) if every descending link $L_c(v)$ is finite.

Condition (F) is implied by local finiteness and by c having order type at most ω with finite fibres, but is weaker than either.

Theorem 7.8 (Reindexing under (F))**.** *Let (X, c) be a countably infinite exchange complex with c injective and satisfying* (F)*. Then there is a bijection $h\colon X^{(0)} \to \mathbb{N}$ such that $\mathrm{lk}_h^{\downarrow}(v) = L_c(v)$ for every vertex v.*

Proof. Orient each edge $\{u, w\}$ of X from the c-smaller to the c-larger end and let P be the reflexive transitive closure of this relation; P is a partial order because c is strictly increasing along oriented edges. Fix a vertex v and consider the tree T_v of finite descending paths $v = u_0, u_1, \ldots, u_k$ with $u_{i+1} \in L_c(u_i)$. Each node has finitely many children by (F), and T_v has no infinite branch because c decreases strictly along a branch and the values are well-ordered. By König's lemma T_v is finite, so v has finitely many P-predecessors.

A countable partial order in which every element has finitely many predecessors admits a bijective linear extension onto $\mathbb{N}$: enumerate $X^{(0)}$ as $w_0, w_1, \ldots$ and define h by recursion, at stage n assigning the least unused natural numbers, in any P-compatible order, to those P-predecessors of w_n not yet assigned and then to w_n itself. This is possible because the set of P-predecessors of w_n is finite and its own predecessors are already handled; every vertex is reached by stage n at the latest, so h is onto, and it is injective and P-increasing by construction.

For a vertex v and a neighbour u: if $c(u) < c(v)$ then $u\,P\,v$ and $h(u) < h(v)$; if $c(u) > c(v)$ then $v\,P\,u$ and $h(u) > h(v)$. Hence $\mathrm{lk}_h^{\downarrow}(v) = L_c(v)$. ∎

Corollary 7.9. *Let (X, c) be an exchange complex with c injective which is either finite, or countably infinite and satisfying* (F)*. Then X is contractible. The proof uses Theorem 7.6, Lemma 6.9 and an induction on the number of vertices; it does not use Theorem 7.1.*

Proof. Finite case, by induction on the number n of vertices. For $n = 1$ there is nothing to prove. Let $n \geq 2$ and let $h\colon Z^{(0)} \to \{0, \ldots, n-1\}$ be the rank function of c on a finite exchange complex Z with c injective. Its image is finite, hence closed and discrete in $\mathbb{R}$, and h is injective, so h is a Morse function; and $\mathrm{lk}_h^{\downarrow}(v) = L_c(v)$ because h and c induce the same order. For each v with $h(v) > 0$ the descending link $L(v)$ is a non-empty exchange complex by Lemma 6.9(2), with at most $n-1$ vertices, hence contractible by the inductive hypothesis. Since $Z^{h\leq 0}$ is a single vertex, the second half of Theorem 7.6 with $t = 0$ makes $Z^{h\leq 0} \hookrightarrow Z$ a homotopy equivalence, so Z is contractible.

Countably infinite case. By Theorem 7.8 there is a bijection $h\colon X^{(0)} \to \mathbb{N}$ with $\mathrm{lk}_h^{\downarrow}(v) = L_c(v)$; since $\mathbb{N} \subset \mathbb{R}$ is closed and discrete and h is injective, h is a Morse function. Here $h^{-1}(0)$ is the c-least vertex v_0, which is not something Theorem 7.8 says: it gives a bijection matching descending links, not one matching least elements. It does follow. The vertex $u := h^{-1}(0)$ has $\mathrm{lk}_h^{\downarrow}(u) = \emptyset$, hence $L_c(u) = \emptyset$; and the last paragraph of the proof of Lemma 6.9 produces a c-smaller neighbour for every vertex other than v_0, so $u = v_0$. The point matters because Lemma 6.9(2) asserts non-emptiness of the descending link only away from the least vertex, and the empty complex is not contractible. For each v with $h(v) > 0$ we therefore have $v \neq v_0$, so the descending link $L(v)$ is a non-empty exchange complex by Lemma 6.9(2) and is finite by (F), hence contractible by the finite case. Again $X^{h\leq 0}$ is a single vertex, and the second half of Theorem 7.6 with $t = 0$ gives the result. (In the finite case above the same point is free: h is the rank function of c, so $h(v_0) = 0$ by construction.) □

So under (F) the combinatorial theorem is Zaremsky's Morse lemma together with the heredity of the axioms. We state the converse plainly, because it delimits what is left.

Proposition 7.10. *The conclusion of Theorem 7.8 implies* (F)*.*

Proof. If $h\colon X^{(0)} \to \mathbb{N}$ is injective with $\mathrm{lk}_h^{\downarrow}(v) = L_c(v)$, then $L_c(v)$ injects into $\{0, 1, \ldots, h(v) - 1\}$. □

So the reindexing route is available exactly under (F), and the residual content of Theorem 7.1 is confined to exchange complexes that possess an infinite descending link *or are uncountable*; the second class has to be named separately because Corollary 7.9 assumes the complex finite or countably infinite, and it is non-empty, as the cone on ω_1 vertices over a single vertex v_0 shows. Two things should be said about that residue.

Remark 7.11 (The residue is not obviously empty). Proposition 7.10 shows only that the height function produced by Theorem 7.8 is unavailable when (F) fails, not that Theorem 7.6 is inapplicable. What would be needed is a Morse function h whose descending links are all contractible *and* a level t with $X^{h\leq t}$ contractible. The order-type obstruction blocks $h = c$; and an h unbounded below, which is what would allow a vertex infinitely many descending neighbours, makes the second requirement harder, there being then no bottom level to start from. We can neither produce such an h nor rule one out.

Remark 7.12 (Where our application sits). We do not know whether $\mathrm{IS}(K)$ satisfies (F). The condition must be asked of the tie-broken complexity of Lemma 6.6, and for a general exchange complex the choice of tie-break matters; for $\mathrm{IS}(K)$ it does not, because the fibres of c are finite by Lemma 4.2(i), so the descending link differs from $\{u \in \mathrm{lk}(v) : c(u) < c(v)\}$ by a finite set. Thus

(F) is equivalent to a statement about $c = (g, A)$ itself: *for every vertex v, the set of $u \in \mathrm{lk}(v)$ with $c(u) \le c(v)$ is finite.* Concretely, can a vertex of $\mathrm{IS}(K)$ have infinitely many neighbours of strictly smaller genus? That is the only way the set can be infinite. Kakimizu's example of non-local-finiteness [12, p. 231] produces infinitely many neighbours but does not control their genus. If the answer is no then, $\mathrm{IS}(K)$ having countably many vertices by Lemma 4.2(iv), Corollary 7.9 makes the combinatorial half of this paper Theorem 7.6 plus bookkeeping, and the contribution is the geometry of §§2–5 alone.

8. The Kakimizu complex, and what the method reaches

8.1. **Proof of Theorem A.**

Proof of Theorem A. Let K be a non-trivial knot. We check that $\mathrm{IS}(K)$, with the complexity $c(v) = (g(v), A(v)) \in \mathbb{N} \times \mathbb{R}_{>0}$ of §4 ordered lexicographically, is an exchange complex in the sense of Definition 6.1.

$\mathrm{IS}(K)$ is non-empty, and it is a flag complex by Proposition 3.3: a set of classes admits simultaneously pairwise disjoint representatives as soon as it does so pairwise. It is connected by Kakimizu's Theorem A [12, p. 226], in the incompressible case of that statement, and its edge-path metric is the distance d [12, Proposition 3.1(1), p. 231].

The complexity takes values in a well-ordered set: this is Lemma 4.2, which is where (E) and the finiteness argument of §4 are consumed. Axiom (N1) then holds because $\mathrm{IS}(K)$ is connected — two vertices at distance 2 have a common neighbour — together with Proposition 5.11, which produces an apex of the interval from the flagness of $\mathrm{IS}(K)$ and the exchange lemma. Axiom (N2) is Theorem 5.4, whose proof consumes (U), (E), (R1) — that is, Theorem 2.10 — and the collar conditions (34), (33) of §5.3.

Theorem 7.1 now applies, and $\mathrm{IS}(K)$ is contractible. □

8.2. **Truncations: proof of Theorem C.**

Proof of Theorem C. Fix $\ell \ge g(K)$ and let $D_\ell = \{v : g(v) \le \ell\}$. Since the order on $\mathbb{N} \times \mathbb{R}_{>0}$ is lexicographic with genus in the first coordinate, $c(u) \le c(v)$ and $g(v) \le \ell$ force $g(u) \le \ell$; so D_ℓ is downward closed. It is non-empty because $\ell \ge g(K)$. By Corollary 6.7 the full subcomplex $\mathrm{IS}_\ell(K)$ on D_ℓ is isometrically embedded in $\mathrm{IS}(K)$ and connected, and by Lemma 6.9(1) it is an exchange complex; contractibility is Theorem 7.1.

For the second family, fix in addition $a_0 > 0$ and let $D_{\ell,a_0} = \{v : g(v) < \ell\} \cup \{v : g(v) = \ell,\ A(v) \le a_0\}$. This is again downward closed for the lexicographic order, and non-empty as soon as $\ell > g(K)$, or $\ell = g(K)$ and a_0 is at least the least relative area in genus $g(K)$. The same two citations apply verbatim. □

Taking $\ell = g(K)$ recovers [15, Theorem 1.1, p. 1490], which is stated there for the complex of spanning surfaces of minimal Thurston norm; we do not claim that case. The truncations $\mathrm{IS}_\ell(K)$ for $\ell > g(K)$, and the second family, are new, and they cost nothing: the proof above adds no hypothesis, no citation and no construction to the proof of Theorem A. The reason is structural. The argument of §7 never uses the complex as a whole; it only ever uses initial segments of the complexity, and a truncation *is* an initial segment. A proof organised around a global projection would not have this property.

Remark 8.1. The *connectedness* of $\mathrm{IS}_\ell(K)$ is part of Theorem C and is obtained in the proof above from Corollary 6.7, hence ultimately from the connectedness of $\mathrm{IS}(K)$ itself; it also has a direct source, [4, Proposition 5.15, p. 9 of v4], which we do not use.

8.3. **Links: proof of Theorem D.** Kakimizu's complexes are defined for links, and it is natural to ask how much of the above survives. The obstacle is not the combinatorics but the definition of a vertex.

Before the definition, one point of vocabulary has to be settled, because for $n \geq 2$ the boundary slope of a spanning surface is not the preferred longitude. For $n \geq 2$ the intersection $\lambda_i := \Sigma \cap \partial N(L_i)$ of a spanning surface with the i-th boundary torus is the *surface-framed longitude*: since $\Sigma \subset E(L)$ misses L_i, its boundary bounds in the complement of L_i, so $\mathrm{lk}(\partial\Sigma, L_i) = 0$ and therefore

$$\mathrm{lk}(\lambda_i, L_i) \;=\; -\sum_{j \neq i} \mathrm{lk}(L_j, L_i), \tag{49}$$

a slope depending only on L. The foliation J_i of T_i used in §2 is taken by curves of this slope. The total linking number defines a homomorphism $\varphi\colon H_1(E(L)) \to \mathbb{Z}$ sending every meridian to 1, and, $[c] = \sum_j \mathrm{lk}(c, L_j)\,[m_j]$ for a curve c in $E(L)$,

$$\varphi(\lambda_i) = \mathrm{lk}(\lambda_i, L_i) + \sum_{j \neq i} \mathrm{lk}(L_i, L_j) = 0$$

by (49). So each λ_i lifts to a closed curve in the infinite cyclic cover determined by φ, and the lifting argument of Lemma 4.5 applies verbatim.

Definition 8.2. A link $L = L_1 \cup \cdots \cup L_n$ in S^3 is *linking-indecomposable* if for every partition $\{1, \ldots, n\} = I \sqcup I^c$ into two non-empty parts there is an index m with $\mathrm{lk}(L_m, L_J) \neq 0$, where J is the part not containing m and $L_J = \bigcup_{j \in J} L_j$. For $n = 1$ the condition is vacuous.

Lemma 8.3. *Let L be linking-indecomposable. Then every spanning surface for L in $E(L)$ — that is, every properly embedded orientable surface whose boundary is the union of the n longitudes — is connected, and has exactly n boundary curves. Moreover the property is inherited by the surfaces produced by the frontier construction of §5.*

Proof. Suppose F is a spanning surface with a decomposition $F = F' \sqcup F''$ into non-empty unions of components. A spanning surface has no closed component, so every component of F carries at least one longitude; let I be the set of indices whose longitude lies in $\partial F'$ and I^c the set for F''. These are non-empty and partition $\{1, \ldots, n\}$, the components of F being disjoint.

Write $L_I = \bigcup_{i \in I} L_i$. The chain F' lies in $E(L)$, hence misses L, and $\partial F'$ is the union of the longitudes λ_i, $i \in I$, each of which is isotopic to L_i inside the solid torus $N(L_i)$ by an isotopy disjoint from every other component. Therefore

$$[L_I] = [\partial F'] = 0 \quad \text{in } H_1(S^3 \setminus L_{I^c}).$$

For $j \in I^c$ the linking number $\mathrm{lk}(L_I, L_j)$ is the image of $[L_I]$ under $H_1(S^3 \setminus L_j) \cong \mathbb{Z}$, and the inclusion $S^3 \setminus L_{I^c} \hookrightarrow S^3 \setminus L_j$ carries the vanishing class to it; so $\mathrm{lk}(L_j, L_I) = \mathrm{lk}(L_I, L_j) = 0$ for every $j \in I^c$. The same argument applied to F'' gives $\mathrm{lk}(L_m, L_{I^c}) = 0$ for every $m \in I$. As I and I^c are the two parts of a partition into non-empty sets, this says $\mathrm{lk}(L_m, L_J) = 0$ for *every* index m, with J the part not containing m, contradicting linking-indecomposability.

Two things about this deserve emphasis, since each is a place where a shorter argument would fail. First, the conclusion needs *both* halves: F' alone yields $\mathrm{lk}(L_j, L_I) = 0$ for $j \in I^c$ and nothing about the indices $m \in I$, and it is the second application, to F'', that supplies those. Only with the two together does one have $\mathrm{lk}(L_m, L_J) = 0$ for every m, which is the negation of linking-indecomposability. Second, note which factor carries the surface: the bounding chain F' controls the class of the *union* L_I, and it is paired with a *single* component on the other side; the pairing in the opposite direction is not available, since a component of F' may carry several longitudes and then bounds only their sum.

So F is connected, its boundary consists of the n longitudes, and there are exactly n of them. The frontier construction outputs surfaces which are again spanning surfaces for L, so the same argument applies to them. □

The compression step of Lemma 3.6 needs a companion in the link setting, for the same reason: "keep the component with boundary" has to be unambiguous. We number it beside Lemma 8.3, which it uses.

Lemma 8.3′ (Compression of a spanning surface for a link)**.** *Let L be linking-indecomposable and let F be a connected spanning surface for L, with n boundary curves. Compressing F along an essential circle that separates F produces two disjoint surfaces F_1, F_2. Exactly one of them carries boundary, and it carries all n boundary curves; so "keep the component with boundary" is unambiguous, and the genus of that component strictly decreases. Compression along a non-separating circle keeps the surface connected and again lowers the genus.*

Proof. If both F_1 and F_2 carried boundary curves, then $F_1 \sqcup F_2$ would be a disconnected spanning surface for L — it is properly embedded, orientable, and its boundary is the union of the n longitudes, compression having altered F only in the interior — contradicting Lemma 8.3. So exactly one side, say F_1, carries boundary, and it carries all n curves; F_2 is closed and is discarded. The compressing circle being essential, F_2 is not a sphere, so $\chi(F_2) \leq 0$ and $\chi(F_1) = \chi(F) + 2 - \chi(F_2) \geq \chi(F) + 2$; with n fixed and $\chi = 2 - 2g - n$ this makes the genus drop by at least one. The non-separating case is that of Lemma 3.6 verbatim. □

Lemma 8.3 is what makes the link case tractable, and it does so by removing a discrepancy rather than by proving something about surfaces. Kakimizu's vertices are isotopy classes of spanning surfaces that are allowed to be disconnected and are required only to be incompressible componentwise; ours are connected. On a linking-indecomposable link the two notions coincide, and with them the two edge conventions: the convention of [12, (1.3)(b)], that disjointness is equivalent to distance 1, is false for links in general precisely because of disconnected surfaces, and here that failure cannot occur.

Proof of Theorem D. Let L be non-split, linking-indecomposable and not the unknot. By Lemma 8.3 the vertex set of $\mathrm{IS}(L)$ is the set of isotopy classes of connected incompressible spanning surfaces. Connectedness of $\mathrm{IS}(L)$ is Kakimizu's Theorem A for a non-split link [12, p. 226] — this is where non-splitness is used, through the irreducibility of $E(L)$ — while flagness is Proposition 3.3, whose proof consumes only (U) and (E) and runs unchanged once every spanning surface is known to be connected. The two are separate inputs and neither is a corollary of the other.

The complexity is again (g, A), now with g the genus of the connected spanning surface. The geometric inputs are available in n-component form: (U) and (E) are stated in [16] with the constraint $\partial F \subset J$ and with ∂M allowed to be a union of tori, the second standing hypothesis of the appendix being imposed separately on each boundary component [16, Theorem 4, p. 885 = Theorem A.5, p. 899, for (U); Theorem 2, p. 885 = Corollary A.4, p. 898, for (E)]. Both results rest on the appendix, which assumes that the surface is connected and that it meets each boundary torus in a single curve [16, p. 897]. Both hold here: by Lemma 8.3 every spanning surface representing a vertex of $\mathrm{IS}(L)$ is connected, and a connected spanning surface for an n-component link meets each of the n tori in exactly one curve. So the two inputs transfer without change, and so does (R1) in the n-component form recorded after Theorem 2.10. The argument of §5 then runs with three modifications:

(1) *The exchange lemma on n boundary tori.* Everything in §§5.3–5.5 is local to a single boundary torus apart from finitely many global constants, and the bookkeeping is as follows.

On each torus T_i the curves $\partial_i A$ and $\partial_i B$ are leaves of J_i, hence equal or disjoint; let $E \subseteq \{1, \ldots, n\}$ be the set of indices at which they are equal. The trichotomy of §5.3 becomes: $E = \emptyset$ and $A \pitchfork B$, which is (T0); $E = \emptyset$ with interior tangencies, which is (T1) and needs no change whatever, the proof there never mentioning the boundary except through (44); and $E \neq \emptyset$, which is (T2).

In the last case Proposition 5.5(1) is unchanged, its proof being local at an interior tangency point. Clause (2) is applied on each T_i with $i \in E$ separately, in the collar chart (s, r_i, y) of that torus, the collars being taken disjoint; it yields $r_0^{(i)} > 0$ and $\mathcal{T} \subseteq \bigcap_{i \in E}\{r_i \geq r_0^{(i)}\}$, and clause (3) then gives $\mathcal{T}$ finite as before. Lemma 5.7 gives on each such torus a sign ς_i, an arc $I_i \subset L_i$ and constants $a_0^{(i)}, r_1^{(i)}$. The perturbation (38) becomes

$$A_t = \Big\{ y = f + t \sum_{i \in E} \varsigma_i \, \chi_i(r_i) \; - \; t \sum_{p \in \mathcal{T}} \psi_p \Big\},$$

with χ_i supported in the collar of T_i; on a torus with $i \notin E$ nothing is done, $\partial_i A_t = \partial_i A$ being already disjoint from $\partial_i B$. The sign of the interior bumps is immaterial, (33) being a statement about both signs at once, and that is what lets the boundary signs be chosen independently torus by torus.

Lemma 5.6 and Lemma 5.8(i),(ii) are proved one collar at a time. In Lemma 5.8(iii) the one genuinely new point is that a closed component γ of Γ_t may visit several collars, so the three cases must be read per torus. If γ meets no set $\{r_i < r_1^{(i)}\}$, Case A applies and gives length$(\gamma \cap G) \geq 3\rho_0/8$. Otherwise fix an i with $\gamma \cap \{r_i < r_1^{(i)}\} \neq \emptyset$; the collars being disjoint, either γ stays inside the collar of T_i, which is Case C for that torus, or it leaves it and then $\max_\gamma r_i > r_0^{(i)}/2$, which is Case B for that torus. Both cases use only the behaviour of γ inside that one collar, so the per-torus statement suffices, with $c_\star$ the minimum of $3\rho_0/8$, of the numbers $r_0^{(i)}/2$ and of the arclengths $\ell(I_i)$, $i \in E$, divided by the largest of the n bi-Lipschitz constants of the collar charts. Corollary 5.9 is unchanged, its constants being the minima of the per-torus ones over finitely many tori, and the ramp (43) acquires one collar term for each $i \in E$, the single entry $d\big(x, \{r \leq r_1/2\}\big)$ being replaced by $\min_{i \in E} d\big(x, \{r_i \leq r_1^{(i)}/2\}\big)$. With these readings (41) and the proof of Theorem 5.4 go through word for word.

Lemma 5.3 needs no change: of A and B its proof uses only that each is connected, which is Lemma 8.3, that each has non-empty boundary on $\partial E(L)$, and that $E(L)$ is irreducible. In Lemma 4.5 the boundary count is made torus by torus: over T_i the lifted boundary is an annulus containing the two disjoint circles $\partial_i S_0^+$ and $\partial_i S_0^-$, one above the other, and the argument given there sends them to opposite frontiers. Which frontier receives $\partial_i S_0^+$ may depend on i, and nothing requires it not to; either way each frontier meets each T_i in exactly one curve, hence is a spanning surface for L once its closed components are discarded, hence is connected by Lemma 8.3 and carries all n boundary curves.

(2) Lemma 8.3 is consumed at four places, and it is worth listing them rather than gesturing at them:
   (i) in Proposition 3.4, where the alternative "coincide" is turned into $x = u$ — a step that needs the surface to be connected, since otherwise two distinct vertices may share a component;

(ii) in Lemma 4.5, where each frontier output must have exactly one component with boundary and that component must carry all n boundary curves;
(iii) in Lemma 8.3′, where a separating compression is not allowed to separate the boundary curves, so that "keep the component with boundary" names a single surface;
(iv) in aligning our vertex set and edge convention with Kakimizu's, as explained after Lemma 8.3.

(3) The knot argument invokes, through Lemma 4.5, the assertion that a Seifert surface represents a primitive generator of $H_2(E(K), \partial E(K))$. For a link $H_2(E(L), \partial E(L)) \cong \mathbb{Z}^n$ and the assertion is *false* as stated. It is replaced by the following, which is what the argument actually needs: the total linking number defines a homomorphism $\pi_1(E(L)) \to \mathbb{Z}$, and a spanning surface for L is Poincaré dual to a primitive class, namely to the image of the generator under the induced map on H^1. Primitivity holds because the homomorphism is onto, each meridian mapping to 1.

Axiom (N1) and axiom (N2) are then obtained exactly as in the proof of Theorem A, and Theorem 7.1 gives the result. ☐

Remark 8.4 (The hypothesis is load-bearing). Linking-indecomposability is not a convenience. It is used in Lemma 8.3, and through it at the four points listed in the proof of Theorem D; removing it breaks each of them separately. The last of the four is what aligns the two vertex definitions and the two edge conventions. For connected sums, the structure of IS is governed by [1] rather than by anything proved here.

8.4. **Where the method stops.** *General links.* For a link which is not linking-indecomposable the failing step is not the connectedness of IS(L), which Kakimizu proves for every non-split link, nor the combinatorics of §§6–7. It is the hypothesis $S \cap (S_+ \cup S_-) = \emptyset$ of Lemma 4.5, supplied for knots by (U) through Proposition 3.4. For a link, two distinct vertices may share the isotopy class of one component of the surface; the minimisers then *coincide* on that component rather than being disjoint from it, and the opening sentence of the proof of Theorem 5.4 is false. This is an obstruction to *this proof* only: we have constructed no example realising the failure, and we have not checked whether the appendix of [16] yields the stronger dichotomy "disjoint or identical", which would repair the step.

Equivariance. Let G be a finite group of symmetries of K acting on IS(K). One would like IS$(K)^G$ to be contractible, since [15, §8] obtains a symmetric minimal-genus Seifert surface in that way. The obstruction here is structural. To run Theorem 7.1 on IS$(K)^G$ one needs, for u, w fixed by G at distance 2, a G-fixed apex of smaller complexity. The set of least-complexity apices spans a G-invariant simplex, by flagness, but G may permute its vertices without fixing one; and a G-invariant simplex is not a G-fixed vertex. This is the exact mirror of the paper's advantage over [15]: their projection is canonical, hence equivariant, and that canonicity is what confines them to MS(K); ours requires existence and not selection, which is what reaches IS(K). One cannot have both by this route. Nor is there a shortcut through fixed-point theory: Floyd and Richardson [5, §3, pp. 74–75] construct a simplicial A_5-action without fixed points on a finite complex $|L|$ which they prove acyclic and simply connected, hence contractible by the Hurewicz theorem and [9, Theorem 4.5, p. 346]; the general picture is Oliver's [14].

Haken manifolds. For a Haken manifold M with a boundary pattern one may define IS(M, γ, α) in the same way, and every step of §§4–7 that we have checked goes through with one exception: we know of no source for the *connectedness* of IS(M, γ, α), Kakimizu's proof being written for links in S^3 and using the infinite cyclic cover. Since Definition 6.1 requires connectedness outright, we state no theorem; the missing input is one sentence, and if it is supplied the rest of this paper applies verbatim.

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